

Weather Shocks and Structural Transformation: Micro Evidence from Africa

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Abstract

Structural transformation, the movement of labor from agriculture to more productive non-agricultural activities, is a central driver of economic development, yet it has remained slow in Sub-Saharan Africa. Existing theories emphasize different stages of this process, including labor release from agriculture (Lewis, 1954), labor reallocation under transition frictions, and labor absorption into non-agricultural employment (Harris and Todaro, 1970). This paper develops a structural framework that unifies these perspectives by decomposing structural transformation into three sequential stages: labor release, labor reallocation, and labor absorption. Using household panel data from Niger, Malawi, and Ethiopia, I first employ a Random Forest algorithm to construct a measure of weather-induced agricultural productivity and exploit the resulting productivity shocks to estimate the causal effect of agricultural productivity on structural transformation outcomes. I then estimate the structural model using the simulated method of moments to identify the economic constraints governing each stage of the transformation process and conduct counterfactual policy analysis. The results show that agricultural productivity consistently releases labor from agriculture across all three countries, but the subsequent transition differs markedly. Subsistence constraints are the primary barrier in Niger and Malawi, whereas labor reallocation frictions and limited labor absorption constrain structural transformation in Ethiopia. These findings demonstrate that labor release alone is insufficient for successful structural transformation and that effective policy depends on identifying the country-specific binding constraint at each stage of the transformation process.

1 Introduction

Structural Transformation, the movement of labor from low-productivity agriculture to more productive non-agriculture, is a central driver of economic growth in developing countries. (Lewis, 1954; Syrquin, 1988) Yet, the pace of structural transformation has been considerably slower in Sub-Saharan Africa than in other developing regions (Morsy et al., 2023; Mensah et al., 2018; Fox et al., 2017). Agriculture continues to employ the majority of workers in many African countries despite its relatively low productivity (Christiaensen and Maertens, 2022; Herrendorf et al., 2022; Barrett et al., 2017; McMillan and Headey, 2014). This persistence concentration of labor in agriculture limits productivity growth and slows economic development. Therefore,

understanding why labor does not move successfully into more productive non-agricultural activities is a central question for development economics.

Agricultural productivity is viewed as a key driver of structural transformation in developing countries (Lewis, 1954; Gollin et al., 2002; Ngai and Pissarides, 2007; Gollin et al., 2007). However, the response of labor allocation to productivity improvements remains theoretically ambiguous. Higher agricultural productivity may relax subsistence constraints and release labor from agriculture (Benjamin, 1992; Alvarez-Cuadrado and Poschke, 2011; Bustos et al., 2016; Gollin, 2023), but it may also increase the returns to farming and encourage households to retain labor in agricultural production (Bustos et al., 2016; Karayalcin and Pintea, 2022). Even when productivity improvements release labor from agriculture, workers may still face barriers to entering productive non-agricultural employment. As a result, similar improvements in agricultural productivity may generate very different patterns of structural transformation across countries.

These different responses correspond to distinct stages emphasized in existing theories of structural transformation. Lewis (1954) argues that improvements in agricultural productivity relax subsistence constraints and release surplus labor from agriculture. Under this view, structural transformation is primarily limited by the ability of households to meet their subsistence needs while reducing agricultural labor. More recent studies emphasize labor reallocation, showing that mobility costs (Bryan et al., 2014; Donaldson, 2018), land institutions (De Janvry et al., 2015; Adamopoulos et al., 2024), and other frictions can prevent workers released from agriculture from entering non-agricultural activities. Harris and Todaro (1970) show that releasing labor from agriculture is not sufficient for structural transformation because workers must also be successfully absorbed into the non-agricultural sector. In this framework, the movement depends on the probability of obtaining non-agricultural employment rather than simply leaving agriculture. Together, these perspectives suggest that structural transformation depends on three sequential stages: labor release, labor reallocation, and labor absorption.

Despite these theoretical insights, it remains unclear which mechanism is the primary constraint to structural transformation in practice. This paper answers two related questions. First, when agricultural productivity increases, does labor actually leave agriculture, consistent with the labor-push mechanism proposed by Lewis, 1954, or does higher productivity increase the return to farming and retain labor in agriculture? Second, conditional on labor being released from agriculture, what determines whether workers successfully move into productive non-agricultural employment? Do mobility barriers prevent workers from entering non-agricultural activities, or do they reach non-agricultural labor markets but fail to obtain productive employment? Answering these questions is essential for understanding how agricultural productivity shocks translate into structural transformation and why countries respond differently to similar productivity improvements.

To answer these questions, I first exploit weather-induced agricultural productivity shocks to identify the

causal effect of agricultural productivity on household labor allocation in three Sub-Saharan African countries (Niger, Malawi, and Ethiopia). This approach provides exogenous variation in agricultural productivity and allows me to test whether higher productivity releases labor from agriculture, or instead encourages households to retain labor in farming. I find that positive productivity shocks consistently reduce agricultural labor allocation across all three countries, indicating that the labor-release effect dominates the tendency for higher agricultural productivity to retain labor in farming.

I next examine how labor released from agriculture is reallocated across sectors. This decomposition allows me to distinguish whether structural transformation is limited by labor releases themselves or by subsequent processes into productive non-agricultural employment. Although agricultural productivity shocks release labor from agriculture in all three countries, the subsequent transition differs markedly across countries. *In Niger and Malawi, released workers primarily enter non-agricultural employment, whereas in Ethiopia, they are substantially more likely to remain idle rather than move into productive non-agricultural activities.* Does the population size matter? Ethiopia has a larger population and more unemployment.

To explain these heterogeneous responses, I develop a structural model that decomposes structural transformation into three sequential stages: labor release, labor reallocation, and labor absorption. The model jointly identifies the constraints operating at each stage and allows the binding constraint to differ across countries. Estimating the model using a simulated method of moments reveals substantial heterogeneity in these constraints. In Niger and Malawi, subsistence constraints primarily limit labor release, whereas in Ethiopia, barriers to entering non-agricultural activities and limited labor demand constrain labor reallocation and absorption. Counterfactual policy experiments further show that the effectiveness of policy interventions depends critically on which stage constitutes the binding constraint.

This paper makes several main contributions to the literature on structural transformation and development policy. First, I show that the release of labor from agriculture and successful structural transformation are distinct economic processes. Existing studies have generated important insights into structural transformation but typically focus on a single stage of the process. Studies of labor release examine whether agricultural productivity growth shifts labor out of agriculture (Bustos et al., 2016; Krishna and Meenakshi, 2022; Wang, 2025; Vicente Cateia et al., 2024). Studies of labor reallocation emphasize mobility costs, migration frictions, and land institutions that affect workers' ability to enter non-agricultural activities (Bryan et al., 2014; Donaldson, 2018; Adamopoulos and Restuccia, 2020; Deininger and Jin, 2008; Deininger and Jin, 2006). Studies of labor absorption focus on whether non-agricultural sectors generate sufficient productive employment for released workers (Harris and Todaro, 1970; McMillan and Headey, 2014; Diao et al., 2019; Diao et al., 2025; Rodrik, 2016). By integrating all three stages within a common empirical and structural framework, I show that similar productivity shocks can produce very different structural transformation outcomes because countries face different binding constraints along the transformation process.

Second, I identify a broader measure of labor underutilization, hereafter referred to as *idle labor*, as an overlooked margin of structural transformation. The concept is closely related to the International Labour Organization’s (ILO) notion of labor underutilization, which captures individuals whose need for employment is not fully met because they are unemployed, underemployed, or only weakly attached to the labor market (International Labour Organization, 2013). In rural low-income economies, however, the standard distinction between employment and non-employment may overstate successful labor absorption because individuals engaged only in intermittent work or unpaid family agricultural activities are classified as employed under the ILO definition. To better capture the outcome of structural transformation, I define *idle labor* as workers who are not durably engaged in productive employment, including individuals who report no regular work as well as those engaged only in unpaid household agricultural work or temporary and casual employment.

While a growing literature shows that workers leaving agriculture often move into informal or low-productivity employment (McMillan and Headey, 2014; Diao et al., 2019; Diao et al., 2025; Rodrik, 2016), I show that agricultural productivity growth may also increase idle labor rather than productive non-agricultural employment. This additional margin reveals that labor can be released from agriculture without being successfully reallocated or durably absorbed into productive employment.

Thirdly, I develop a structural framework that separately identifies the economic constraints underlying observed labor allocation decisions. Existing structural models typically focus on a subset of the mechanisms underlying structural transformation, including agricultural productivity and subsistence constraints (Lagakos and Waugh, 2013; Adamopoulos and Restuccia, 2014), migration and mobility frictions (Bryan et al., 2014; Tombe and Zhu, 2019), or labor market distortions (Diao et al., 2019; McMillan et al., 2014). My framework jointly identifies the mechanisms governing labor release, labor reallocation, and labor absorption. This allows me to determine which constraint is binding in each country and to evaluate how alternative policy interventions affect structural transformation under different institutional environments.

Fourth, I leverage a Random Forest machine learning algorithm to construct a generalizable measure of weather-induced agricultural productivity, contributing to the empirical literature on climate and structural transformation.¹ Existing studies typically identify agricultural productivity shocks using either weather-based proxies, such as extreme deviations in rainfall or temperature (Alfano and Görlach, 2024; Colmer, 2021; Branco and Féres, 2021; Emerick, 2018; Jessoe et al., 2018; Cattaneo and Peri, 2016; Hirvonen, 2016), or technology-induced changes, such as the introduction of new agricultural technologies (Gollin and Rogerson, 2014; Bustos et al., 2016; Adamopoulos and Restuccia, 2020; Gollin et al., 2021; Rajaonarison, 2014). My approach uses a Random Forest algorithm to model satellite-observed vegetation conditions (NDVI) as

¹Instead of using rainfall shocks, temperature anomalies, or drought indices such as SPEI directly, I estimate the component of agricultural productivity that is systematically driven by weather. Weather variables measure climatic conditions, whereas the economic object of interest in this paper is agricultural productivity. The same weather realization may generate different productivity responses across crops, locations, and years because of differences in crop composition, crop calendars, soil characteristics, and production conditions. Recovering the weather-induced component of realized agricultural productivity provides a more direct measure of the economic shock affecting household labor allocation. Appendix C discusses this construction and its advantages in detail.

a nonlinear function of rainfall and temperature, thereby recovering the weather-driven component of agricultural productivity. By combining climatic inputs with the realized response of vegetation, the resulting measure captures the productivity effects of both favorable and unfavorable weather realizations and can be readily applied across countries and time periods.

Finally, I demonstrate that effective structural transformation policies depend on the country-specific constraints that impede different stages of the transformation process. Existing studies emphasize policies that improve agricultural productivity (Barrett et al., 2019; Suri and Udry, 2022; Byerlee et al., 2009), expand non-agricultural employment opportunities (Diao et al., 2025; McKenzie, 2017), or reduce mobility barriers (Donaldson, 2018; Bryan et al., 2014). Using counterfactual policy simulation, I show that interventions targeting subsistence constraints, mobility barriers, household land rights, and labor absorption have markedly different effects across countries because the binding constraint varies across institutional environments. These findings imply that there is no one-size-fits-all strategy for promoting structural transformation. Instead, effective policy design requires identifying and relaxing the country-specific constraint that most limits labor release, labor reallocation, or labor absorption.

The remainder of the paper is organized as follows. Section 2 describes the data sources and the construction of the weather productivity measure. Section 3 outlines the empirical strategy and presents the reduced-form results. Section 4 introduces the theoretical model. Section 5 estimates the structural parameters. Section 6 calibrates the policies and conducts the policy counterfactual analysis. Section 7 concludes and discusses the policy implications.

2 Data and Measurement

Answering the questions requires measuring both agricultural productivity and household labor allocation over time. I combine nationally representative household panel surveys with high-resolution geospatial weather data to construct a panel of agricultural productivity and labor allocation in Niger, Malawi, and Ethiopia. This section first describes the household and geospatial data sources before introducing the measure of weather-induced agricultural productivity used throughout the analysis.

2.1 Country Selection and Survey Design

This paper uses household panel data from the World Bank Living Standards Measurement Study–Integrated Surveys on Agriculture (LSMS-ISA) for Niger, Malawi, and Ethiopia. The selection of these countries is primarily driven by the requirements of the empirical design rather than representativeness. The analysis examines changes in household labor allocation over time and requires continuously tracked households that can be followed across multiple survey waves. Moreover, because the empirical analysis compares changes in labor allocation following weather-induced agricultural productivity shocks, the observation interval should

be comparable across countries.

The three selected countries satisfy these requirements. In each country, the LSMS-ISA surveys re-interview the same households after approximately three years, allowing the construction of comparable household panels. The common survey interval ensures that changes in household labor allocation are measured over similar time horizons, improving the comparability of transition dynamics across countries. In addition, all three surveys provide detailed labor allocation modules and georeferenced household locations, making it possible to construct comparable weather-productivity measures using satellite and climate data.

Table 1 summarizes the survey design. Although the survey years differ across countries, each panel spans approximately three years and follows the same households between survey waves. Because the empirical analysis exploits within-household changes rather than cross-sectional differences, the exact calendar years are less important than the comparable observation interval.

Table 1: Household Panel Surveys Used in the Analysis

Country	Survey Waves	Panel Interval	Household Tracking	GPS Available
Niger	2011, 2014	3 years	Yes	Yes
Malawi	2016, 2019	3 years	Yes	Yes
Ethiopia	2018, 2021	3 years	Yes	Yes

Notes: The table summarizes the household panel surveys used in this paper. All three surveys are part of the World Bank Living Standards Measurement Study–Integrated Surveys on Agriculture (LSMS-ISA). Households are continuously tracked across survey waves, allowing the construction of comparable household panels. Each panel spans approximately three years, ensuring that household labor transitions are measured over similar observation windows across countries. Interview periods differ because surveys are conducted to coincide with each country’s agricultural calendar.

Finally, these countries provide useful variation for studying structural transformation. All three are Sub-Saharan African economies where agriculture remains the dominant source of employment and agricultural production is highly dependent on weather conditions. The purpose of selecting these countries is not to obtain a representative sample of Sub-Saharan Africa, but to exploit comparable household panels while examining whether similar agricultural productivity shocks generate different patterns of labor allocation. This variation allows the paper to identify which stage of structural transformation, labor release, labor reallocation, or labor absorption, constitutes the binding constraint in each setting.

2.2 Background of the Countries Studied

Figure 1 highlights the three countries included in this study: Niger, Malawi, and Ethiopia. These countries provide a useful setting for studying structural transformation because they differ substantially in their economic development, institutional environments, and geographic contexts, while sharing a common feature, that is, agriculture remains the primary source of employment for a large share of the population.

The three countries represent different stages of economic development. In 2024, Ethiopia had the highest GDP per capita (916 USD), followed by Niger (585 USD) and Malawi (552 USD). These differences are important because the severity of subsistence constraints, the availability of non-agricultural employment,

and the feasibility of labor reallocation are likely to vary across countries at different income levels.

The sample also spans three major regions of Sub-Saharan Africa: Niger in West Africa, Ethiopia in East Africa, and Malawi in Southern Africa. Beyond geography, the countries differ in their historical and institutional environments, including colonial legacies, market development, and rural livelihood systems. This diversity provides useful variation for examining whether similar productivity shocks lead to different patterns of structural transformation under different institutional settings.

A particularly important institutional difference concerns land tenure. Ethiopia operates under a state-owned land system in which rural households hold use rights that are closely tied to residence and cultivation. As a result, migration into non-agricultural employment may increase the risk of losing access to land, potentially discouraging labor mobility (Abate and Kiros, 2010; Manyazewal and Shiferaw, 2019). In contrast, land institutions in Niger and Malawi generally provide greater scope for customary land rental and transfer. These institutional differences motivate the paper's focus on the role of household land rights as a key mechanism through which agricultural productivity shocks translate into labor reallocation across countries. Stronger household land rights reduce the opportunity cost of leaving agriculture by allowing households to better preserve the value and security of their land assets when pursuing non-agricultural opportunities.

Overall, these differences make the three countries well-suited for examining whether similar agricultural productivity shocks generate different patterns for labor release, labor reallocation, and labor absorption.



Figure 1: Niger, Malawi, and Ethiopia

2.3 Household Survey Data

The empirical analysis uses household panel data from the World Bank's Living Standards Measurement Study-Integrated Surveys on Agriculture (LSMS-ISA), which provides nationally representative panel sur-

veys for developing countries (Grosh and Glewwe, 1998; Bentze and Wollburg, 2024). The LSMS-ISA surveys collect detailed information on household demographics, labor allocation, agricultural production, consumption, expenditures, and food security, making them particularly well suited for studying structural transformation (Lagakos and Shu, 2023).

To facilitate cross-country comparison, I construct balanced household panels with a common three-year interval between survey waves. The final sample consists of Niger (2011, 2014), Malawi (2016, 2019), and Ethiopia (2018, 2021). Restricting attention to similar observation windows reduces differences arising from survey frequency while improving the comparability of estimated productivity responses across countries.

This matching procedure allows us to construct household labor allocations across four mutually exclusive categories: agriculture, manufacturing, services, and idle activities. Agricultural labor includes both on-farm self-employment and agricultural wage employment. Manufacturing and services include both wage employment and self-employment in the corresponding sectors. Idle activities refer to working-age household members who report neither agricultural nor non-agricultural employment during the survey reference period. This category includes individuals who are not engaged in productive work, regardless of whether they are actively searching for employment. Accordingly, it should be interpreted as a measure of unabsorbed labor rather than unemployment in the conventional labor force sense. Because the key explanatory variable captures weather-induced agricultural productivity, the estimation sample is restricted to households exposed to agricultural production.²

To capture differences in land institutions, I construct a categorical measure of household land-rights index based on households' reported land rights in LSMS surveys. The index captures whether households possess inheritance and collateral rights over agricultural land. Type 0 denotes households whose land can neither be pledged as collateral nor bequeathed, representing the lowest degree of transferability. Type 1 denotes households whose land possesses one of these land rights, either collateral or inheritance, but not both. Type 2 denotes households whose land can both be pledged as collateral and freely bequeathed, representing the highest degree of transferability. This measure is designed to capture heterogeneity in the security and transferability of household land rights within each country rather than differences in national land-tenure regimes. Households with stronger inheritance or collateral rights face a lower opportunity cost of reallocating labor because they can better preserve the value of their land assets while pursuing non-agricultural opportunities.

2.4 Climate Data

To measure weather-driven variation in agricultural productivity, I combine the LSMS household surveys with three complementary geospatial datasets capturing vegetation conditions, precipitation, and temperature. These variables provide plausibly exogenous measures of local growing conditions and form the basis

²I define the estimation sample based on agricultural participation rather than administrative rural–urban status, including all households engaged in agricultural production regardless of location.

for constructing the weather-induced agricultural productivity measure introduced in the next subsection.

Household locations are obtained from the LSMS enumeration-area (EA) GPS coordinates. Following the LSMS disclosure protocol, these coordinates are randomly displaced to protect respondent confidentiality. To account for this displacement, climate variables are extracted by averaging raster values within 2 km buffers for urban enumeration areas and 5 km buffers for rural enumeration areas, consistent with the LSMS-ISA recommendations.

Vegetation conditions are measured using the Normalized Difference Vegetation Index (NDVI) derived from the MODIS MOD13Q1 satellite product at 250-meter spatial resolution.³ NDVI is one of the most widely used remote-sensing measures of vegetation health and has been extensively employed as a proxy for agricultural productivity in economics, agriculture, and environmental science. Unlike rainfall and temperature, which measure climatic inputs into production, NDVI captures the realized response of vegetation to weather conditions and local growing environments. Because it reflects the cumulative effects of precipitation, temperature, soil conditions, and crop growth, NDVI provides a more direct indicator of agricultural productivity than weather variables alone. Annual NDVI is constructed by averaging all MODIS observations within each year and then averaging across all pixels contained within the household buffer.

Rainfall data are obtained from the Climate Hazards Group InfraRed Precipitation with Station data (CHIRPS), which has a spatial resolution of approximately 0.05° (about 5.5 km). The buffered extraction described above is applied to these raster data to reduce measurement error arising from GPS displacement. Rainfall is included because precipitation is one of the primary determinants of agricultural production in the predominantly rain-fed farming systems of Sub-Saharan Africa.

Temperature data are obtained from the ERA5-Land reanalysis dataset at approximately 0.1° (about 10 km) spatial resolution. The annual mean temperature is assigned to each household according to its buffered location. Temperature complements rainfall by capturing thermal conditions that influence crop growth, including heat stress and growing-season conditions. Although the ERA5-Land resolution is coarser than the household buffers, it accurately reflects local climatic conditions relevant for agricultural production.

Because the household surveys are conducted at three-year intervals, climate variables are constructed annually throughout the observation period, including the three years preceding the first survey wave. These annual measures provide continuous information on local growing conditions and allow agricultural productivity to be measured consistently over time. In the next subsection, I combine NDVI, rainfall, and temperature to construct a weather-induced measure of agricultural productivity that serves as the key explanatory variable throughout the empirical analysis.

³NDVI is calculated as $(NIR - Red)/(NIR + Red)$, where *NIR* and *Red* denote near-infrared and red spectral reflectance, respectively. Healthy vegetation absorbs visible red light for photosynthesis while strongly reflecting near-infrared light, resulting in higher NDVI values.

2.5 Measuring Weather-Induced Agricultural Productivity

2.5.1 Identification Strategy

Estimating the effect of agricultural productivity on labor allocation raises an important identification challenge because observed agricultural productivity is jointly determined with household production decisions. Households simultaneously choose labor allocation, technology adoption, fertilizer use, and other management practices, while unobserved local economic conditions may influence both agricultural productivity and labor supply decisions (Branco and Féres, 2021; Emerick, 2018; Jessoe et al., 2018). Consequently, directly using observed agricultural output or productivity as an explanatory variable would generally produce biased estimates of the effect of agricultural productivity on structural transformation.

To address this challenge, I construct a measure of weather-induced agricultural productivity, defined as the component of vegetation growth explained solely by weather conditions. Figure 2 illustrates the identification strategy. Observed annual NDVI growth contains two sources of variation. The first is weather-driven variation, arising from rainfall and temperature, which is plausibly exogenous to household production decisions. The second reflects human and other non-weather factors, including labor allocation, technology adoption, fertilizer use, management practices, and other local conditions. The objective is to isolate only the weather-driven component and use it as the treatment variable throughout the empirical analysis.

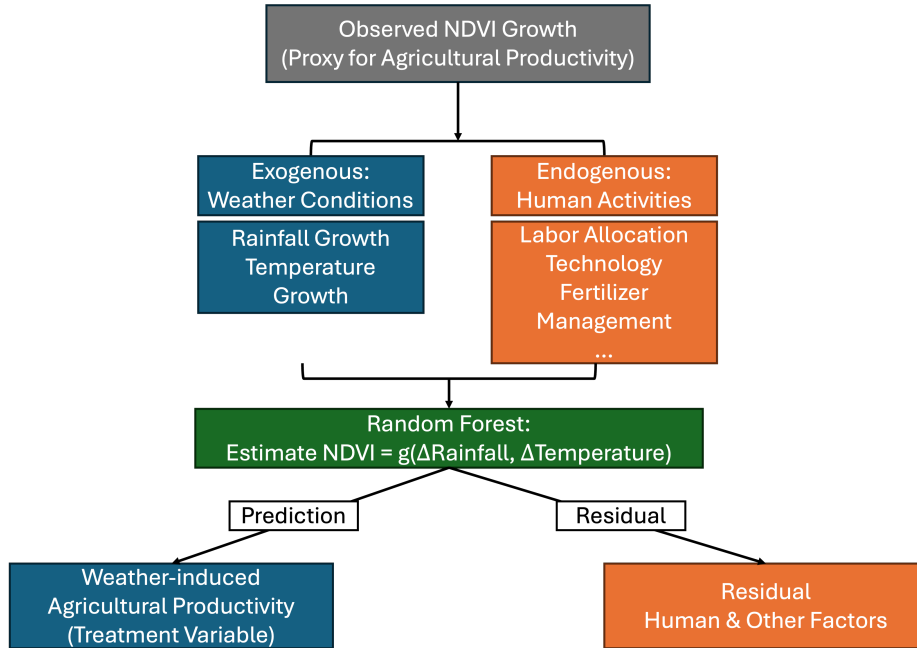


Figure 2: Weather-Induced Agricultural Productivity Identification Strategy

This decomposition can be motivated by a standard agricultural production function in which agricultural output depends on both weather conditions and household production decisions. Appendix A provides the formal derivation. The resulting treatment variable is defined as

$$\widehat{\Delta\theta}_{it} = g(\Delta\text{Rainfall}_{it}, \Delta\text{Temperature}_{it}), \quad (1)$$

where $\widehat{\Delta\theta}_{it}$ denotes the predicted weather-induced agricultural productivity growth, and $g(\cdot)$ is the Random Forest prediction function that maps annual weather changes to annual vegetation growth.

2.5.2 Construction

To estimate the weather-driven component, I use annual NDVI growth as a proxy for changes in agricultural productivity. NDVI has been shown to closely track agricultural production and crop yields in both economics and remote-sensing studies (Donaldson and Storeygard, 2016; Lobell et al., 2015). Because NDVI reflects the realized response of vegetation to local growing conditions, it provides an objective measure of agricultural productivity that is observed independently from household survey responses.

LSMS-ISA randomly displaces rural enumeration-area coordinates by up to 5 km to protect respondent confidentiality. Following the World Bank disclosure protocol, the remote-sensing variables (NDVI, temperature, and rainfall) used for weather-productivity construction are extracted using a 5km buffer around each rural GPS point, ensuring that the actual enumeration area remains within the extraction window despite the displacement. Because the agricultural production is measured for households residing within the sampled enumeration area, and plot locations are generally referenced relative to the household homestead in LSMS-ISA surveys, this buffering strategy substantially reduces measurement error arising from the displacement while preserving the relevant local production environment. (Ali et al. (2019))

The relationship between weather conditions (temperature and rainfall) and vegetation growth (NDVI) is inherently nonlinear. Crop growth depends on interactions between rainfall and temperature rather than monotonic changes in either variable alone. For example, additional rainfall may improve agricultural productivity under moderate temperatures but reduce productivity when temperatures are already low or when excessive rainfall causes flooding. Linear regression, therefore, provides an inadequate representation of the weather-productivity relationship.

To address the problem, we estimate the following relationship:

$$\Delta NDVI_{it} = g(\Delta Rainfall_{it}, \Delta Temperature_{it}) + \varepsilon_{it} \quad (2)$$

where $g(\cdot)$ is a function estimated using a Random Forest algorithm. The predicted component,

$$g(\Delta Rainfall_{it}, \Delta Temperature_{it})$$

, represents the variation in vegetation growth systematically explained by weather conditions and is interpreted as the weather-induced agricultural productivity shock used in the subsequent analysis.⁴ The residual

⁴The weather-productivity shock is a generated regressor obtained from out-of-sample Random Forest predictions. The reported second-stage standard errors are conditional on the estimated prediction function and therefore do not account for first-stage prediction uncertainty.

component, ε_{it} , captures the remaining variation in vegetation growth arising from non-weather factors, such as labor allocation, technology, fertilizer, land management practices, or other human activities. By using only the predicted component, the treatment variable isolates exogenous weather-induced productivity variation rather than changes in vegetation associated with household production decisions or other local factors.

The relationship between weather and vegetation is highly nonlinear and involves complex interactions between rainfall and temperature that are difficult to capture using conventional linear specifications. Random Forest is well-suited for estimating $g(\cdot)$ because it flexibly models nonlinearities and high-order interactions without imposing a parametric functional form, while its ensemble-learning framework reduces prediction variance by averaging across a large number of decision trees. We use growth rates rather than levels because changes in agricultural productivity, rather than productivity levels, drive households' labor reallocation decisions in our theoretical model.

The prediction model is estimated using 5-fold cross-validation. Specifically, the remote sensing dataset is randomly partitioned into 5 mutually exclusive folds. For each fold, the Random Forest is trained on the remaining 4 folds and used to predict NDVI growth for the held-out fold. Because each predicted value is generated by a model that excludes the corresponding observation from the training sample, the resulting weather-productivity measure is not affected by in-sample overfitting. All prediction statistics reported below are calculated from these out-of-sample predictions.

The out-of-sample predictions isolate the component of annual vegetation growth that is systematically explained by weather conditions and are used as the weather-induced agricultural productivity shock in the subsequent empirical analysis. The out-of-sample (R^2) equals 0.29 for Niger, 0.60 for Malawi, and 0.30 for Ethiopia, while the mean absolute prediction error remains low across all three countries. These results suggest that rainfall and temperature explain a meaningful share of annual vegetation dynamics while leaving the remaining variation to household production decisions and other non-weather factors.

2.5.3 Distribution of Weather-Induced Agricultural Productivity

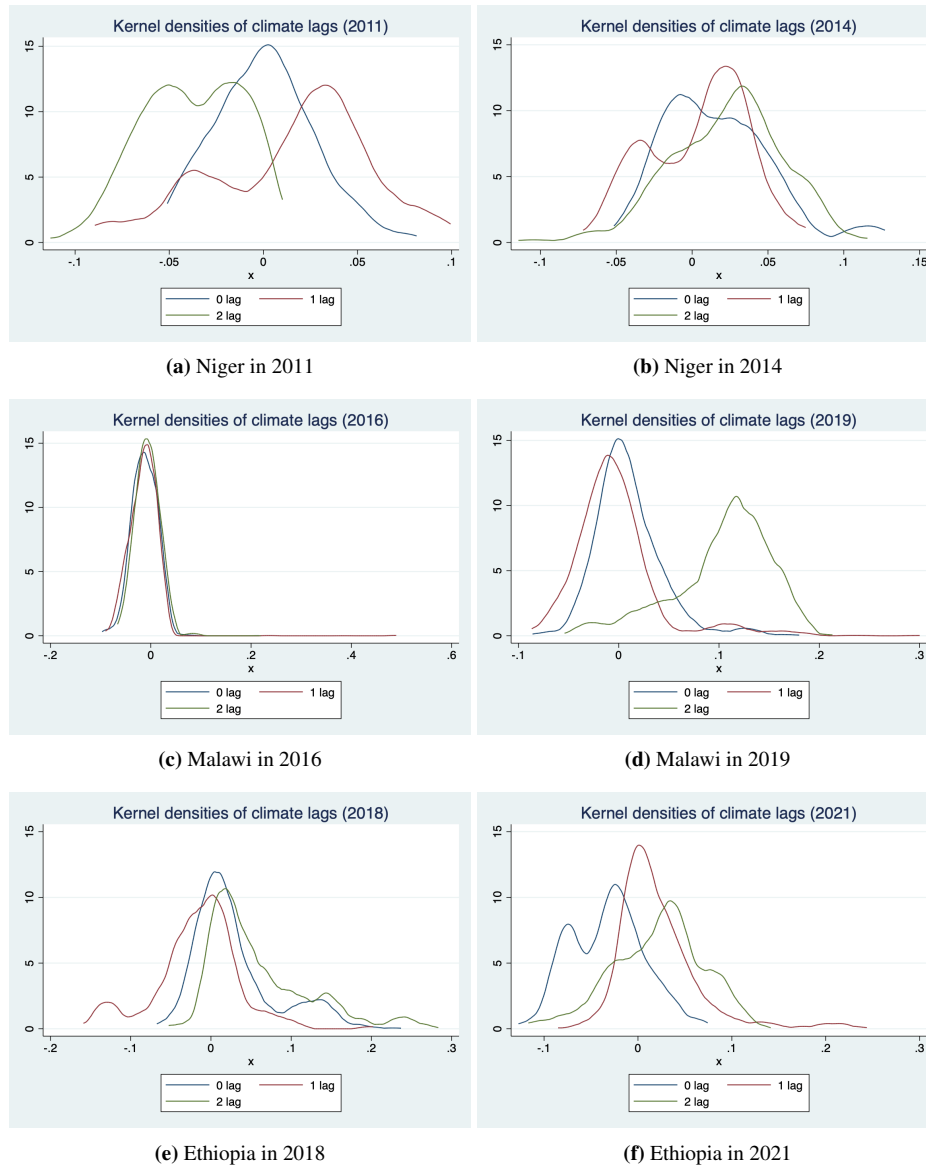
Figure 3 presents the distribution of weather-induced agricultural productivity across enumeration areas (EAs) for each country and year. Positive values indicate weather conditions that improve agricultural productivity relative to the previous year, whereas negative values correspond to less favorable growing conditions. Different colors represent different crop years within each country.

The distributions exhibit substantial variation across both space and time. Some years display distributions centered above zero, indicating broadly favorable weather conditions (for example, 2010 in Niger and 2019 in Malawi), while others are centered below zero, reflecting widespread adverse growing conditions. In several years, the distribution is concentrated around zero, suggesting relatively normal agricultural conditions. These patterns demonstrate that weather-induced agricultural productivity varies considerably across enu-

meration areas even within the same year, providing the cross-sectional and temporal variation required for identification.

Importantly, the weather-productivity shock represents a temporary, exogenous change in agricultural productivity rather than a permanent difference in agricultural potential across locations. This feature makes it well suited for identifying households' short-run labor reallocation responses to productivity changes while avoiding confounding long-run differences in economic development, technology changes, or infrastructure. The estimated responses capture how households adjust labor allocation following an exogenous productivity shock, providing the behavioral relationships that underpin the subsequent structural analysis. Although the weather shock is transitory, the resulting labor reallocation can be long-term if transitions across sectors involve fixed costs or other adjustment frictions. We will return to this distinction when motivating the structural model in Section 5.

Figure 3: Distribution of Weather Productivity



2.6 Summary Statistics

Table 2 reports summary statistics for the variables used in both the reduced-form and structural analyses. The dependent variables are the share of working-age household members (ages 15-85) allocated to agriculture, manufacturing, services, and underutilized (idle) labor. These shares are calculated using all working-age household members as the denominator. Consequently, the four reported shares do not necessarily sum to one because the remaining share consists of working-age individuals outside the labor force, such as students, retirees, homemakers, or individuals not participating in economic activities.

Agriculture remains the dominant source of employment in all three countries, accounting for 63.4 percent of household labor in Niger, 77.6 percent in Malawi, and 66.0 percent in Ethiopia. Non-agricultural employment is concentrated primarily in services, while manufacturing accounts for only a small share of household labor. A notable difference across countries is the prevalence of idle labor. Idle labor (underutilized labor) refers to working-age individuals who are unemployed or only marginally attached to productive employment through unpaid household agricultural work or temporary and casual activities (see Appendix A for the harmonized construction). Ethiopia exhibits a substantially larger idle labor share (18.2 percent) than Niger (2.8 percent) and Malawi (0.4 percent). Throughout the paper, idle labor refers to working-age household members who are not engaged in either agricultural or non-agricultural employment and therefore measures the extent to which labor is not absorbed into productive sectors. The high incidence of idle labor in Ethiopia provides preliminary evidence that labor released from agriculture may not always translate into productive non-agricultural employment.

The table also reports the key variables used to examine the mechanisms underlying structural transformation. Food subsistence serves as a proxy for subsistence constraints, minimum distance to the nearest market captures mobility barriers, and land type classifies households according to differences in household land-rights index. These variables exhibit substantial variation across countries, providing useful heterogeneity for the empirical and structural analyses. Food subsistence is most prevalent in Malawi (70.3 percent of households), followed by Niger (48.4 percent), while Ethiopia has the lowest incidence (15.4 percent). Average distance to markets ranges from 3.7 kilometers in Malawi to approximately 19 kilometers in Niger, indicating considerable cross-country differences in market accessibility.

The estimation sample consists of 3,780 households from 269 communities in Niger, 1,910 households from 81 communities in Malawi, and 2,240 households from 446 communities in Ethiopia. Together, these descriptive statistics reveal substantial cross-country heterogeneity in labor allocation, household constraints, and market access despite a common dependence on agriculture. The following section examines whether these differences translate into heterogeneous responses of household labor allocation to weather-induced agricultural productivity shocks.

Table 2: Summary Table

	(1)	(2)	(3)	(4)	(5)	(6)
	Niger		Malawi		Ethiopia	
	Mean	SD	Mean	SD	Mean	SD
HH ag share	0.634	0.406	0.776	0.357	0.660	0.462
HH manuf share	0.118	0.276	0.052	0.188	0.023	0.134
HH serv share	0.208	0.333	0.154	0.301	0.094	0.265
HH idle share	0.028	0.135	0.004	0.054	0.182	0.371
Weather Productivity Shock (1-lag)	0.016	0.037	-0.011	0.027	0.004	0.048
Food Subsistence	0.484	0.500	0.703	0.457	0.154	0.361
Min-distance	18.993	24.166	3.724	3.196	16.786	18.284
Household Land-rights Index	0.081	0.345	1.108	0.670	1.394	0.895
Household Obs	3780		1910		2240	
Community Obs	269		81		446	

Labor shares are calculated as the proportion of all working-age household members (ages 15–85) allocated to agriculture, manufacturing, services, and underutilized labor (idle). The shares do not necessarily sum to one because working-age individuals outside the labor force are not included in these four categories.

3 Reduced Form Evidence

3.1 Empirical Strategy

This section examines two central questions motivating the paper. First, does higher agricultural productivity release labor from agriculture, as predicted by Lewis (1954)? Second, once labor is released, does it transition into productive non-agricultural employment or remain outside productive sectors, consistent with the mechanisms emphasized by Harris and Todaro (1970)? To answer these questions, I exploit annual variation in weather-induced agricultural productivity together with household panel data observed at three-year intervals.

Although households are interviewed only once every three years, the weather-induced productivity measure is available annually for each enumeration area (EA). To align the timing of productivity shocks with household labor allocation decisions, I use the one-year lagged weather-induced agricultural productivity as the main treatment variable.⁵

⁵In the theoretical model, weather-induced productivity is realized at the end of the agricultural season, after which households update their economic conditions and make labor allocation decisions for the subsequent period. The one-year lag therefore matches the timing of the household decision problem. Results using contemporaneous and two-year lagged productivity shocks are reported as robustness checks.

The baseline reduced-form specification is

$$s_{ijt} = \beta_{tot} \Delta \theta_{gt-1} + \delta_i + \lambda_t + \varepsilon_{it} \quad (3)$$

where s_{ijt} denotes the share of household labor allocation to sector $j \in A, M, S, I$ for household i in year t , corresponding to agriculture, manufacturing, services, and idle activities, respectively. The variable $\Delta \theta_{gt-1}$ is the one-year lagged weather-induced agricultural productivity shock in enumeration area g . The coefficient β_{tot} measures the overall effect of weather-induced productivity on household labor allocation. Throughout the paper, I estimate Equation 3 separately for each sectoral labor share, allowing agricultural productivity to have different effects on agriculture, manufacturing, services, and idle labor.

Identification comes from within-household variation in exposure to weather-induced productivity shocks over time. Household fixed effects (δ_i) absorb all time-invariant household characteristics, such as productive assets, household preferences, and persistent differences in production capacity. They also absorb persistent differences in baseline labor-force participation across households. For example, households may consistently contain members who do not participate in the labor market because of chronic health limitations, long-standing caregiving responsibilities, or persistent social norms governing labor supply. Consequently, the estimated effects on labor allocation, especially on idle labor, capture within-household changes associated with weather-induced productivity shocks rather than permanent differences in baseline labor-force participation. Year fixed effects (λ_t) control for aggregate shocks common to all households within each survey wave, including national macroeconomic conditions and policy changes. Standard errors are clustered at the enumeration-area level because weather-induced productivity is measured at the EA level, implying that households within the same EA are exposed to common weather shocks.

Finally, I examine whether the effects of weather-induced productivity differ across the mechanisms emphasized by the theoretical framework. Specifically, I study four mechanisms: subsistence constraints, mobility barriers, household land-rights, and local labor absorption capacity. These mechanisms correspond to different stages of structural transformation. Subsistence constraints determine whether households release labor from agriculture, mobility barriers, and household land-rights influence the reallocation of released workers across sectors, and labor absorption capacity determines whether released workers successfully enter productive non-agricultural employment. The following subsections describe the empirical evidence for each mechanism before the structural model jointly quantifies their relative importance.

3.2 Does Agricultural Productivity Release Labor?

The first empirical question is whether improvements in agricultural productivity release labor from agriculture. Lewis (1954) argues that higher agricultural productivity relaxes subsistence constraints, allowing households to reduce agricultural labor and reallocate workers to other sectors. An alternative possibility,

however, is that higher agricultural productivity raises the return to farming and induces households to retain more labor in agriculture. Whether agricultural productivity ultimately releases or retains labor is therefore an empirical question.

Table 3 and 4 report the baseline reduced-form estimates. The intensive-margin specification examines how weather-induced productivity affects the share of household labor allocated to each sector. The extensive margin specification instead examines whether households are employed in manufacturing, services, or idle activities. Together, the two specifications capture both changes in the distribution of labor within households and changes in households' entry into or exit from different sectors.

The results provide consistent evidence that higher agricultural productivity releases labor from agriculture. Across all three countries, positive weather-induced productivity shocks significantly reduce the household agricultural labor share. A one-standard-deviation increase in weather-induced productivity reduces the agricultural labor share by approximately 2.4 percentage points in Niger, 2.2 percentage points in Malawi, and 4.9 percentage points in Ethiopia. The extensive-margin estimates yield the same conclusion: households become significantly less likely to allocate all of their workers to agriculture following positive productivity shocks. The consistency between the intensive- and extensive-margin estimates indicates that the observed decline in agricultural labor reflects genuine labor reallocation rather than changes in labor intensity among existing workers.

These findings indicate that the labor-release effect dominates the agricultural-return effect. Although favorable weather conditions increase agricultural income and the return to farming, households do not respond by retaining additional labor in agriculture. Instead, improved agricultural productivity relaxes binding subsistence constraints, enabling households to reduce agricultural labor allocation. This finding provides direct empirical support for the labor-release mechanism emphasized by Lewis (1954).

On the other hand, labor leaving agriculture represents only the first stage of structural transformation. Whether these released workers subsequently enter productive non-agricultural employment remains an open question. Therefore, in the next subsection, I examine where released workers go after leaving agriculture.

3.3 Where Does Released Labor Go?

The second empirical question is where workers go after leaving agriculture. While the Lewis model explains how labor is released, Harris and Todaro (1970) emphasizes that structural transformation also depends on whether released workers can be successfully absorbed into productive non-agricultural employment. Labor release is therefore a necessary but not sufficient condition for structural transformation.

The results in Table 3 and 4 reveal substantial across-country heterogeneity in the destination of released labor. In Niger, positive weather-induced productivity shocks significantly increase manufacturing employment

while having no statistically significant effect on service employment or idle labor. A one-standard-deviation increase in weather-induced productivity raises the manufacturing labor share by approximately 2.2 percentage points, with essentially no increase in idle labor. The extensive-margin estimates confirm this pattern: households become significantly more likely to participate in manufacturing by exhibiting no corresponding increase in idle labor. These findings indicate that workers released from agriculture are successfully absorbed into productive non-agricultural employment.

Malawi exhibits a similar pattern. A one-standard-deviation increase in weather-induced productivity increases the manufacturing labor share by approximately 1.6 percentage points, while the increase in idle labor is economically small, at roughly 0.5 percentage points. The extensive-margin estimates likewise show that households become more likely to enter manufacturing following positive productivity shocks. Although some workers remain idle, the dominant response is a transition into productive non-agricultural employment.

Ethiopia presents a markedly different picture. Although higher agricultural productivity releases labor from agriculture, the increase in manufacturing employment is modest, amounting to only 0.8 percentage points for a one-standard-deviation productivity shock. Instead, the largest response is an increase in idle labor of approximately 4.5 percentage points, nearly matching the decline in agricultural labor. The extensive-margin estimates reinforce this finding: households become less likely to specialize entirely in agriculture but substantially more likely to have idle workers. Unlike in Niger and Malawi, labor released from agriculture in Ethiopia is not successfully absorbed into productive non-agricultural activities.

These contrasting responses demonstrate that labor release and successful structural transformation are distinct processes. Positive agricultural productivity shocks consistently release labor from agriculture across all three countries, yet the destination of released workers differs substantially. In Niger and Malawi, released workers primarily transition into manufacturing, whereas in Ethiopia, they are disproportionately absorbed into idle labor. This distinction represents the central empirical contribution of the paper. Existing studies typically examine whether agricultural productivity shifts labor out of agriculture. The evidence here shows that labor release alone is insufficient to generate structural transformation.

This empirical puzzle motivates the remainder of the paper. The next subsection examines four mechanisms, subsistence constraints, mobility barriers, household land-rights, and labor absorption capacity, which may explain these cross-country differences. The structural model then jointly identifies the relative importance of these mechanisms and evaluates the policy interventions most likely to promote structural transformation under different institutional environments.

3.4 Mechanisms Along the Three Stages of Structural Transformation

Previous sections establish two empirical findings. First, positive weather-induced productivity shocks consistently release labor from agriculture across all three countries. Second, the destination of released workers

differs substantially across countries. These results suggest that structural transformation involves multiple sequential stages rather than a single transition from agriculture to non-agriculture.

To understand the sources of this cross-country heterogeneity, this section examines the mechanisms operating at each stage of structural transformation. Following the conceptual framework developed in this paper, structural transformation is decomposed into three sequential stages: labor release, labor reallocation, and labor absorption. Different constraints are expected to operate at different stages. Subsistence constraints influence whether households release labor from agriculture, mobility barriers and household land-rights affect the allocation of released workers across sectors, and labor absorption determines whether released workers are ultimately absorbed into productive non-agricultural employment.

Empirically, each mechanism is examined by interacting the weather-induced productivity shock with a measure of the corresponding constraint. Therefore, the interaction coefficient captures whether the effect of agricultural productivity varies systematically with the severity of that constraint. A statistically significant interaction indicates that the mechanism alters the transmission of productivity improvements into labor reallocation, providing reduced-form evidence on its role in structural transformation. To mitigate endogeneity concerns, all household-level constraint measures are fixed at their baseline values measured in the first survey wave. The only exception is the unemployment measure, which is matched to each survey wave using the most recent pre-period observation so that it reflects predetermined local labor market conditions.

Table 3: Household Shares and Weather Productivity Growth (Intensive Margin)

	Agriculture	Manufacturing	Service	Idle
Niger				
$\Delta\theta_{gt-1}$	-0.640**	0.585***	0.0295	0.0825
	(-2.06)	(3.37)	(0.12)	(0.75)
Mean dep. var	0.634	0.118	0.209	0.028
Observations	3780	3780	3780	3780
Malawi				
$\Delta\theta_{gt-1}$	-0.824**	0.589**	-0.175	0.196*
	(-2.10)	(2.54)	(-0.67)	(1.85)
Mean dep. var	0.776	0.052	0.154	0.005
Observations	1910	1910	1910	1910
Ethiopia				
$\Delta\theta_{gt-1}$	-1.016**	0.167*	-0.250	0.935**
	(-1.98)	(1.73)	(-1.29)	(2.05)
Mean dep. var	0.660	0.023	0.094	0.182
Observations	2240	2240	2240	2240

t statistics in parentheses. All regressions include household and year fixed effects. Standard errors are clustered at the community level. Constant terms are included but not reported.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4: Household Shares and Weather Productivity Growth (Extensive Margin)

	All in Agriculture	Any Manufacturing	Any Service	Any Idle
Niger				
$\Delta\theta_{gt-1}$	-0.739*	0.857***	-0.0547	0.117
	(-1.75)	(3.27)	(-0.15)	(0.62)
Mean dep. var	0.484	0.181	0.332	0.046
Observations	3780	3780	3780	3780
Malawi				
$\Delta\theta_{gt-1}$	-0.773*	0.718**	-0.259	0.407**
	(-1.88)	(2.05)	(-0.79)	(2.48)
Mean dep. var	0.668	0.087	0.251	0.008
Observations	1910	1910	1910	1910
Ethiopia				
$\Delta\theta_{gt-1}$	-1.046**	0.279**	-0.284	1.115**
	(-2.07)	(2.07)	(-1.23)	(2.05)
Mean dep. var	0.633	0.032	0.129	0.205
Observations	2240	2240	2240	2240

t statistics in parentheses. All regressions include household and year fixed effects. Standard errors are clustered at the community level. Constant terms are included but not reported.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

3.4.1 Stage I: Labor Release

The first stage of structural transformation is the release of labor from agriculture. The Lewis model predicts that households operating close to subsistence must retain sufficient labor in agriculture to satisfy minimum consumption requirements. Consequently, improvements in agricultural productivity are expected to generate larger labor-release effects among households that are already above subsistence, whereas subsistence-constrained households may continue allocating labor to agriculture despite higher productivity.

To test this prediction, I interact the weather-induced productivity shock with a household-level indicator for food subsistence constraints in Table 5. The interaction coefficient measures whether the effect of agricultural productivity differs between households below and above the subsistence threshold. A positive interaction in the agricultural labor regression implies that subsistence-constrained households reduce agricultural labor by less following positive productivity shocks, indicating that subsistence constraints limit labor release.

The results strongly support the prediction that subsistence constraints limit labor release from agriculture. In both Niger and Malawi, the interaction term is positive and statistically significant in the agricultural labor regression, implying that households experiencing food insecurity respond less to positive productivity shocks by reducing agricultural labor. In Malawi, this weaker labor release is accompanied by a significantly smaller increase in manufacturing employment among subsistence-constrained households. In Niger, although the manufacturing interaction is not statistically significant, the estimates are directionally consistent with the hypothesis. Overall, the evidence indicates that productivity gains among subsistence-constrained households are initially used to satisfy basic consumption needs before facilitating labor reallocation.

By contrast, the interaction terms are statistically insignificant in Ethiopia. This result is consistent with the lower prevalence of food insecurity documented in the summary statistics and suggests that subsistence constraints play a more limited role in determining labor release. Overall, the evidence indicates that subsistence constraints primarily operate during the labor-release stage and are particularly important in poorer economies such as Niger and Malawi.

Table 5: Household Shares Changes and Subsistence Constraints

	Agriculture	Manufacturing	Service	Idle
Niger				
$\Delta\theta_{gt-1}$	-0.908*** (-2.65)	0.603*** (3.01)	0.191 (0.68)	0.0664 (0.59)
$\Delta\theta_{gt-1} \times \text{subsistence}(= 1)$	0.590* (1.88)	-0.0396 (-0.17)	-0.355 (-1.33)	0.0353 (0.24)
Mean dep. var	0.634	0.118	0.209	0.028
Observations	3780	3780	3780	3780
Malawi				
$\Delta\theta_{gt-1}$	-1.633*** (-3.74)	1.054*** (3.05)	-0.114 (-0.40)	0.221 (1.10)
$\Delta\theta_{gt-1} \times \text{subsistence}(= 1)$	1.413** (2.49)	-0.814* (-1.99)	-0.106 (-0.24)	-0.0427 (-0.17)
Mean dep. var	0.776	0.052	0.154	0.005
Observations	1910	1910	1910	1910
Ethiopia				
$\Delta\theta_{gt-1}$	-1.148** (-2.21)	0.202** (2.01)	-0.178 (-0.90)	0.931** (2.01)
$\Delta\theta_{gt-1} \times \text{subsistence}(= 1)$	0.706 (0.55)	-0.187 (-1.03)	-0.382 (-0.87)	0.0206 (0.02)
Mean dep. var	0.660	0.023	0.094	0.182
Observations	2240	2240	2240	2240

t statistics in parentheses. All regressions include household and year fixed effects. Standard errors are clustered at the community level. Constant terms are included but not reported.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Subsistence is a dummy variable where the below food subsistence group equals 1.

3.4.2 Stage II: Labor Reallocation

Released labor from agriculture does not automatically generate structural transformation. Once labor leaves agriculture, households must be able to reallocate workers into productive non-agricultural activities. Therefore, the second stage of structural transformation concerns whether released workers can successfully access non-agricultural opportunities. In this stage, households face both spatial and institutional constraints. Spatial mobility barriers may limit access to non-agricultural employment opportunities, while land institutions may

increase the opportunity cost of leaving agriculture if households risk losing valuable land rights.

Mobility Barriers Spatial mobility barriers increase the cost of accessing non-agricultural employment. Households located farther from economic centers may face higher transportation costs. Consequently, even if agricultural productivity releases labor, geographically isolated households may be less able to reallocate workers into productive non-agricultural sectors.

To examine this mechanism, I interact the weather-induced productivity shock with the household's minimum distance to the nearest economic center. The interaction coefficient measures whether the effect of agricultural productivity varies with geographic isolation. A negative interaction for manufacturing employment or a positive interaction for idle labor indicates that greater mobility barriers impede successful labor reallocation.

The results in Table 6 provide limited evidence that spatial mobility barriers influence the reallocation of released labor. In both Niger and Malawi, the interaction terms are statistically insignificant across sectors, suggesting that geographic isolation does not substantially alter where workers move following positive productivity shocks.

By contrast, the Ethiopian results suggest that mobility barriers influence the destination of released workers. The interaction between weather-induced productivity and distance is positive and statistically significant for idle labor, indicating that households located farther from economic centers are more likely to have released workers remain idle following positive productivity shocks. These findings suggest that mobility barriers primarily affect the destination of released workers rather than the initial decision to leave agriculture, and that this mechanism is important in Ethiopia.

Table 6: Household Shares Changes and Spatial Mobility Barriers

	Agriculture	Manufacturing	Service	Idle
Niger				
$\Delta\theta_{gt-1}$	-0.280	0.502**	-0.158	-0.0168
	(-0.76)	(2.34)	(-0.50)	(-0.16)
$\Delta\theta_{gt-1} \times \text{Min} - \text{distance}$	-0.0160	0.00364	0.00839	0.00432
	(-1.62)	(0.52)	(1.57)	(0.99)
Mean dep. var	0.634	0.118	0.208	0.028
Observations	3776	3776	3776	3776
Malawi				
$\Delta\theta_{gt-1}$	-1.079*	0.596*	-0.163	0.299*
	(-1.80)	(1.78)	(-0.43)	(1.94)
$\Delta\theta_{gt-1} \times \text{Min} - \text{distance}$	0.0880	-0.00265	-0.00416	-0.0356
	(0.76)	(-0.03)	(-0.05)	(-1.35)
Mean dep. var	0.776	0.052	0.154	0.005
Observations	1910	1910	1910	1910
Ethiopia				
$\Delta\theta_{gt-1}$	-0.254	0.244	-0.135	-0.0514
	(-0.39)	(1.63)	(-0.52)	(-0.09)
$\Delta\theta_{gt-1} \times \text{Min} - \text{distance}$	-0.0492	-0.00499	-0.00745	0.0638**
	(-1.38)	(-1.02)	(-0.72)	(2.07)
Mean dep. var	0.660	0.023	0.094	0.182
Observations	2240	2240	2240	2240

t statistics in parentheses. All regressions include household and year fixed effects. Standard errors are clustered at the community level. Constant terms are included but not reported.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Min-distance is a continuous variable indicating the distance between the household and the nearest economic nodes.

Household Land Rights Land institutions represent another important determinant of labor reallocation. Once labor is released from agriculture, households must decide whether to move workers into non-agricultural

activities. This decision depends not only on the availability of non-agricultural opportunities but also on the opportunity cost of leaving agriculture. When land rights are weak, households may hesitate to shift labor away from farming because doing so could jeopardize future access to land or reduce the value of their agricultural assets. This mechanism is particularly relevant in Ethiopia, where rural households hold state-owned land through use rights that are closely tied to continued cultivation. By contrast, stronger household land rights lowers the opportunity cost of leaving agriculture by allowing households to retain the economic value of their land while reallocating labor into non-agricultural employment.

The household land rights variable measures whether individual households possess inheritance or collateral rights. It is therefore intended to capture within-country variation in land rights, not to rank countries according to the overall transferability of their land-tenure systems. Ethiopia may exhibit higher average inheritance or collateral rights than Niger in the survey data while still operating under a more restrictive national land-tenure system in which migration can jeopardize continued land use. The empirical analysis exploits within-country variation in household land rights to examine whether stronger land rights facilitate labor reallocation under different institutional environment.

To examine the mechanism, I interact the weather-induced productivity shock with the household's household land-rights category. As described in Section 2.2, household land rights index is measured using three categories based on households' reported land rights: Type 0 denotes land that can neither be pledged as collateral nor be inherited, Type 1 denotes land with one of the two land rights, and Type 2 denotes land possessing both land rights. The interaction coefficients test whether households with stronger land rights respond differently to positive productivity shocks than households facing more restrictive land institutions. If household land-rights facilitates structural transformation, the productivity-induced transition into non-agricultural employment should become more transferable.

The results indicate substantial cross-country heterogeneity in Table 7. In Niger and Malawi, the interaction terms are generally small and statistically insignificant, suggesting that household land rights do not substantially alter the reallocation of released workers. In Ethiopia, however, households with stronger household land rights respond differently to positive productivity shocks. In particular, households with the strongest household land rights (Land Type 2) exhibit significantly larger transitions into manufacturing, while the estimated effect for Land Type 1 is positive but statistically insignificant. Moreover, both Land Types 1 and 2 experience significantly smaller increases in idle labor following positive productivity shocks than households with the weakest household land rights (Land Type 0). These results suggest that stronger household land rights, especially Land Type 2, facilitate the movement of released workers into productive non-agricultural employment and reduce the accumulation of idle labor relative to households with weak land rights.

Overall, the evidence suggests that land institutions primarily influence the destination of released workers rather than the initial release of labor from agriculture. In settings where land rights are closely tied to con-

tinued agricultural use, such as Ethiopia, stronger household land rights substantially lowers the institutional barriers to labor reallocation and promotes successful structural transformation.

Table 7: Household Shares Changes and Household Land Rights

	Agriculture	Manufacturing	Service	Idle
Niger				
$\Delta\theta_{gt-1}$	-0.650*	0.625***	0.0275	0.00194
	(-1.84)	(3.33)	(0.09)	(0.09)
$\Delta\theta_{gt-1} \times \text{landtype}(=1)$	1.057	-1.353	0.0640	0.0462
	(0.96)	(-1.41)	(0.10)	(0.58)
$\Delta\theta_{gt-1} \times \text{landtype}(=2)$	0.286	-0.179	-0.118	0.00187
	(0.12)	(-0.11)	(-0.12)	(0.08)
Mean dep. var	0.675	0.107	0.217	0.001
Observations	2660	2660	2660	2660
Malawi				
$\Delta\theta_{gt-1}$	-1.437**	0.501	0.277	0.119
	(-2.57)	(1.25)	(0.60)	(0.59)
$\Delta\theta_{gt-1} \times \text{landtype}(=1)$	0.818	0.175	-0.576	0.154
	(1.00)	(0.29)	(-0.94)	(0.68)
$\Delta\theta_{gt-1} \times \text{landtype}(=2)$	1.425	0.172	-1.074	-0.0537
	(1.57)	(0.31)	(-1.17)	(-0.25)
Mean dep. var	0.776	0.052	0.154	0.005
Observations	1910	1910	1910	1910
Ethiopia				
$\Delta\theta_{gt-1}$	-1.719*	-0.113	-0.0738	1.831**
	(-1.92)	(-0.81)	(-0.15)	(2.47)
$\Delta\theta_{gt-1} \times \text{landtype}(=1)$	2.535	0.390	-1.164	-2.047**
	(1.64)	(0.69)	(-1.20)	(-2.06)
$\Delta\theta_{gt-1} \times \text{landtype}(=2)$	1.031	0.398**	-0.137	-1.342*
	(1.11)	(2.05)	(-0.25)	(-1.78)
Mean dep. var	0.660	0.023	0.094	0.182
Observations	2240	2240	2240	2240

t statistics in parentheses. All regressions include household and year fixed effects. Standard errors are clustered at the community level. Constant terms are included but not reported.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Land type equals 0 if there is no land right, equals 1 if the land is either collateral or can be bequeathed, and equals 2 if both are allowed.

3.4.3 Stage III: Labor Absorption

The final stage of structural transformation concerns whether workers released from agriculture are successfully absorbed into productive non-agricultural employment. Harris and Todaro (1970) argue that structural transformation depends not only on the release of labor from agriculture but also on the capacity of the non-agricultural economy to create jobs. When local labor demand is weak, workers leaving agriculture may be unable to secure productive employment and instead remain idle. The final stage therefore asks whether released workers can be successfully absorbed into the labor market.

To examine this mechanism, I interact the weather-induced productivity shock with regional measures of labor market conditions. The primary measure is constructed from the World Bank Enterprise Surveys as the share of firms within each region that expand employment, capturing labor absorption from the demand side of the labor market. Because Enterprise Survey waves do not coincide exactly with the LSMS survey years, I assign the closest available Enterprise Survey wave to each LSMS wave and examine alternative matching schemes as robustness checks.

To further assess the robustness of the results, I also construct an alternative measure based on changes in regional unemployment rates from Ethiopia's Urban Employment and Unemployment Survey (UEUS). Unlike the Enterprise Survey, which measures firms' hiring behavior, the UEUS captures labor market conditions from the worker side and provides an independent measure of the economy's ability to absorb labor. Across both measures, the interaction coefficient captures whether regional labor-market conditions affect the economy's ability to absorb workers released from agriculture following positive weather productivity shocks. The expected sign depends on how labor demand is measured. For the share of firm expanding employment (Table 8 Column (1) and (2)), which is a positive measure of labor demand, a negative interaction indicates that stronger labor demand reduces the accumulation of idle labor. By contrast, the change in the unemployment rate is an inverse measure of labor demand (Table 8 Column (3) and (4)), so a positive interaction implies that worsening labor-market conditions increase the accumulation of idle labor.

The results in Table 8 provide consistent evidence supporting the labor-absorption mechanism. Unlike the baseline reduced-form specifications, which focus on the rural LSMS sample, Table 8 uses the combined rural and urban LSMS sample because labor absorption is measured at the regional labor-market level rather than through agricultural production alone. Including both rural and urban households therefore allows the labor-demand measures to capture the broader employment opportunities available within each regional labor market.

Using the Enterprise Survey measure, regions with a larger share of firms expanding employment experience significantly smaller increases in idle labor following positive productivity shocks. The negative interac-

tion indicates that stronger labor demand improves the economy's ability to absorb workers released from agriculture. Similar conclusions emerge when labor-market conditions are measured using changes in regional unemployment rates. The positive interaction coefficient indicates that regions experiencing larger increases in unemployment, that is, worsening labor-market conditions, experience larger increases in idle labor following positive productivity shocks. Conversely, regions with smaller increases (or larger declines) in unemployment exhibit weaker increases in idle labor. These results are consistent with those based on the Enterprise Survey measure and indicate that the findings are robust to the choice of labor-demand proxy.

Taken together, these findings complement the evidence from the previous stages. Positive agricultural productivity shocks consistently release labor from agriculture, while mobility barriers and land institutions influence where released workers move. The final stage demonstrates that successful structural transformation ultimately depends on whether local labor markets are capable of absorbing released workers into productive non-agricultural employment. This evidence provides reduced-form support for the Harris-Todaro mechanism and motivates the structural model developed in the next section, which jointly identifies the relative importance of labor release, labor reallocation, and labor absorption.

Table 8: Labor Demand and Surplus Labor in Ethiopia

	(1)	(2)	(3)	(4)
Dependent Variable: The Share of Surplus Labor in Each Household				
$\Delta\theta_{gt-1}$	7.585** (1.97)	8.980** (2.10)	0.669** (1.97)	0.670* (1.96)
$\Delta\theta_{gt-1} \times \text{Share of Firms Expanding Employment}$	-10.671* (-1.73)	-11.524* (-1.69)		
$\Delta\theta_{gt-1} \times \text{Share of Firms Expanding Employment} \times \text{Urban}(=1)$		-2.504* (-1.75)		
$\Delta\theta_{gt-1} \times \Delta\text{Unemployment Rate}$			0.181** (2.49)	0.161 (1.52)
$\Delta\theta_{gt-1} \times \Delta\text{Unemployment Rate} \times \text{Urban}(=1)$				0.045 (0.31)
Household Fixed Effect	Yes	Yes	Yes	Yes
Year Fixed Effect	Yes	Yes	Yes	Yes
Observations	4,456	4,456	5,152	5,152
Clusters	290	290	360	360
Within R^2	0.041	0.051	0.045	0.046

t-statistics are reported in parentheses.

Columns (1) and (2) measure labor absorption using Enterprise Survey employment growth. The 2014 Enterprise Survey is matched to the 2018 LSMS wave, and the 2024 Enterprise Survey is matched to the 2021 LSMS wave. Column (2) additionally interacts the absorption measure with an urban indicator.

Columns (3) and (4) measure labor absorption using changes in regional unemployment rates constructed from the 2016, 2018, and 2020 Urban Employment and Unemployment Surveys. Column (4) additionally interacts the unemployment measure with an urban indicator.

The estimation sample includes both rural and urban LSMS households because labor-demand measures are defined at the regional labor-market level.

Standard errors are clustered at the LSMS community level.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

3.5 Summary of Reduced-Form Evidence

Table 9 summarizes the main findings from the reduced-form analysis. Positive weather-induced productivity shocks consistently release labor from agriculture across all three countries, providing empirical support for the labor-release mechanism emphasized by Lewis (1954). However, successful structural transformation requires more than labor release alone. While workers released from agriculture in Niger and Malawi transition primarily into manufacturing employment, those in Ethiopia are substantially more likely to remain idle, suggesting that labor release does not necessarily translate into productive non-agricultural employment.

The mechanism analysis further demonstrates that different constraints operate at different stages of struc-

tural transformation. In Niger and Malawi, subsistence constraints primarily limit the release of labor from agriculture. In contrast, Ethiopia’s slower structural transformation is associated with barriers arising after labor has been released, including household land-rights constraints, spatial mobility barriers, and limited labor demand. These findings indicate that the constraints governing structural transformation differ systematically across countries, implying that a common policy prescription is unlikely to be effective in all settings.

Although the reduced-form evidence identifies the relevant mechanisms, it cannot quantify their relative importance or evaluate the effects of relaxing each constraint. The next section therefore develops a structural model that jointly identifies the mechanisms operating at the three stages of structural transformation, labor release, labor reallocation, and labor absorption, and uses the estimated model to evaluate alternative policy interventions across countries.

Table 9: Summary of Reduced-Form Evidence Across the Three Stages of Structural Transformation

Stage	Empirical Question	Niger	Malawi	Ethiopia
Stage I: Labor Release	Does higher agricultural productivity release labor from agriculture?	✓	✓	✓
	Subsistence constraints weaken labor release	✓	✓	—
Stage II: Labor Reallocation	Where do most released workers go?	Manufacturing	Manufacturing	Idle
	Mobility barriers affect reallocation	—	—	✓
	household land-rights affects reallocation	—	—	✓
Stage III: Labor Absorption	Are released workers absorbed into productive employment?	Yes	Yes	No
	Local labor demand matters	N/A	N/A	✓

This table summarizes the main findings from the reduced-form analysis. “✓” indicate that the corresponding mechanism is supported by the empirical evidence. “—” indicate that the mechanism was examined but not found to play a statistically significant role. N/A indicates that the mechanism was not directly evaluated because comparable regional labor-demand data are unavailable for that country.

3.6 Persistence of Structural Transformation

The reduced-form estimates identify households’ labor reallocation responses to temporary, weather-induced productivity shocks over the interval between two survey waves. An important question, however, is whether these responses represent only short-run cyclical adjustments or whether they contribute to persistent structural transformation. Although the LSMS-ISA panels contain only two survey waves for each country, the approximately three-year interval provides useful evidence on the persistence of labor market transitions. Therefore, we construct individual-level transition matrices across agriculture, non-agriculture, and idle labor. These transition matrices characterize the persistence of labor market outcomes and help interpret how temporary productivity shocks can generate more persistent changes in labor allocation.

Recent work emphasizes that temporary shocks can have long lasting effect on sectoral reallocation (Dinkelmann et al., 2024), especially when workers faces adjustment frictions (Porzio et al., 2022). Consistent with this view, Table 10 reports the transition probabilities across agriculture, non-agriculture, and idle labor over the three-year survey interval. Approximately one-half of workers entering non-agricultural employment

remain in the sector over the subsequent three-year survey interval, indicating that labor reallocation is not merely a short-lived response to temporary productivity shocks. By contrast, the persistence of idle labor differs sharply across countries. Only 22.6 percent of idle labor in Niger and 19.7 percent in Malawi remain idle three years later, whereas the corresponding figure reaches 66.4 percent in Ethiopia. Moreover, workers initially employed outside agriculture are substantially more likely to transition into underutilized labor in Ethiopia than in Niger or Malawi. These patterns suggest that while weather-induced productivity shocks are temporary, the employment states reached following those shocks exhibit meaningful persistence, consistent with adjustment frictions governing labor reallocation and labor absorption.

The transition matrices show that employment states themselves exhibit persistence once workers have allocated across sectors. Moreover, the heterogeneous responses documented throughout Section 4 indicate that labor reallocation depends systematically on structural constraints, including subsistence constraints, household land-rights, and labor absorption capacity, rather than reflecting only short-run cyclical conditions. Together, the reduced-form estimates identify households' behavioral responses to exogenous productivity changes, while the transition evidence demonstrates that these responses can generate persistent labor market outcomes. This distinction motivates the structural model developed in the next section, which uses the estimated behavioral responses to evaluate how persistent changes in structural constraints affect household labor allocation under alternative policy environments.

Table 10: Labor Market Transition Matrices

Panel A. Niger (2011–2014)			
2011 \ 2014	Agriculture	Non-agriculture	Idle
Agriculture	81.67	16.83	1.50
Non-agriculture	47.08	50.43	2.49
Idle	46.23	31.13	22.64

Panel B. Malawi (2016–2019)			
2016 \ 2019	Agriculture	Non-agriculture	Idle
Agriculture	83.92	15.13	0.94
Non-agriculture	38.24	58.93	2.82
Idle	66.20	14.08	19.72

Panel C. Ethiopia (2018–2021)			
2018 \ 2021	Agriculture	Non-agriculture	Idle
Agriculture	74.59	7.59	17.82
Non-agriculture	35.75	45.08	19.17
Idle	24.83	8.72	66.44

Entries report row percentages. Each row sums to 100 percent. Non-agriculture combines manufacturing and service employment.

4 Structural Model

4.1 Overview

The reduced-form evidence identifies the stages at which structural transformation is constrained, but it does not quantify the relative importance of the underlying mechanisms or evaluate how changes in structural constraints affect household labor allocation. To address these questions, I developed a structural model that decomposes structural transformation into three sequential stages: labor release, labor reallocation, and labor absorption. By modeling each stage separately, the framework isolates the economic mechanisms governing the labor movement out of agriculture and provides a unified framework for evaluating how alternative structural constraints shape household labor allocation.

The reduced-form analysis uses temporary weather-induced productivity shocks to identify households' labor-reallocation behavior in response to exogenous changes in agricultural productivity. Therefore, the structural model does not assume that temporary weather shocks themselves persist. As shown in the previous section, although the productivity shocks are temporary, employment states exhibit substantial persistence once workers have reallocated across sectors. The structural model is parameterized using these estimated behavioral responses to evaluate how persistent changes in structural constraints affect household labor allocation under alternative policy environments.

Figure 4 illustrates the three-stage framework. The model is motivated by exogenous changes in agricultural productivity that reveal households' labor-reallocation behavior. In the first stage, higher agricultural productivity relaxes households' food subsistence constraints and determines the amount of labor that can be released from agriculture. This stage captures the labor-release mechanism emphasized by Lewis (1954). In the second stage, released workers face barriers to entering non-agricultural activities. Spatial mobility costs and land institutions jointly determine whether households can successfully reallocate labor outside agriculture. Finally, in the third stage, workers who gain access to non-agricultural opportunities must be absorbed by the local labor market. The availability of productive employment determines whether released workers obtain non-agricultural jobs or remain idle, consistent with the labor-absorption mechanism emphasized by Harris and Todaro (1970).

The model integrates the key insights of Lewis (1954) and Harris and Todaro (1970) within a single framework while explicitly introducing labor reallocation as an intermediate stage between labor release and labor absorption. This decomposition is motivated by the reduced-form evidence, which shows that workers released from agriculture do not necessarily transition directly into productive non-agricultural employment. Instead, structural transformation depends on a sequence of decisions and constraints that operate after agricultural productivity improves.

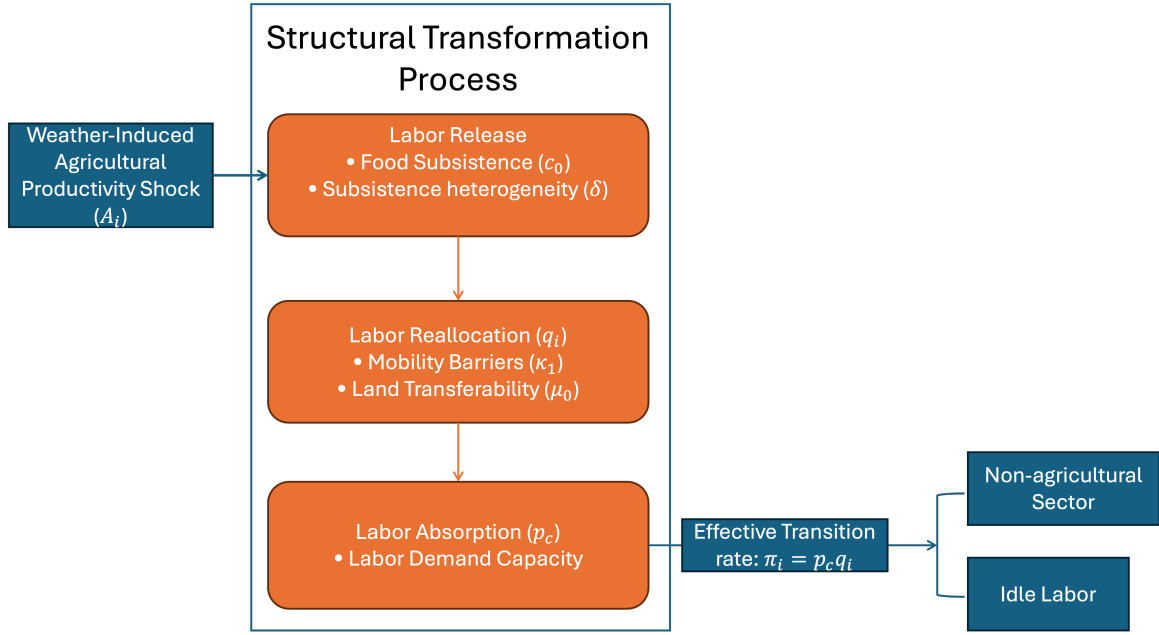


Figure 4: Sequential Structure of the Structural Transformation Model

4.2 Environment

The economy consists of households indexed by i . Each household is endowed with one unit of labor, which can be allocated across three activities: agricultural production, non-agricultural employment, and idle labor. Let

$$s_{Ai} + s_{Mi} + s_{Ni} = 1, \quad (4)$$

where s_{Ai} denotes the share of household labor allocated to agriculture, s_{Mi} denotes the share allocated to non-agricultural employment, and s_{Ni} denotes the share that remains idle. Idle labor represents household members who are not engaged in either agricultural or non-agricultural economic activities.

Households are heterogeneous in their agricultural productivity, household size, land institutions, and geographic location. Agricultural productivity is determined by a weather-induced productivity shock,

$$A_i = A(\theta_i), \quad (5)$$

where θ_i denotes the weather-induced agricultural productivity measure constructed in Section 2. A higher value of θ_i increases agricultural productivity and relaxes the household's production constraint.

The timing of the model follows the agricultural production cycle. At the beginning of each period, households observe the realized weather-induced productivity shock. They then determine the amount of labor required for agricultural production, decide whether released labor can access non-agricultural opportunities, and finally participate in the local labor market. Consequently, the final allocation of labor across agriculture, non-agricultural employment, and idle activities is determined sequentially through the three stages illustrated in Figure 4.

4.3 Stage I: Labor Release

The first stage of the model determines how much labor households can release from agriculture following a weather-induced productivity shock. Following Lewis (1954), households must first allocate sufficient labor to agriculture to satisfy their food subsistence requirement before supplying labor to non-agricultural activities. Improvements in agricultural productivity increase output for a given labor input, allowing households to meet their food needs with fewer agricultural workers and thereby releasing labor for non-agricultural activities.

Agricultural production is given by

$$y_{Ai} = A_i s_{Ai}^\gamma, \quad (6)$$

where A_i denotes weather-induced agricultural productivity and $\gamma \in (0, 1)$ captures diminishing marginal returns to agricultural labor.

Each household faces a minimum food subsistence requirement,

$$\bar{c}_i = c_0(1 + \delta Z_i)h_i, \quad (7)$$

where h_i denotes household size, Z_i is an indicator for whether the household is food-subsistence constrained, c_0 captures the baseline subsistence requirement, and δ measures the additional food requirement faced by subsistence-constrained households.⁶ When $\delta = 0$, all households face the same subsistence requirement regardless of food-security status.

Households choose agricultural labor to maximize utility from agricultural consumption and labor available for non-agricultural activities:

$$\max_{s_{Ai} \in [0,1)} \left\{ \log(A_i s_{Ai}^\gamma - \bar{c}_i) + \beta \log(1 - s_{Ai}) \right\}, \quad (8)$$

subject to

$$A_i s_{Ai}^\gamma \geq \bar{c}_i. \quad (9)$$

The first term represents agricultural consumption net of the household's subsistence requirement, while the second term captures the value of labor available for non-agricultural activities. The parameter $\beta > 0$ governs the relative value households place on labor released from agriculture.

The subsistence requirement implies a minimum feasible agricultural labor allocation,

$$s_{Ai} = \left(\frac{\bar{c}_i}{A_i} \right)^{1/\gamma}. \quad (10)$$

⁶The parameter δ allows subsistence requirements to differ across households by capturing the additional food requirement faced by subsistence-constrained households. This heterogeneity reflects that households experiencing more severe food insecurity must allocate more labor to agriculture before participating in non-agricultural activities.

For an interior solution, the optimal allocation satisfies the first-order condition

$$\frac{A_i \gamma s_{Ai}^{\gamma-1}}{A_i s_{Ai}^{\gamma} - \bar{c}_i} = \frac{\beta}{1 - s_{Ai}}. \quad (11)$$

Let s_{Ai}^{int} denote the corresponding interior solution. The household's optimal agricultural labor allocation is therefore

$$s_{Ai}^* = \max \{1, \max \{s_{Ai}, s_{Ai}^{int}\}\}. \quad (12)$$

If the subsistence requirement cannot be met even when the household allocates its entire labor endowment to agriculture, the household reaches a corner solution with $s_{Ai}^* = 1$. In this case, no labor is released from agriculture ($S_i = 0$)

The amount of labor released from agriculture is

$$S_i = 1 - s_{Ai}^*, \quad (13)$$

where S_i denotes the labor available for non-agricultural activities after satisfying the household's food subsistence requirement.

Equation (13) summarizes the labor-release stage of the model. A positive weather-induced productivity shock raises agricultural productivity A_i , reducing the minimum labor required to satisfy subsistence needs and increasing the amount of labor released from agriculture. Conversely, adverse weather shocks reduce agricultural productivity, requiring households to retain more labor in agriculture and leaving less labor available for structural transformation.

4.4 Stage II: Labor Reallocation

The second stage of the model determines whether labor released from agriculture can successfully access non-agricultural opportunities. Releasing labor from agriculture does not automatically imply successful structural transformation. Even after agricultural productivity has increased and households have labor available to leave farming, workers may face barriers that prevent them from entering non-agricultural activities. These barriers arise from both geographic frictions and institutional constraints, which jointly determine the opportunity cost of leaving agriculture.

Let $q_i \in [0, 1]$ denote the probability that labor released from agriculture successfully reaches the non-agricultural labor market. This probability captures households' ability to overcome reallocation barriers before employment opportunities can be realized. The reallocation probability is specified as

$$q_i = \Lambda [\mu_0^{NT} (1 - T_i) + \mu_0^T T_i + \kappa_1 dist_i], \quad (14)$$

where $\Lambda(\cdot)$ denotes the logistic cumulative distribution function, T_i is an indicator equal to one if the household processes stronger land rights and zero otherwise. For the structural estimation, the three-category land-rights measure used in the reduced-form analysis is collapsed into this binary indicator, with land Type 1 and 2 classified as strong land rights ($T_i = 1$) and land Type 0 classified as weak land rights ($T_i = 0$). The variable $dist_i$ measures the household's mobility barrier, proxied by the distance to the nearest economic center.

The parameters μ_0^{NT} and μ_0^T capture the baseline probability of accessing non-agricultural opportunities under weak land-rights ($T_i = 0$) and strong land-rights ($T_i = 1$) institutions, respectively. The parameter κ_1 measures the effect of spatial mobility barriers on labor reallocation. A larger value of μ_0^T relative to μ_0^{NT} implies that stronger land rights lowers the opportunity cost of leaving agriculture and facilitates labor reallocation, while a more negative value of κ_1 implies that greater mobility barriers reduce access to non-agricultural opportunities.

Conditional on labor being released from agriculture, the amount of labor successfully reallocated to the non-agricultural labor market is

$$R_i = q_i S_i \quad (15)$$

where S_i denotes labor released from agriculture in Stage I. The variable R_i summarizes the amount of labor that has successfully overcome reallocation barriers and therefore becomes available for non-agricultural employment. Whether this reallocated labor is ultimately employed or remains idle is determined by local labor demand in the next stage of the model.

4.5 Stage III: Labor Absorption

The final stage of the model determines whether workers who have successfully accessed non-agricultural opportunities obtain productive employment. Access to the labor market does not guarantee employment. Following Harris and Todaro (1970), workers compete for a limited number of non-agricultural jobs, and the availability of these jobs depends on local labor demand. Consequently, even after overcoming reallocation barriers, some workers may remain outside productive employment if labor demand is insufficient.

Let $p_i \in [0, 1]$ denote the probability that workers entering the non-agricultural labor market are successfully absorbed into productive employment. The parameter p_i summarizes the labor absorption capacity of the local economy.

Conditional on reaching the non-agricultural labor market, the amount of labor employed outside agriculture is

$$s_{Mi} = p_i R_i \quad (16)$$

where $R_i = q_i S_i$ is the amount of labor successfully reallocated in Stage II.

Idle labor consists of two groups of workers: those who fail to access non-agricultural opportunities and those who access the labor market but fail to obtain employment. Therefore,

$$s_{Ni} = (S_i - R_i) + (1 - p_i)R_i \quad (17)$$

The model interprets idle labor as the component of labor allocation generated by unsuccessful transitions during the structural transformation process. It captures workers who remain idle because they are unable to reach productive non-agricultural opportunities or fail to obtain employment after entering the labor market. The model is not intended to explain all sources of observed idle labor, such as persistent baseline non-participation unrelated to agricultural productivity shocks.

which can be equivalently written as

$$s_{Ni} = (1 - p_i q_i) S_i \quad (18)$$

For notational convenience, define

$$\pi_i = p_i q_i \quad (19)$$

where π_i denotes the overall probability that labor released from agriculture ultimately enters productive non-agricultural employment. The employment and idle labor allocations can therefore be written as

$$s_{Mi} = \pi_i S_i \quad (20)$$

And

$$s_{Ni} = (1 - \pi_i) S_i \quad (21)$$

Equations (16)–(19) summarize the labor-absorption stage of the model. A higher labor-absorption probability increases the share of reallocated workers entering productive non-agricultural employment, while a lower absorption probability results in a larger share remaining idle following labor release. Thus, successful structural transformation depends not only on releasing labor from agriculture or enabling labor reallocation, but also on the capacity of local labor markets to absorb workers into productive employment.

In the theoretical framework, the absorption probability p_i is allowed to vary across households and locations to reflect heterogeneity in local labor demand. For empirical tractability, however, the structural estimation assumes a common within-country absorption probability ($p_i = p$), so that the estimated parameter captures the average labor absorption capacity of the national economy while abstracting from within-country regional variation.

4.6 Model Implication

The model characterizes structural transformation as a sequential process in which households move through three distinct stages. Weather-induced agricultural productivity first determines the amount of labor released from agriculture through the household's subsistence constraint. Released labor must then overcome reallocation barriers to access non-agricultural opportunities. Finally, workers who successfully reach the non-agricultural labor market are either absorbed into productive employment or remain idle, depending on local labor demand.

Taken together, the three stages determine the household labor allocation implied by the model:

$$S_i = 1 - s_{Ai}^*, \quad (22)$$

$$R_i = q_i S_i, \quad (23)$$

$$s_{Mi} = p_i R_i, \quad (24)$$

$$s_{Ni} = (S_i - R_i) + (1 - p_i) R_i. \quad (25)$$

The model contains four groups of structural parameters corresponding to the three stages of structural transformation. The first-stage parameters, c_0 and δ , govern households' subsistence requirements and therefore determine the amount of labor released from agriculture following a productivity shock. The second-stage parameters, μ_0^{NT} , μ_0^T , and κ_1 , determine the probability that released labor can access non-agricultural opportunities by capturing the effects of land institutions and spatial mobility barriers. The third-stage parameter, p , captures the probability that workers who reach the non-agricultural labor market are successfully absorbed into productive employment.

These structural parameters are not directly observed in the data. Instead, they are inferred by matching the model-implied labor allocation generated by the three-stage framework to the corresponding labor allocation observed in the household survey data. The next section estimates these parameters using the simulated method of moments and then uses the estimated model to evaluate the effects of alternative policy interventions on structural transformation.

5 Structural Estimation

5.1 Parameterization

The structural model developed in the previous section contains two types of parameters: calibrated parameters that are fixed prior to estimation and structural parameters that are estimated from the data. The calibrated parameters determine the agricultural production technology and the mapping from weather-induced produc-

tivity shocks to agricultural productivity. Specifically, agricultural productivity is constructed as

$$A_i = \exp(a_0 + a_1 \Delta \theta_{gt}), \quad (26)$$

where the calibration parameters a_0 and a_1 are obtained from the reduced-form relationship between household agricultural income and weather-induced productivity growth reported in Appendix. The remaining production parameters, including the agricultural labor elasticity $\gamma = 0.5$, and the utility parameter $\beta = 0.95$ appearing in the household optimization problem, are calibrated prior to estimation.

Conditional on these calibrated parameters, the structural estimation focuses on recovering the behavioral parameters governing the three stages of structural transformation. These parameters summarize households' subsistence requirements, the barriers to labor reallocation, and the ability of local labor markets to absorb workers into productive employment. While the model specifies how these parameters jointly determine household labor allocation, they are not directly observed in the data and must be estimated.

Table 11 summarizes the structural parameters estimated in the model and their corresponding economic interpretations. Consistent with the three-stage framework, each group of parameters governs a distinct stage of structural transformation.

Table 11: Structural Parameters

Stage	Parameter	Economic Interpretation
Labor Release	c_0	Baseline subsistence requirement
	δ	Additional subsistence requirement for food-constrained households
Labor Reallocation	μ_0^{NT}	Baseline access to non-agricultural opportunities under weaker land rights
	μ_0^T	Baseline access to non-agricultural opportunities under stronger land rights
	κ_1	Effect of spatial mobility barriers on labor reallocation
Labor Absorption	p	Probability that reallocated workers obtain productive employment

5.2 Estimation Procedure

I estimate the structural parameters using simulated method of moments (SMM). For a candidate parameter vector, the model is solved for each household using its observed characteristics to predict labor allocation across agriculture, productive non-agricultural employment, and idle labor. These household-level predictions are then aggregated into a set of simulated moments that are directly comparable to the corresponding moments computed from the household survey. The structural parameters are chosen to minimize the distance between the simulated and empirical moments, thereby selecting the parameter values that best reproduce the observed labor-allocation patterns.

The moment conditions are organized according to the three stages of the model (see Table 12), allowing each stage to be disciplined by moments directly related to its underlying economic mechanism. Conditional

on the fixed preference and technology parameters β and γ , the labor-release parameters, c_0 and δ , are identified using agricultural labor-share moments, including the overall agricultural labor share, agricultural labor shares across four climate-productivity-shock bins, and agricultural labor shares by subsistence status. These moments identify the amount of labor released from agriculture in response to productivity changes.

The labor-reallocation parameters, μ_0^{NT} , μ_0^T , and κ_1 , are identified from transition-probability moments across land-transferability regimes and distance bins. These moments exploit systematic variation in households' ability to access non-agricultural opportunities arising from land institutions and spatial mobility barriers, thereby identifying heterogeneity in the household-specific reallocation probability, q_i .

Finally, the labor-absorption parameter, p , is identified from the aggregate allocation of released labor between productive non-agricultural employment and underutilized labor. Conditional on the estimated distribution of household-specific reallocation probabilities, the remaining split between successful non-agricultural employment and underutilized labor disciplines the common labor-absorption probability. Thus, labor reallocation is identified from cross-household heterogeneity in structural barriers, whereas labor absorption is identified from the aggregate employment allocation.

The moment conditions are organized according to the three stages of the model (see Table 12). The labor-release parameters, c_0 and δ , are identified using agricultural labor-share moments, including the overall agricultural labor share, agricultural labor shares across four climate-productivity-shock bins, and agricultural labor shares by subsistence status. The labor-reallocation parameters, μ_0^{NT} , μ_0^T , and κ_1 , are identified using transition-probability moments, which compare the model-implied probability that released labor enters productive non-agricultural employment, $\pi_i = p_i q_i$, with the corresponding transition probabilities observed across land regimes, four distance bins, and their interactions. Finally, the labor-absorption parameter, p , is identified using moments based on the allocation of released workers between productive non-agricultural employment and idle labor.

Let $\Theta = (c_0, \delta, \mu_0^{NT}, \mu_0^T, \kappa_1, p)$ denote the vector of structural parameters to be estimated. For each candidate parameter vector Θ , the model generates a vector of simulated moments $m(\Theta)$. Let $\hat{m}$ denote the corresponding empirical moments computed from the household survey. The SMM estimator solves

$$\hat{\Theta} = \arg \min_{\Theta} [m(\Theta) - \hat{m}]' W [m(\Theta) - \hat{m}],$$

where W is a weighting matrix. In the baseline specification, I use an identity weighting matrix so that the estimator minimizes the unweighted squared distance between the model-implied and empirical moments.

Although productive non-agricultural employment depends jointly on labor reallocation and labor absorption, the two mechanisms are identified using different sources of variation. Cross-household heterogeneity in distance and household land-rights identifies the household-specific reallocation probability through

observed differences in transition behavior. Conditional on this estimated heterogeneity, the remaining aggregate allocation between productive non-agricultural employment and idle labor identifies the common labor-absorption probability. This separation therefore relies jointly on the transition moments, the labor-allocation moments, and the maintained structural assumptions governing the reallocation and absorption processes.

The estimation proceeds in two steps. In the first step, I estimate the labor-release parameters, c_0 and δ , using only the agricultural labor-share moments because these parameters govern households' decisions to release labor from agriculture. Holding these estimates fixed, I jointly estimate the labor-reallocation and labor-absorption parameters, μ_0^{NT} , μ_0^T , κ_1 , and p , by matching the transition-probability moments together with the observed allocation of released labor between productive non-agricultural employment and idle labor. This sequential procedure follows the economic structure of the model, allowing the labor-release mechanism to be identified independently before estimating the subsequent reallocation and absorption stages.

Inference is based on a household-cluster bootstrap. For each bootstrap replication, households are resampled with replacement, the complete estimation procedure is repeated, and the resulting parameter estimates are recorded. Standard errors are calculated from the bootstrap distribution, and confidence intervals are constructed using bootstrap percentiles. The baseline results are based on 100 bootstrap replications.

Table 12: Moments Used in Simulated Method of Moments Estimation

Model Stage	Parameters	Primary Identifying Moments	Identification
Labor Release	c_0, δ	$E[s_A], E[s_A \mid \text{climate bin}], E[s_A \mid D]$	Variation in agricultural labor shares identifies the amount of labor released from agriculture under different productivity and subsistence conditions.
Labor Reallocation	$\mu_0^{NT}, \mu_0^T, \kappa_1$	$E[\pi^{pred} - \pi^{obs} \mid \text{distance bin}],$ $E[\pi^{pred} - \pi^{obs} \mid T],$ $E[\pi^{pred} - \pi^{obs} \mid T, \text{distance bin}]$	Cross-household variation in distance and household land-rights identifies heterogeneity in the household-specific reallocation probability q_i .
Labor Absorption	p	$E[s_M^{pred} - s_M^{obs}],$ $E[s_N^{pred} - s_N^{obs}]$	Conditional on the estimated distribution of q_i , the aggregate allocation between productive non-agricultural employment and underutilized labor identifies the common labor-absorption probability.

a. s_A , s_M , and s_N denote agricultural labor, productive non-agricultural employment, and underutilized labor shares, respectively.

b. Climate-productivity bins and distance bins are constructed using quartiles of the corresponding household-level distributions.

c. D denotes household subsistence status. $T = 1$ indicates stronger land rights and $T = 0$ indicates weaker land rights.

d. $\pi_i = pq_i$ denotes the probability that labor released from agriculture successfully enters productive non-agricultural employment, where q_i is household-specific and varies with household land-rights and distance, while p is a country-specific labor-absorption probability.

e. Aggregate labor-allocation moments are primarily informative about π_i . Variation across land-transferability regimes and distance bins disciplines the household-specific reallocation probability q_i , while the remaining aggregate split between productive non-agricultural employment and underutilized labor identifies p .

5.3 Structural Parameter Estimates

Table 13 reports the estimated structural parameters for Niger, Malawi, and Ethiopia. The estimates quantify the mechanisms governing the three stages of structural transformation and complement the reduced-form evidence presented in Section 3 by jointly estimating the relative importance of multiple frictions within a unified structural framework.

The estimated labor-release parameters indicate substantial cross-country differences in the importance of subsistence constraints. Malawi exhibits the largest baseline subsistence requirement ($c_0 = 1.125$), followed by Niger ($c_0 = 0.895$), while Ethiopia has the lowest value ($c_0 = 0.343$). The estimated food subsistence heterogeneity parameter, δ , is positive and statistically significant in Niger and Malawi, implying that households below the food subsistence threshold require substantially more agricultural labor satisfy minimum consumption needs. By contrast, the estimate of δ is statistically insignificant in Ethiopia, suggesting that food subsistence constraints play a much smaller role in determining labor release.

The estimated labor-reallocation parameters also differ markedly across countries. The migration-cost parameter, κ_1 , is negative and statistically significant in Niger and Ethiopia, indicating that greater distance from economic centers reduces households' access to non-agricultural opportunities. However, as shown in Panel B, the implied probability effect is substantially larger in Ethiopia than in Niger, indicating that mobility costs constitute a quantitatively more important barrier to labor reallocation in Ethiopia. In contrast, the migration-cost parameter is close to zero and statistically insignificant in Malawi, implying that spatial mobility barriers are relatively weak.

The latent land effect is positive and statistically significant in Malawi and Ethiopia but insignificant in Niger, suggesting that household land rights substantially improves households' ability to access non-agricultural opportunities only in the former two countries. This pattern is consistent with institutional differences across countries. Ethiopia's state ownership of rural land and restrictions on stronger land rights increase the importance of secure and stronger land rights for households considering non-agricultural employment. Malawi also exhibits a statistically significant household land-rights effect, although its magnitude is considerably smaller than in Ethiopia. By contrast, the insignificant land effect in Niger suggests that the household land-rights measure used here does not substantially explain heterogeneity in labor allocation. This does not necessarily imply that broader land institutions are unimportant, but rather that variation in inheritance and collateral rights appears less consequential for household labor responses in Niger sample.

Because the latent land effect is expressed on the logistic index, Panel B reports the corresponding average effects on economically meaningful probabilities. Stronger household land rights increases the probability of accessing non-agricultural opportunities by approximately 2.3 percentage points in Malawi and 29.6 percentage points in Ethiopia, while having essentially no effect in Niger. Although the structural model identifies statistically significant household land-rights effects in both Malawi and Ethiopia, the much larger probabil-

ity effect in Ethiopia is consistent with the reduced-form evidence that downstream institutional barriers are considerably more important there.

Finally, the estimated labor-absorption probability highlights substantial differences in local labor demand across countries. Niger and Malawi exhibit nearly complete labor absorption, with estimated probabilities exceeding 0.97, indicating that workers who successfully access non-agricultural opportunities are almost always absorbed into productive employment. Ethiopia, on the other hand, has an estimated absorption probability of only 0.735, implying that roughly one-quarter of workers who successfully reach the non-agricultural labor market remain idle rather than obtaining productive employment.

Overall, the structural estimates closely mirror the reduced-form evidence. Structural transformation in Niger and Malawi is primarily constrained by labor released from agricultural through subsistence requirements, whereas structural transformation in Ethiopia is constrained by both labor reallocation and labor absorption. These estimated mechanisms provide the foundation for the policy counterfactual analysis in the next section.

Table 13: Structural Parameter Estimates and Model-Implied Land Effects

	Niger	Malawi	Ethiopia
Panel A. Structural Parameters			
Subsistence parameter c_0	0.895*** (0.030)	1.125*** (0.094)	0.343*** (0.019)
Subsistence heterogeneity δ	0.128** (0.055)	1.230*** (0.217)	-0.105 (0.171)
Migration cost κ_1	-1.879*** (0.036)	0.010 (0.185)	-0.774*** (0.139)
Latent land effect ($\mu_0^T - \mu_0^{NT}$)	-0.349 (0.206)	2.487*** (0.449)	1.642*** (0.335)
Absorption probability p	0.973*** (0.003)	0.999*** (0.0004)	0.735*** (0.034)
Panel B. Model-Implied Land Effects			
$\Delta q = E[q T = 1] - E[q T = 0]$	-0.028	0.023	0.296
$\Delta \pi = E[\pi T = 1] - E[\pi T = 0]$	-0.027	0.023	0.218

Bootstrap standard errors (100 household-cluster bootstrap replications) are reported in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The latent land effect ($\mu_0^T - \mu_0^{NT}$) measures the difference in the transition index between strong and weak land rights. Panel B reports the corresponding average effects on the probability of accessing non-agricultural opportunities (q) and the overall transition probability into productive non-agricultural employment ($\pi = pq$), which are more directly interpretable.

$T = 1$ denotes stronger land rights and $T = 0$ denotes weaker land rights.

6 Policy Counterfactual Analysis

The structural estimates quantify the key mechanisms governing labor release, labor reallocation, and labor absorption. However, parameter estimates alone do not indicate which policies are most effective at accelerating structural transformation. The structural model provides a natural framework for answering this question because it allows individual mechanisms to be varied while holding all other features of the economy fixed.

The objective of the counterfactual analysis is to quantify the relative importance of the structural frictions identified by the model. Accordingly, each policy experiment perturbs one structural parameter while holding the remaining mechanisms fixed. Therefore, these counterfactuals should be interpreted as mechanism-specific comparative statics rather than comprehensive policy evaluation.

Using the estimated parameters, I conduct a series of counterfactual policy experiments corresponding to the three stages of structural transformation. Specifically, I examine policies that relax subsistence constraints, improve agricultural productivity, reduce mobility barriers, strengthen household land-rights, and increase absorption in the non-agricultural labor market. Comparing the simulated labor allocations across these counterfactual scenarios reveals both the relative importance of each mechanism and the policy interventions most likely to promote structural transformation in each country.

6.1 Counterfactual Design

The estimated structural model provides a framework for evaluating how alternative policies affect structural transformation. I conduct counterfactual simulations by perturbing one policy-relevant structural parameter at a time while holding all other estimated parameters fixed. This allows the simulations to isolate how each intervention affects household labor allocation through labor release and labor reallocation.

To compare policies across countries, I calibrate each intervention using a common fiscal budget of USD 500 million per country. The calibration translates the same policy budget into changes in the structural model through three steps. First, I define the target population for each policy. Second, I convert the budget into either a transfer per eligible household, the share of households reached, or the share of an implementation gap closed. Third, I map this policy intensity into the corresponding structural parameter. Detailed country-specific calculations are reported in Appendix A.

Policy 1: Relaxing Subsistence Constraints. The first policy provides targeted food assistance to working-age individuals in rural households below the food-subsistence threshold. Because the policy directly relaxes minimum consumption requirements, it is mapped into the subsistence component of the model.

Let B denote the common policy budget. Let POP_c denote the national population in country c , and let $\hat{s}_c^T$ denote the share of the baseline population that is simultaneously working age, rural, and living in a household below the food-subsistence threshold, estimated from the nationally representative LSMS survey. This joint population share differs from the household-level food-subsistence share reported in Table 2 because it is calculated over the baseline population rather than over households.

The number of targeted individuals is

$$N_c^T = \hat{s}_c^T \times POP_c.$$

The transfer per targeted individual is

$$T_c^F = \frac{B}{N_c^T}. \quad (27)$$

I express this transfer relative to the annual food-subsistence benchmark F_c , measured on a per-person basis using the national food poverty line reported by the World Bank. The resulting policy intensity is

$$\lambda_c^F = \min \left\{ 1, \frac{T_c^F}{F_c} \right\}.$$

The policy is implemented by proportionally reducing the additional subsistence requirement for households below the food-subsistence threshold:

$$\bar{c}_i' = c_0 [1 + (1 - \lambda_c^F) \delta Z_i] h_i, \quad (28)$$

where $Z_i = 1$ for households below the food-subsistence threshold and $Z_i = 0$ otherwise.

Policy 2: Improving Agricultural Productivity. The second policy represents agricultural input programs, such as fertilizer subsidies and improved seed distribution. These policies increase agricultural productivity and allow households to satisfy subsistence needs with less agricultural labor.

Let C_A denote the cost of providing an agricultural input package to one household, and let N_c^R denote the number of rural households in country c .⁷ The share of households reached by the policy is

$$s_c^A = \min \left\{ 1, \frac{B}{C_A N_c^R} \right\}. \quad (29)$$

If treated households experience a productivity gain g_A , the average productivity increase is

$$\lambda_c^A = s_c^A g_A. \quad (30)$$

The policy is implemented as a proportional increase in agricultural productivity:

$$A_i' = (1 + \lambda_c^A) A_i. \quad (31)$$

Policy 3: Reducing Mobility Barriers. The third policy represents investments in rural roads and transport infrastructure. Improved transportation lowers the effective mobility barriers faced by rural households and facilitates access to non-agricultural opportunities.

Let C_R denote the country-specific construction cost per kilometer of rural roads, obtained from the World

⁷ Similar to the N_c^T , I approximate the number of rural households using national population data and the rural share observed in the household survey.

Bank's Road Costs Knowledge System (ROCKS). Under the common policy budget B , the number of kilometers financed is

$$K_c^R = \frac{B}{C_R}. \quad (32)$$

Let $\bar{K}_c^R$ denote the country-specific rural road gap, measured as the additional kilometers of rural roads required to achieve full rural accessibility using country-specific World Bank transport assessments. The share of the road gap closed by the policy is

$$s_c^R = \min \left\{ 1, \frac{K_c^R}{\bar{K}_c^R} \right\}. \quad (33)$$

Following the transport literature, I assume that completely closing the road gap removes a fraction $\bar{\rho}^R$ of the estimated mobility barrier.⁸ The resulting policy intensity is

$$\lambda_c^R = s_c^R \bar{\rho}^R. \quad (34)$$

The policy is implemented as a proportional reduction in the migration-cost parameter:

$$\kappa'_{1c} = (1 - \lambda_c^R) \kappa_{1c}. \quad (35)$$

Policy 4: Strengthening household land-rights. The fourth policy represents land certification and tenure reforms that strengthen land rights. These reforms reduce institutional barriers to labor reallocation by allowing households to retain or transfer the value of agricultural land while participating in non-agricultural activities.

Let C_L denote the cost of providing land certification or tenure reform to one household, and let N_c^{NT} denote the number of households with weaker land rights in country c .⁹ Under the common policy budget B , the share of weak land-rights households covered by the reform is

$$\lambda_c^L = \min \left\{ 1, \frac{B}{C_L N_c^{NT}} \right\}. \quad (36)$$

Because the estimated implementation cost of land certification is relatively low, the common policy budget is sufficient to certify all households with weaker land rights in the three study countries.

The policy is implemented by randomly converting a fraction λ_c^L of households from weaker land rights

⁸In the benchmark calibration, we set $\bar{\rho}^R = 0.5$, implying that completely closing the rural road gap removes one-half of the estimated mobility barrier. This assumption reflects that improvements in road infrastructure reduce, but do not eliminate, all barriers to labor mobility, such as information frictions, relocation costs, and social constraints.

⁹I estimate N_c^{NT} by multiplying the total number of households by the share of households with weaker land rights observed in the nationally representative LSMS survey.

($T_i = 0$) to stronger land rights ($T_i = 1$):

$$T'_i = \begin{cases} 1, & \text{if household } i \text{ is selected for certification and } T_i = 0, \\ T_i, & \text{otherwise.} \end{cases} \quad (37)$$

The reallocation probability is then recalculated as

$$q'_i = \Lambda [\mu_0^{NT} (1 - T'_i) + \mu_0^T T'_i + \kappa_1 dist_i]. \quad (38)$$

Instead of altering the estimated structural parameters, the policy changes households' institutional environment by expanding the share of households with stronger land rights. The estimated reallocation function is then re-evaluated using the updated land status, increasing households' probability of accessing non-agricultural opportunities.

Policy 5: Increasing Labor Absorption Capacity. The fifth policy represents industrial investment that expands the ability of the non-agricultural sector to absorb workers. This policy targets the labor-absorption stage of the model by increasing the probability that workers who successfully access the non-agricultural labor market obtain productive employment.

Because the structural estimates indicate that labor-demand constraints are primarily binding in Ethiopia, I calibrate this policy using evidence from Ethiopia's industrial park investments. The World Bank Competitiveness and Job Creation Project supported the Bole Lemi II and Kilinto industrial parks and generated approximately 19,000 direct jobs with total financing of USD 425 million. Scaling this benchmark to the common policy budget of USD 500 million implies approximately 22,353 new jobs. Because both industrial parks are located in Addis Ababa, I express the job creation effect relative to the unemployed population in Addis Ababa reported in the 2020 Urban Employment and Unemployment Survey.

Let J denote the number of jobs created by the policy and let U_{AA} denote the unemployed population in Addis Ababa. The implied increase in labor absorption capacity is

$$\lambda^D = \frac{J}{U_{AA}}. \quad (39)$$

The policy is implemented as an increase in the labor absorption probability:

$$p' = \min \{1, p + \lambda^D\}. \quad (40)$$

For Niger and Malawi, I retain the estimated labor absorption probabilities because the structural estimates indicate that labor-demand constraints are already weak in these two countries. Consequently, the labor absorption policy is evaluated only for Ethiopia, where it represents a realistic industrial investment designed to

relax the binding demand-side constraint.

Table 14 summarizes the four counterfactual policies and the model objects they modify. The resulting labor allocations are compared with the baseline allocations generated by the estimated structural model.

Table 14: Calibrated Policy Intensities

Country	Food Subsidy (λ_c^F)	Ag. Productivity (λ_c^A)	Roads (λ_c^R)	Land Reform Coverage (λ_c^L)	Labor Absorption (λ_D)
Niger	0.887	0.150	0.500	1.000	0.000
Malawi	0.397	0.150	0.500	1.000	0.000
Ethiopia	0.463	0.063	0.052	1.000	0.056

Food subsidy, agricultural productivity, and roads report the calibrated proportional changes in the corresponding model objects. Land reform reports the share of households with weaker land rights that are converted to stronger land rights under the common policy budget. Because the estimated cost of land certification is relatively low, the common budget is sufficient to certify all households with weaker land rights in the three study countries. Labor absorption reports the increase in the labor absorption probability calibrated from the job creation effect of a USD 500 million industrial investment. This policy is calibrated only for Ethiopia because the structural estimates indicate that labor-demand constraints are binding only in Ethiopia.

To examine complementarity across stages of structural transformation, I also simulate combinations of policies. For each combination, the total budget is fixed at USD 500 million and divided equally among the selected policies. The corresponding parameter changes are implemented simultaneously, and the model is re-solved to obtain the resulting household labor allocation.

6.2 Niger

Figures 5a and 5b present the counterfactual policy simulations for Niger. Consistent with the structural estimates in Section 5, the results indicate that structural transformation in Niger is primarily constrained during the labor-release stage, while labor absorption is already close to its frontier.

Among the single-policy interventions, improving agricultural productivity generates the largest structural transformation effect. Agricultural labor declines by 4.6 percentage points, accompanied by a 4.4 percentage point increase in non-agricultural employment. Targeted food subsidies also substantially relax the subsistence constraint, reducing agricultural labor by 1.8 percentage points and increasing non-agricultural employment by a similar magnitude. Neither policy changes the transition probability (π) or the reallocation probability (q), indicating that their effects operate entirely through the first stage of the model by releasing labor from agriculture.

Policies targeting the second stage produce considerably smaller effects. Reducing mobility barriers increases the transition probability by 1.5 percentage points and raises non-agricultural employment by only 0.5 percentage points, suggesting that although mobility frictions exist, they are not the primary obstacle to structural transformation. By contrast, strengthening household land-rights has virtually no effect on labor allocation and slightly reduces the estimated transition probability. This result is consistent with the insignificant land-transferability effect estimated in the structural model, indicating that land institutions do not constitute a

major barrier to labor reallocation in Niger.

Finally, increasing labor absorption capacity produces essentially no change in labor allocation. Because the estimated labor absorption probability is already very high ($p \approx 0.96$), expanding labor demand generates little additional structural transformation. This finding confirms that insufficient labor demand is not the binding constraint in Niger.

The combined-policy simulations further reinforce these conclusions. Under a fixed total policy budget of USD 500 million, allocating resources to both food subsidies and agricultural productivity improvements generates the largest reduction in agricultural labor. Redistributing part of this budget toward mobility improvements produces only modest additional gains through higher transition probabilities. Similarly, allocating resources to land reform or labor absorption policies yields relatively limited additional improvements beyond those achieved through the agricultural interventions. Even when the fixed budget is distributed across all five policy instruments, most of the increase in non-agricultural employment continues to be attributable to relaxing subsistence constraints rather than to additional investments in labor reallocation or labor absorption.

Overall, the counterfactual results closely mirror the structural estimates. In Niger, structural transformation is primarily constrained by households' ability to release labor from agriculture. Policies that improve agricultural productivity and relax food subsistence constraints therefore generate the largest gains, whereas policies targeting later stages of structural transformation have comparatively limited effects.

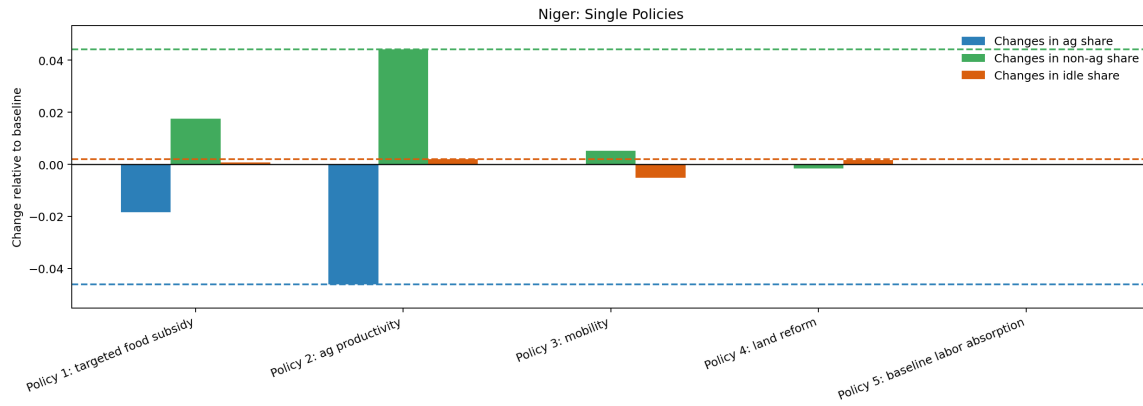
6.3 Malawi

Figures 6a and 6b present the counterfactual policy simulations for Malawi. Consistent with the structural estimates in Section 5, the results indicate that structural transformation in Malawi is overwhelmingly constrained during the labor-release stage. Once labor is released from agriculture, households face relatively small barriers to reallocation, while labor absorption is already close to complete.

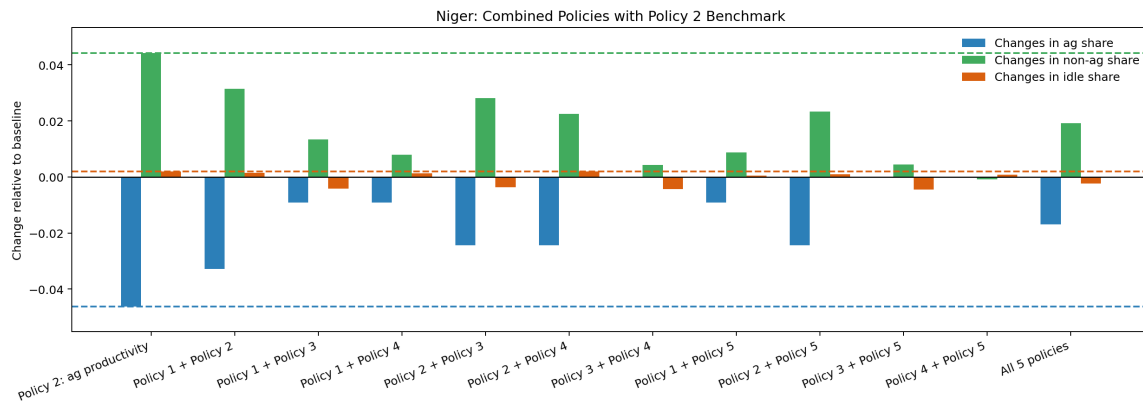
Among the single-policy interventions, targeted food subsidies generate the largest improvement in structural transformation, reducing the agricultural labor share by 5.5 percentage points and increasing non-agricultural employment by 5.4 percentage points. Improving agricultural productivity produces a similarly large effect, reducing agricultural labor by 4.1 percentage points and increasing non-agricultural employment by 4.1 percentage points.

Policies targeting later stages of structural transformation generate substantially smaller effects. Reducing mobility barriers produces virtually no change in labor allocation or transition probabilities, consistent with the insignificant migration-cost parameter estimated in the structural model. Strengthening household land-rights modestly increases the transition probability by 0.7 percentage points, but raises non-agricultural em-

Figure 5: Counterfactual Analysis in Niger



(a) Single Policy



(b) Combined Policy

The counterfactual simulations indicate that policies targeting the labor-release stage generate the largest improvements in structural transformation in Niger. Agricultural productivity improvements and targeted food subsidies substantially reduce agricultural labor and increase non-agricultural employment. Reducing mobility barriers provides modest additional gains by improving access to non-agricultural opportunities, whereas strengthening household land-rights and labor absorption produce relatively limited effects. These results are consistent with the structural estimates, which identify subsistence constraints as the primary barrier to structural transformation and mobility frictions as a secondary constraint in Niger.

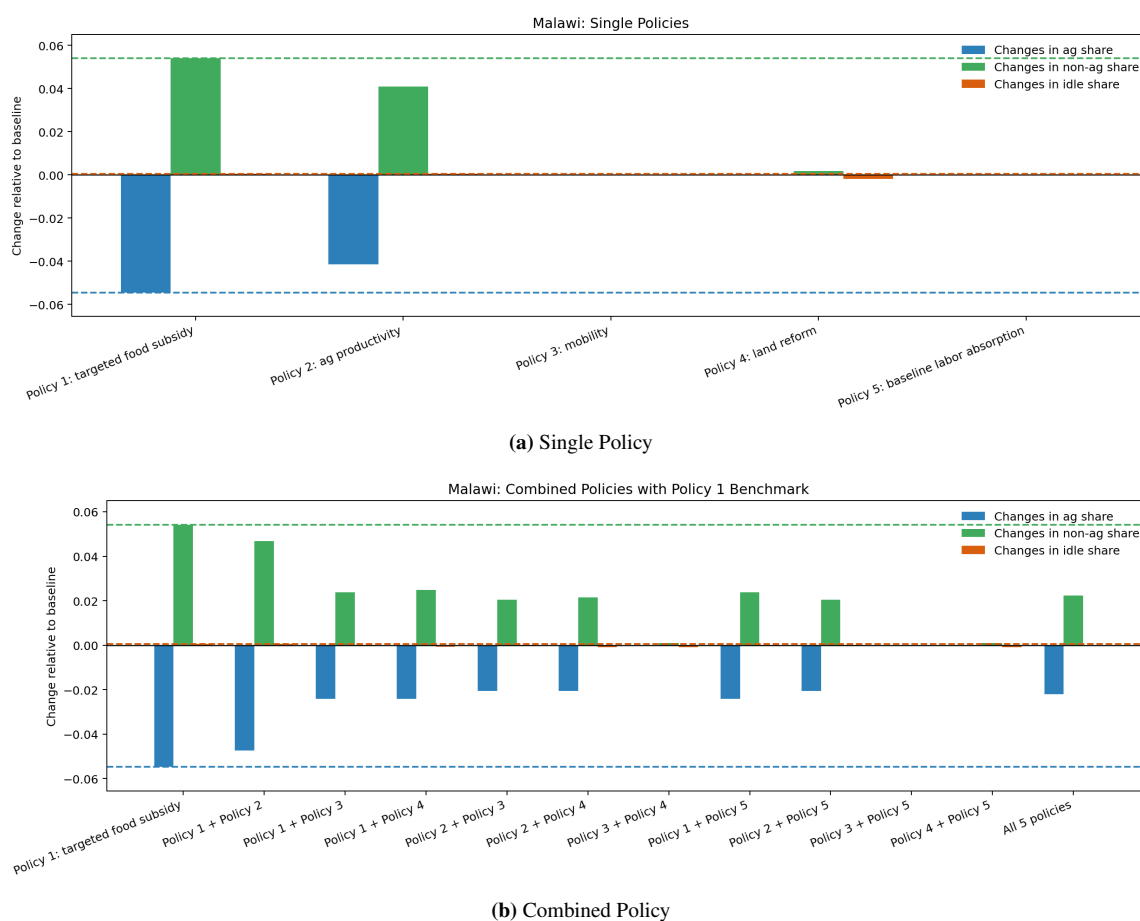
ployment by only 0.2 percentage points. Likewise, increasing labor absorption capacity produces no measurable improvement because the estimated labor absorption probability is already close to one.

The combined-policy simulations reinforce these findings. Combining targeted food subsidies and agricultural productivity improvements generates the largest reduction in agricultural labor among the two-policy combinations, while adding mobility or labor-absorption policies contributes almost no additional gains. Land reform provides only a small incremental improvement through a higher transition probability. Even when all five policies are implemented simultaneously, nearly all of the increase in non-agricultural employment is attributable to policies that relax the subsistence constraint rather than interventions targeting labor reallocation or labor absorption.

Overall, the Malawi counterfactuals strongly support the structural interpretation of the model. Structural transformation is constrained primarily by households' inability to release labor from agriculture rather than by barriers to labor reallocation or insufficient labor demand. Consequently, policies that relax subsistence constraints through food support or improvements in agricultural productivity generate substantially larger

gains than policies targeting later stages of structural transformation.

Figure 6: Counterfactual Analysis in Malawi



The counterfactual simulations indicate that policies targeting the labor-release stage generate the largest improvements in structural transformation in Malawi. Targeted food subsidies and agricultural productivity improvements substantially reduce agricultural labor and increase non-agricultural employment, whereas policies targeting labor reallocation and labor absorption produce only modest additional gains. These results are consistent with the structural estimates, which identify subsistence constraints as the primary barrier to structural transformation in Malawi.

6.4 Ethiopia

Figures 7a and 7b present the counterfactual policy simulations for Ethiopia. In contrast to Niger and Malawi, the structural estimates indicate that Ethiopia's primary constraints arise after labor is released from agriculture. Although improvements in agricultural productivity can release additional labor, successful structural transformation depends critically on households' ability to access and be absorbed into non-agricultural employment.

Among the single-policy interventions, improving agricultural productivity reduces the agricultural labor share by 2.6 percentage points and increases non-agricultural employment by 1.5 percentage points. However, nearly half of the released labor remains idle, increasing the idle labor share by 1.1 percentage points. Similarly, targeted food subsidies have virtually no effect on aggregate labor allocation. These results indicate that relaxing subsistence constraints alone does not generate substantial structural transformation because re-

leased workers continue to face barriers after leaving agriculture.

Policies targeting the second and third stages of the model generate substantially larger gains. Reducing mobility barriers increases the transition probability by 1.1 percentage points and raises non-agricultural employment by 0.5 percentage points while reducing idle labor by a similar amount. Strengthening household land-rights produces the largest single-policy improvement, increasing the transition probability by 6.4 percentage points and raising non-agricultural employment by 2.7 percentage points while reducing idle labor by the same magnitude. Increasing labor absorption capacity also generates large gains, increasing non-agricultural employment by 1.9 percentage points and lowering idle labor by 1.9 percentage points without affecting agricultural labor. These findings suggest that improving access to non-agricultural opportunities and expanding labor demand are substantially more effective than further relaxing agricultural constraints.

The combined-policy simulations highlight strong complementarities across stages of structural transformation. Combining agricultural productivity improvements with land reform increases non-agricultural employment by 2.1 percentage points, substantially more than agricultural productivity alone. Likewise, combining land reform with labor absorption produces the largest improvement among the two-policy combinations, increasing non-agricultural employment by 3.3 percentage points while reducing idle labor by an equal amount. When all five policies are implemented simultaneously, non-agricultural employment increases by 3.3 percentage points, idle labor declines by 2.6 percentage points, and the transition probability increases by 6.4 percentage points. These gains substantially exceed those generated by policies targeting labor release alone.

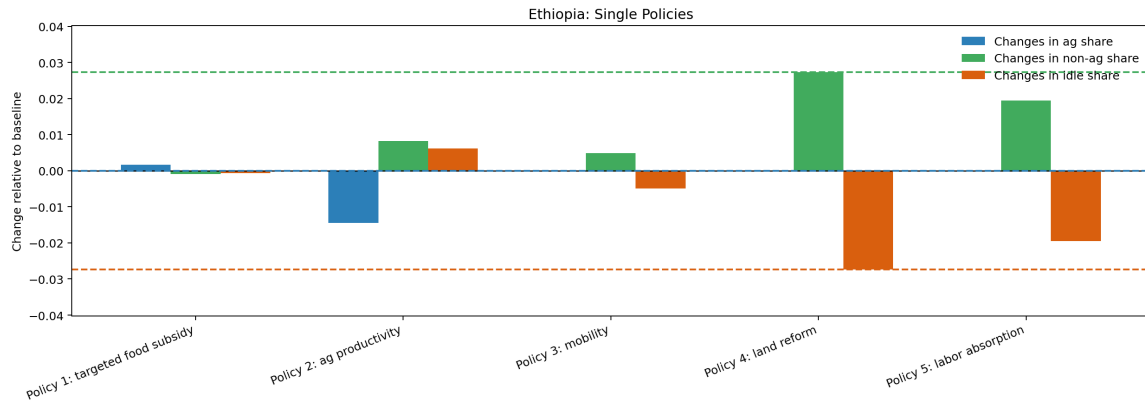
Overall, the Ethiopia counterfactuals provide strong support for the structural estimates. Unlike Niger and Malawi, releasing labor from agriculture is not sufficient to promote structural transformation. Instead, policies that strengthen land institutions and expand labor absorption capacity generate the largest improvements by enabling released workers to successfully transition into productive non-agricultural employment. These findings suggest that demand-side and institutional constraints, rather than agricultural productivity alone, constitute the primary barriers to structural transformation in Ethiopia.

6.5 Cross-Country Comparison

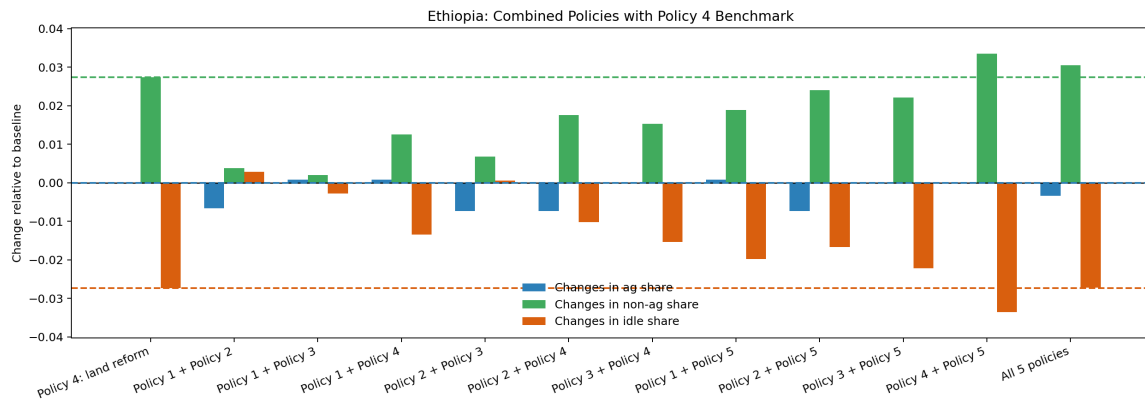
The counterfactual analysis reveals substantial heterogeneity in the policy drivers of structural transformation across the three countries. These differences closely mirror the structural estimates presented in Section 5 and highlight the importance of identifying the stage at which labor reallocation is constrained.

In both Niger and Malawi, the largest gains arise from policies that relax subsistence constraints and increase the amount of labor released from agriculture. Improvements in agricultural productivity and targeted food subsidies consistently generate the largest reductions in agricultural labor and corresponding increases in non-agricultural employment. By contrast, policies targeting labor reallocation and labor absorption produce relatively small additional gains, reflecting the high estimated transition and absorption probabilities in these

Figure 7: Counterfactual Analysis in Ethiopia



(a) Single Policy



(b) Combined Policy

The counterfactual simulations indicate that policies targeting the labor-reallocation and labor-absorption stages generate the largest improvements in structural transformation in Ethiopia. Strengthening land transferability and increasing labor absorption capacity produce the largest single-policy improvement and increase non-agricultural employment, whereas policies targeting labor release produce only modest additional gains. These results are consistent with the structural estimates, which identify labor reallocation and labor absorption barriers to structural transformation in Ethiopia.

countries.

Ethiopia exhibits a fundamentally different pattern. Although agricultural productivity improvements release additional labor, a substantial fraction of released workers remains idle unless policies also improve access to non-agricultural opportunities or expand labor demand. household land-rights reforms and industrial investments therefore generate considerably larger improvements than policies targeting agricultural productivity alone. These findings are consistent with the structural estimates, which identify labor reallocation and labor absorption as the primary bottlenecks in Ethiopia.

Taken together, the results demonstrate that effective structural transformation policies depend on the location of the binding constraint within the transformation process. Policies that target the relevant stage of the model consistently generate larger gains than policies aimed at non-binding constraints. The structural model, therefore, provides a useful framework for diagnosing country-specific barriers to structural transformation and evaluating policies that directly address those barriers.

7 Conclusion

Structural transformation is widely recognized as a key driver of long-run economic development. However, labor released from agriculture does not always move successfully into productive non-agricultural employment. This paper argues that understanding structural transformation requires distinguishing between three sequential stages: releasing labor from agriculture, reallocating released labor into non-agricultural opportunities, and absorbing reallocated workers into productive employment. While previous studies typically focus on only one of these margins, I develop a structural framework that explicitly links all three stages and allows the binding constraint to differ across countries.

Using household panel data from Niger, Malawi, and Ethiopia, I estimate the structural model using simulated method of moments. The estimated parameters reveal substantial heterogeneity in the constraints governing structural transformation. In Niger and Malawi, the primary barrier lies in the labor-release stage. Strong subsistence constraints require households to retain labor in agriculture even when productivity opportunities exist elsewhere. In Ethiopia, by contrast, improvements in agricultural productivity release labor from agriculture, but many of the released workers remain idle because they face institutional barriers to entering non-agricultural activities and limited labor demand once they arrive. These findings suggest that similar patterns of slow structural transformation may arise from fundamentally different underlying mechanisms.

The results show that the process of structural transformation extends beyond the mechanisms emphasized in the classic Lewis and Harris–Todaro models. Consistent with Lewis (1954), relaxing subsistence constraints promotes the release of labor from agriculture. Consistent with Harris and Todaro (1970), the ability of the non-agricultural sector to absorb workers remains essential for successful structural transformation. Between these two margins, however, households must overcome reallocation barriers arising from land institutions and spatial mobility constraints. By explicitly modeling this intermediate stage, the paper provides a more complete framework for understanding why similar agricultural productivity gains can lead to different structural transformation outcomes across countries.

The counterfactual policy analysis demonstrates the importance of identifying these country-specific constraints before designing interventions. In Niger and Malawi, policies that relax subsistence constraints, including targeted food subsidies and agricultural productivity improvements, generate the largest increases in non-agricultural employment. Policies aimed at reducing mobility barriers or expanding labor demand produce comparatively small gains because reallocation and absorption are already relatively unconstrained. Ethiopia exhibits the opposite pattern. Policies that strengthen household land-rights and expand labor absorption capacity substantially outperform policies that focus solely on agricultural productivity, highlighting the importance of demand-side and institutional constraints once labor has been released from agriculture. The results show that the effectiveness of development policies depends not only on their scale but also on whether they target the binding stage of structural transformation.

Several limitations suggest directions for future research. The model abstracts from general equilibrium effects, including wage adjustments, endogenous firm entry, and feedback between labor supply and labor demand. The counterfactual analysis captures partial-equilibrium responses rather than economy-wide adjustments. Future work could incorporate endogenous labor demand and firm behavior within a general equilibrium framework to study how policy interventions affect both households and firms simultaneously. In addition, extending the model to account for dynamic human capital accumulation, migration decisions, and gender-specific labor constraints would provide a richer understanding of the long-run process of structural transformation.

Overall, by distinguishing labor release, labor reallocation, and labor absorption as separate stages of structural transformation, this paper demonstrates that similar aggregate outcomes can arise from different underlying constraints. Although the empirical analysis exploits temporary weather-induced productivity shocks, the transition evidence shows that labor reallocation generate persistent employment states, providing a behavioral foundation for evaluating how structural constraints shape household labor allocation under alternative policy environments. Identifying these constraints is essential for designing effective development policies and for understanding why structural transformation proceeds at different speeds across countries.

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A Data and Labor Market Variable Construction

A.1 Survey Timing and Data Collection

A.1.1 Niger

The Niger ECVI survey is conducted in two visits per wave.

- **First visit (planting season):** household characteristics, demographics, labor market participation, and non-farm enterprises
- **Second visit (harvest season):** agricultural production, land, inputs, and output

We combine information from both visits to construct labor allocation and agricultural activity variables.

The 2011 and 2014 survey waves follow this structure, with labor market information collected during the first visit and agricultural information collected during the second visit. As described in Appendix A.2.2, the labor module records individuals' main economic activity during the previous 12 months. Consequently, labor allocation observed during the first (planting-season) visit reflects employment decisions made during and immediately following the preceding agricultural year. This timing is consistent with the empirical specification, which uses the one-year-lagged weather-induced productivity shock as the treatment variable, ensuring that the productivity shock aligns with the observed labor allocation.

A.1.2 Malawi

Malawi IHS data are collected in a single household visit. However, the questionnaire distinguishes agricultural activities across planting and harvest periods through recall questions. Employment and agricultural modules are therefore temporally comparable, although no separate survey visits are conducted.

A.1.3 Ethiopia

The Ethiopia ESS survey uses multiple visits:

- Post-planting visit
- Post-harvest visit
- Household visit

We combine information across visits to construct consistent employment and agricultural variables. For rural households, agricultural modules are administered separately from household employment modules.

A.2 Employment Reference Period

The household surveys record employment using multiple reference periods, including employment during the previous seven days and regular employment during the previous twelve months. This paper uses the

twelve-month reference period because the objective is to study structural transformation, which refers to persistent changes in labor allocation across sectors rather than temporary seasonal fluctuations in employment. Household labor allocation decisions in the theoretical framework represent durable employment choices that evolve over relatively long horizons and correspond more closely to regular employment over the previous year than to employment observed during a single week.

The seven-day measure is well-suited for measuring short-run labor market conditions but is heavily influenced by agricultural seasonality. In predominantly agricultural economies, whether an individual worked during the previous week depends critically on the timing of planting, harvesting, and other seasonal agricultural activities, as well as the survey interview date. Consequently, the seven-day measure captures temporary fluctuations in labor demand rather than persistent occupational choices. Because interviews occur at different times across households and countries, relying on a seven-day reference period would introduce substantial variation unrelated to structural transformation.

Temporary and casual employment nevertheless remains an important source of income in rural Sub-Saharan Africa. Because this paper studies durable labor allocation rather than short-term employment spells, individuals are classified according to their regular employment status over the previous twelve months. Workers who report no regular employment and whose economic activity consists only of temporary or casual work, unpaid household agricultural work, or no work are classified as idle labor. This classification is conceptually related to the ILO notion of labor underutilization, reflecting the idea that intermittent attachment to productive employment represents incomplete labor absorption. From the perspective of structural transformation, these workers have not yet been durably absorbed into productive employment, even though they may participate intermittently in economic activities.

A.3 Definition of Working-age Employment

A.3.1 Definition of Working-Age Population

We define the working-age population as individuals aged 15–85. All labor market variables are constructed at the individual level and then aggregated to the household level.

This broader age range reflects labor market conditions in rural developing economies. Individuals often begin working at younger ages, particularly in family farming and household enterprises, while weak pension systems and limited social safety nets imply that older individuals frequently remain economically active well beyond conventional retirement ages. As a result, labor supply in rural settings includes both younger and older household members.

This approach is consistent with the LSMS-based literature, which often includes all individuals aged 15 and above and does not impose a strict retirement age when measuring labor allocation across sectors. In particular, studies using LSMS and similar microdata emphasize that older individuals continue to participate

in agricultural and informal activities, contributing to household production and labor reallocation decisions (Gollin et al., 2014; Dillon and Barrett, 2017; Young, 2013). Moreover, labor force participation at older ages remains high in low-income countries due to limited pension coverage and continued reliance on family labor (World Bank, 2019).

A.3.2 Definition of Employment

The labor modules of the LSMS-ISA surveys in Niger, Malawi, and Ethiopia all distinguish between agricultural self-employment, household non-agricultural enterprises, wage employment, and temporary or casual work using comparable individual-level questions. Although questionnaire wording and survey implementation differ slightly across countries, we harmonize labor status using a common hierarchy based on reported economic activities rather than survey-specific variable definitions. The same classification procedure is applied consistently across all three countries, ensuring that cross-country differences in underutilized labor reflect differences in reported labor market attachment rather than differences in coding rules.

Employment information is based on individuals' main activity during the last 12 months. We focus on the primary job only; secondary and tertiary jobs are not considered. While the wording of employment questions differs slightly across countries, they are conceptually comparable. The surveys ask about individuals' main occupation during the previous year, such as "Principal employment during the last 12 months" in Niger, "Describe your main employed occupation over the past 12 months" in Malawi, and "Was [NAME] employed in the last 12 months?" in Ethiopia. Individuals reporting a main job are classified as employed workers.

Self-employment is identified using the Non-Agriculture Enterprise modules in the household questionnaires. These modules record up to two owners for each enterprise. We link enterprise owner identifiers to the household roster to identify self-employed individuals. When two owners are reported, enterprise revenue is split equally between them to ensure consistent assignment of labor participation without overstating employment.

An important advantage of the data is that both formal and informal employment are captured. The dataset includes wage employment (formal and informal), household non-farm enterprises, and self-employment. We therefore do not restrict the analysis to formal sector employment.

Labor is first assigned to an employment sector using individual-level information on primary economic activities. The household agricultural information is used only for individuals who are not assigned a regular employment sector. Consequently, living in an agricultural household does not mechanically classify all household members as agricultural workers. Individuals without regular sectoral employment remain classified as underutilized labor when their reported activities indicate only marginal attachment to employment, such as unpaid household agricultural work, temporary or casual work, or unsuccessful labor market participation.

We distinguish regular employment from a broader category of underutilized labor (International Labour Organization, 2013), which corresponds to individuals who are not durably absorbed into regular sectoral employment. This category includes working-age individuals who are unemployed or only marginally attached to economic activity through unpaid household agricultural work or temporary and casual employment. Specifically, underutilized labor includes individuals who report only participation in household agricultural activities, casual or temporary work, active job search, willingness to work despite not actively searching, or only temporary employment during the previous twelve months. Individuals with regular agricultural, household enterprise, wage, or salaried employment are classified as employed rather than underutilized. This definition aligns the empirical measure more closely with the concept of underutilized labor represented in the structural model.

A.4 Absent Household Members and Labor Allocation

A potential concern is that rural workers may work outside the village, particularly in nearby towns or cities. All three surveys explicitly track absent household members, allowing us to account for individuals who are temporarily away but still belong to the household. In Niger, the ECVI survey records all household members, including those temporarily absent, and collects information on the duration and reason for absence over the past 12 months. This allows identification of individuals working outside the village while remaining part of the household. In Malawi and Ethiopia, absent household members are also recorded, and information on their employment and activities is collected through proxy responses from the household head or another adult household member. As a result, individuals working outside the household location remain part of the household roster and are included in labor allocation measures.

The surveys also distinguish between temporary and permanent movers. Temporary movers remain members of the household and are included in the labor-allocation measures, whereas permanent movers, for example, individuals who migrate and establish a new household, are no longer part of the original household roster. Accordingly, the analysis captures labor reallocation through local employment, commuting, seasonal migration, informal non-farm work, participation in household enterprises, and temporary migration, but does not observe permanent migration once individuals leave the household. Consequently, the measured non-agricultural labor allocation reflects both local employment opportunities and non-local opportunities accessed while workers remain household members. While permanent migration is not directly observed, temporary migration and commuting remain important channels through which households respond to spatial mobility barriers, allowing the empirical analysis to identify the role of migration frictions in structural transformation. Permanent migration accompanied by household splitting is not observed and therefore represents a limitation of the LSMS household panel data rather than of the empirical framework.

These data features allow us to capture multiple forms of labor allocation, including local employment, migration to nearby towns, seasonal migration, informal non-farm work, and participation in household en-

terprises. Consequently, measured non-agricultural labor participation reflects both local and non-local job opportunities, consistent with the migration mechanism in the model.

B Visual Validation of Household Exposure to Agricultural Activity

The sample construction differs slightly across countries to reflect differences in the meaning of administrative classifications. In Niger and Ethiopia, we restrict the analysis to households classified as rural. In Malawi, however, the administrative definition of “urban” often includes settlements that remain closely integrated with surrounding agricultural land. We therefore include households located in communities with observable agricultural activity.

To verify that sampled households in Malawi are plausibly exposed to agricultural production conditions, we examine high-resolution satellite imagery surrounding the displaced enumeration-area coordinates provided in the LSMS survey, consistent with the 5 km buffer strategy described in Section 2.4.2. Because the World Bank intentionally displaces survey coordinates to protect respondent confidentiality, the imagery is interpreted at the community level rather than as the exact location of individual households. The satellite images show that sampled settlements are embedded within active agricultural landscapes, supporting the use of surrounding environmental conditions to characterize agricultural exposure.

To complement the visual evidence, we also examine labor allocation patterns across administrative classifications. In Malawi, households classified as urban allocate 47 percent of labor to agriculture, substantially higher than the corresponding urban shares in Niger (12 percent) and Ethiopia (7 percent). This difference suggests that the administrative definition of urban areas in Malawi includes settlements that remain closely integrated with surrounding agricultural production, consistent with the community-level evidence from the satellite imagery.

Figure 8 presents representative examples of household locations in Malawi. The imagery reveals that many households, including those located in relatively dense settlements, are situated in close proximity to cultivated plots and agricultural land. In several cases, residential structures are directly adjacent to or interspersed with visibly cultivated fields, indicating a mixed land-use pattern that combines residential and agricultural activity.

These patterns suggest that administrative classifications such as “urban” do not necessarily correspond to a separation from agricultural production in the Malawian context. Instead, households often remain spatially embedded in environments where agricultural activity is prevalent, and thus are plausibly exposed to local agricultural productivity shocks.

Importantly, the key identifying assumption is that household income opportunities, either directly through own production or indirectly through local labor markets, are affected by agricultural conditions. The visual

evidence presented here supports the plausibility of this assumption.



Figure 8: Satellite imagery of representative household locations in different regions in Malawi. The figures illustrate a dense settlement with nearby cultivated plots and a mixed residential–agricultural landscape where households are located adjacent to farmland.

Finally, we verify that the inclusion of urban households in Malawi (Table 15) does not drive the main results. Re-estimating the baseline specifications using only rural households yields effects that are similar in sign and magnitude, although estimated with less precision. This suggests that the main findings are not sensitive to the inclusion of households classified as urban.

Table 15: Sectoral Choice and Weather Productivity Growth

	(1)	(2)	(3)	(4)
	Agriculture	Manufacturing	Service	Idle
$\Delta\theta_{gt-1}$	-0.585 (-1.21)	0.568** (2.03)	-0.104 (-0.39)	0.121 (0.81)
Constant	0.835*** (74.89)	0.0440*** (7.34)	0.113*** (16.65)	0.00820** (2.60)
Household Fixed Effect	Yes	Yes	Yes	Yes
Year Fixed Effect	Yes	Yes	Yes	Yes
Mean dep. var	0.821	0.049	0.125	0.004
SD dep. var	0.326	0.189	0.276	0.060
Observations	1750	1750	1750	1750
R-squared	0.686	0.615	0.679	0.500

* t statistics in parentheses

** * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

C Weather Productivity Shock Construction

C.1 Exogenous Weather Productivity Growth: Definition

Estimating the effect of agricultural productivity on household labor allocation raises important endogeneity concerns. Observed agricultural productivity is jointly determined with household production decisions, and is affected by unobserved factors such as technology adoption, input use, irrigation, management practices, and local economic conditions (Branco and Féres, 2021; Emerick, 2018; Jessoe et al., 2018). Consequently,

directly using observed agricultural productivity as an explanatory variable would generally produce biased estimates of its effect on labor allocation.

To isolate the exogenous component of agricultural productivity, this paper constructs a weather-induced productivity measure. The idea is to separate productivity changes driven by exogenous weather conditions from those generated by endogenous household decisions. The construction begins with the following agricultural production function at the enumeration-area (EA) level,

$$Y = TFP \cdot \theta \cdot L^{\alpha_1} K^{\alpha_2} T^{\alpha_3} \cdot \varepsilon, \quad (41)$$

where Y denotes agricultural output, TFP represents household-specific productivity, θ captures the productivity effect of weather conditions, L , K , and T denote labor, capital, and land inputs, respectively, and ε is an idiosyncratic error term.

Taking logarithms gives

$$y = \log(TFP) + \log(\theta) + \alpha_1 \log(L) + \alpha_2 \log(K) + \alpha_3 \log(T) + \varepsilon. \quad (42)$$

The empirical analysis focuses on annual changes rather than productivity levels because household labor allocation responds to changes in agricultural conditions over time. Taking first differences yields

$$\Delta y = \Delta \log(\theta) + \Delta H + \Delta \varepsilon, \quad (43)$$

where

$$\Delta H = \Delta \log(TFP) + \alpha_1 \Delta \log(L) + \alpha_2 \Delta \log(K) + \alpha_3 \Delta \log(T). \quad (44)$$

The term ΔH summarizes systematic changes in agricultural productivity arising from non-weather factors, including changes in total factor productivity, labor, capital, and land inputs, while $\Delta \varepsilon$ captures transitory production shocks and measurement error.

Following Colmer (2021), capital and land are assumed to be fixed in the short run. Consequently,

$$\Delta H = \Delta \log(TFP) + \alpha_1 \Delta \log(L), \quad (45)$$

which captures changes arising from household production decisions, including technology adoption, management practices, labor adjustments, crop choices, and other non-weather factors. The objective is therefore to recover the weather productivity component,

$$\Delta \theta \equiv \Delta \log(\theta),$$

which serves as the exogenous treatment variable throughout the paper.

C.2 Exogenous Weather Productivity Growth: Construction

Annual agricultural output is not consistently observed across countries and survey waves. Following Donaldson and Storeygard (2016) and Lobell et al. (2015), annual NDVI is therefore used as a proxy for agricultural productivity because it provides a consistent satellite-based measure of vegetation growth that is highly correlated with crop production.

Using NDVI, Equation 43 can be written as

$$\Delta NDVI_i = \Delta \theta_i + \Delta H_i. \quad (46)$$

The challenge is that only the left-hand side is observed. The weather productivity component $\Delta \theta_i$ must therefore be recovered from exogenous weather variables.

A natural approach would be to estimate

$$\Delta NDVI_i = \beta_1 \Delta Rainfall_i + \beta_2 \Delta Temperature_i + u_i. \quad (47)$$

However, this specification imposes a restrictive linear relationship between rainfall, temperature, and agricultural productivity. Agricultural production depends on nonlinear interactions between rainfall and temperature, including threshold effects and complementarities. Consequently, a flexible nonparametric approach is preferred.

This paper estimates the conditional expectation

$$E(\Delta NDVI_i | \Delta Rainfall_i, \Delta Temperature_i) = g(\Delta Rainfall_i, \Delta Temperature_i) \quad (48)$$

using a Random Forest regression model. The estimated function $\hat{g}(\cdot)$ provides the weather-induced component of productivity,

$$\widehat{\Delta \theta}_i = \hat{g}(\Delta Rainfall_i, \Delta Temperature_i), \quad (49)$$

while the residual

$$\widehat{\Delta H}_i = \Delta NDVI_i - \widehat{\Delta \theta}_i \quad (50)$$

captures the component of productivity changes arising from household production decisions, crop choices, measurement error, and other local factors that is not predicted by rainfall and temperature. Accordingly, the weather-induced component should be interpreted as the portion of productivity that is predictable from weather conditions, which may include both direct biophysical effects of weather and systematic household

responses to those conditions.

Random Forest is particularly suitable because it flexibly captures nonlinear relationships and interactions between rainfall and temperature without requiring a parametric functional form. Each regression tree is estimated using a bootstrap sample of the training data, and predictions are averaged across trees to reduce prediction variance. The model is trained using a random 80% sample of the observations and evaluated on the remaining 20% to assess out-of-sample performance.

Finally, the estimated Random Forest model is applied to the full sample to construct the weather productivity measure used throughout the paper. Because the prediction depends exclusively on exogenous weather variables, the constructed treatment is predetermined with respect to household labor allocation decisions. The out-of-sample prediction performs well across all three countries, with R^2 values ranging from 0.29 in Niger, 0.60 in Malawi, and 0.30 in Ethiopia. These values indicate that weather conditions explain a substantial share of annual vegetation growth while leaving considerable variation attributable to household production decisions and other non-weather factors, which is precisely the variation excluded from the treatment measure.

C.3 Why Annual Weather-Induced Productivity?

Crop responses to weather are inherently crop- and location-specific because different crops follow distinct phenological calendars, with substantial variation in planting, flowering, grain-filling, and harvesting dates across regions and management systems (Sacks et al., 2010; Iizumi et al., 2019; Minoli et al., 2022). This heterogeneity is particularly pronounced in the multi-country household surveys used in this paper, where households cultivate different crop mixes under diverse agroecological conditions. Moreover, the productivity effect of a given weather realization depends critically on when it occurs. Insufficient rainfall during crop growth and excessive rainfall during harvest, for example, may both reduce final output even though they represent very different rainfall realizations. Thus, constructing a direct weather-shock measure would require knowing not only the relevant crop calendar for each household, but also the optimal timing and intensity of rainfall and temperature for each crop and location and then measuring deviations from those crop- and location-specific optima. Such information is generally unavailable and would be difficult to harmonize across countries.

This consideration motivates the use of NDVI as an intermediate measure between weather and agricultural productivity. Rather than defining favorable or unfavorable weather *ex ante*, NDVI measures the realized vegetation outcome after crops have been exposed to the sequence of weather conditions occurring over the production cycle. This is particularly useful in a setting with heterogeneous crop calendars: weather conditions that are beneficial for one crop or at one point in the production cycle may be harmful for another, whereas their consequences are ultimately reflected in realized vegetation growth. The empirical strategy

therefore does not require the researcher to determine whether a particular amount or timing of rainfall constitutes a positive or negative agricultural shock for each crop and location.

The objective of this paper also differs from that of the crop-modeling literature. Rather than estimating the biological response of individual crops to weather at specific phenological stages, this paper seeks to identify the causal effect of agricultural productivity on household labor allocation. The economically relevant treatment is therefore not rainfall, temperature, drought, or another weather realization itself, but the component of realized agricultural productivity that is attributable to exogenous weather. Conceptually, the empirical strategy follows the relationship

$$\text{Weather conditions} \longrightarrow \text{Realized agricultural productivity} \longrightarrow \text{Household labor allocation.}$$

Weather provides the exogenous source of variation, while agricultural productivity is the treatment whose effect on labor allocation is of interest.

The annual frequency follows from this economic object. The analysis focuses on realized agricultural productivity over the production year and on subsequent household labor allocation rather than on the effect of individual weather events. I therefore aggregate rainfall and temperature to the annual level and use annual NDVI to measure realized vegetation growth over the corresponding production period. This provides a common temporal unit that can be applied consistently across households cultivating different crops and across countries with different agricultural calendars. Importantly, this choice does not imply that within-year weather timing is unimportant. Rather, annual aggregation reflects the objective of constructing a comparable measure of annual weather-induced productivity rather than identifying the agronomic effect of weather at each stage of crop development.

To recover the exogenous weather-induced component of annual productivity, I employ a Random Forest machine learning algorithm to estimate the relationship between annual rainfall, annual temperature, and annual NDVI. Existing studies commonly accommodate nonlinear weather effects using temperature bins (Schlenker and Roberts, 2009; Barreca et al., 2016) or polynomial functions of precipitation and temperature (Deschênes and Greenstone, 2007; Burke et al., 2015). These approaches are well suited when the objective is to estimate particular weather-response functions, but they require researchers to choose functional forms, thresholds, and interaction structures *ex ante*. Random Forest instead allows the mapping from rainfall and temperature into realized vegetation productivity to be nonlinear and interactive without imposing a particular parametric response function.

Importantly, the objective of the Random Forest is not to explain all variation in NDVI or to construct the most detailed possible model of crop production. It is to recover the component of annual vegetation productivity that is systematically predictable from exogenous weather. The baseline specification therefore uses annual

rainfall and annual temperature as predictors. More elaborate measures—such as drought indices, temperature bins, rainfall variability, or researcher-defined extreme-weather indicators—require additional choices about thresholds, reference periods, or functional forms. Restricting the baseline predictor set to consistently measured rainfall and temperature minimizes such researcher discretion, while the Random Forest provides flexibility in how these variables map into realized productivity.

Annual aggregation inevitably sacrifices information about the exact timing and intensity of within-year weather events. Field interviews conducted for this study underscore this limitation: farmers frequently described unfavorable weather as rainfall occurring at the wrong time, such as insufficient rainfall during crop growth or excessive rainfall during harvest, rather than simply unusually low or high annual rainfall. This evidence reinforces the motivation for focusing on weather-induced *productivity* rather than treating a particular rainfall realization as the agricultural shock itself. A rainfall-based shock would require the researcher to determine which rainfall realizations are favorable or unfavorable given local crops and production calendars. By instead using realized NDVI as the productivity outcome and weather as the exogenous predictor, the approach allows the consequences of heterogeneous production environments to be reflected in realized vegetation productivity without requiring a common crop calendar or a common definition of good” and bad” weather across households.

The resulting measure should therefore be interpreted as a conservative measure of annual weather-induced agricultural productivity: the component of realized annual vegetation productivity that can be systematically predicted from exogenous annual rainfall and temperature. It is not intended to capture every agronomic channel through which weather affects individual crops. This distinction is central to the identification strategy, which prioritizes a transparent and comparable source of exogenous productivity variation over a complete model of crop-specific weather responses.

C.4 Validation of the Weather-Productivity Shock Using Household Crop Output

To examine whether the predicted weather-productivity shock captures economically meaningful variation in agricultural productivity, we relate the predicted shock to household-level crop output. Household-level crop output is constructed by aggregating production across all plots cultivating the same crop within a household. We estimate the following specification:

$$\ln(Y_{icrt}) = \beta \theta_{it} + \gamma_{cr} + \delta_t + \varepsilon_{icrt}$$

where $\ln(Y_{icrt})$ denotes household-level output of crop c produced by household i in community r and year t , θ_{it} is the predicted weather-productivity shock, γ_{cr} are community *times* crop fixed effects, and δ_t are year fixed effects. Standard errors are clustered at the community level.

Table 16 reports the estimated results. Across all three countries, the estimated coefficients are positive,

indicating that favorable weather-induced productivity shocks are associated with higher household crop output. The relationship is statistically significant at the 1 percent level in Niger and Ethiopia and at the 10 percent level in Malawi. These findings provide direct evidence that the Random Forest prediction captures economically meaningful variation in agricultural productivity rather than generic vegetation dynamics. This evidence complements the household agricultural income validation reported in Table 17 and supports the interpretation of the predicted component of NDVI growth as a weather-induced agricultural productivity shock used throughout the empirical analysis.

Table 16: Validation of the Weather-Productivity Shock Using Household Crop Output

	Niger	Malawi	Ethiopia
$\Delta\theta_{gt-1}$	3.133*** (1.066)	12.218* (6.961)	27.693*** (7.287)
Observations	9,584	4,887	7,248
Community clusters	181	225	213
R^2	0.523	0.584	0.778
Community $\times$ crop fixed effects	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes

Notes: The dependent variable is the logarithm of household-level crop output, constructed by summing production across all plots cultivated with the same crop by a household. The weather-productivity shock is the contemporaneous weather-induced component of NDVI growth predicted by the Random Forest model. All specifications include community $\times$ crop fixed effects and year fixed effects. Standard errors, reported in parentheses, are clustered at the community level. Singleton observations are dropped by the fixed-effects estimator.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

D Weather Productivity Shock

D.1 Weather Productivity Growth as the Treatment Variable

In this section, we use the weather productivity variable as the exogenous treatment for the household. Before we run into the empirical results, we first need to discuss why the weather productivity is a good treatment variable to estimate an unbiased effect on the labor reallocation decision in the household.

Recall that in Section 2.4, we capture the variation instead of the trends of the weather productivity $\Delta\theta_g$ and total factor productivity change TFP . The positive variation $\Delta\theta_g$ means good weather conditions for crops, while a negative variation represents bad weather conditions for crops. The weather productivity can be decomposed into the long-run productivity changes from climate change, which is a trend, and the weather productivity variation we study, which is the variation around the trend.

In Section 2.4, both weather productivity growth $\Delta\theta_g$ and human productivity growth ΔTFP affect the labor reallocation ΔL . Besides, $\Delta\theta_g$ also affects ΔTFP , indicating that $\Delta\theta_g$ has both direct and indirect (through ΔTFP) effects on labor reallocation. Therefore, we first discuss the correlation (ϕ) between $\Delta\theta_g$ and ΔTFP .

Firstly, $\Delta\theta_g$ and ΔTFP can be complementary ($\phi > 0$). Farmers observe good weather during the growing

season (e.g., timely rains, no drought) after planting. Since land and crop choices are fixed, they cannot expand land or choose other crops, but they can take more effort on the land to take advantage of good weather conditions. They can adjust labor timing, allocate fertilizer more effectively, improve monitoring and weeding, and prevent pests/diseases more effectively. These adjustments are not changes in observable inputs, but improvements in unobserved managerial efficiency, captured by the human TFP in our model. In this case, $\Delta TFP(\Delta\theta_g)$ increases in $\Delta\theta_g$. That is, with both higher weather productivity and human productivity, households achieve higher agricultural income and can decide how to reallocate the labor.

Secondly, $\Delta\theta_g$ and ΔTFP can be substitute ($\phi < 0$). When the weather is good, farmers may exert less unobserved effort. For example, they will reduce field monitoring, reduce the adjustment in fertilizers, and reduce task coordination. These operate through unobserved management quality. Observable inputs remain unchanged, and the decline appears in lower human TFP. In this case, $\Delta TFP(\Delta\theta_g)$ decrease in $\Delta\theta_g$. Although human productivity may decline when favorable weather substitutes for household effort, we generally expected the direct biophysical gains from improved weather conditions to outweigh the reduction in effort on average. Consequently, favorable weather is expected to increase agricultural productivity, although the magnitude of this increase depends on the strength of households' behavioral responses.

Finally, it is possible that $\Delta\theta_g$ and ΔTFP are independence ($\phi = 0$). Weather conditions affect only the biophysical component (θ_g), not human TFP. This serves as the baseline benchmark commonly adopted in production economics and Integrated Assessment Models(IAM). Weather affects labor only through the biophysical contribution to agricultural income. The effect is cleanly interpreted as “weather-driven income change $\rightarrow$ labor reallocation.”

The discussion above illustrates that different behavioral responses may alter the magnitude of the relationship between weather productivity and realized agricultural productivity. Nevertheless, under plausible behavioral responses, favorable weather is expected to be positively associated with agricultural productivity on average. Rather than relying on a strict theoretical monotonicity assumption, we empirically validate this relationship in Table 17, which shows that the weather-induced productivity shocks are strongly and positively associated with realized household agricultural income across all three countries.

Table 17: Household Agricultural Income and Weather Productivity Growth

	(1)	(2)	(3)
	Niger	Malawi	Ethiopia
$\Delta\theta_{gt-1}$	4.162** (2.45)	4.681** (2.07)	1.874* (1.67)
Constant	1.090*** (33.25)	1.632*** (29.54)	0.295*** (4.81)
Household Fixed Effect	Yes	Yes	Yes
Year Fixed Effect	Yes	Yes	Yes
Mean dep. var	1.417	1.729	0.508
SD dep. var	1.396	1.810	1.536
Observations	2546	1708	1442
R-squared	0.708	0.538	0.638

* t statistics in parentheses

** $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

D.2 Identification of the Total Effect

The reduced-form analysis estimates the total effect of weather-induced agricultural productivity on household labor allocation. Agricultural productivity is determined by both exogenous weather conditions and household-specific productivity factors, such as technology adoption, farming experience, and management practices. Let agricultural productivity be written as

$$\Delta A_{it} = \Delta\theta_{gt} + \Delta P_{it}, \quad (51)$$

where $\Delta\theta_{gt}$ denotes the weather-induced component determined by rainfall and temperature, and ΔP_{it} denotes all other changes in household productivity.

Although household productivity may respond to weather conditions, it can be decomposed into a weather-related component and a remaining component through the linear projection

$$\Delta P_{it} = \phi \Delta\theta_{gt} + X'_{it} \eta + R_{it}, \quad \mathbb{E}[R_{it} \mid \Delta\theta_{gt}, X_{it}] = 0, \quad (52)$$

where R_{it} captures the component of ΔP_{it} that is orthogonal to weather after conditioning on the household controls and fixed effects included in the empirical specification.

Suppose household labor allocation satisfies

$$s_{ijt} = \beta_1 \Delta\theta_{gt-1} + \beta_2 \Delta P_{it} + X'_{it} \pi + u_{it}, \quad \mathbb{E}[u_{it} \mid \Delta\theta_{gt-1}, X_{it}] = 0, \quad (53)$$

where s_{ijt} denotes the household labor share in sector j , X_{it} contains household-level controls, and u_{it} is an idiosyncratic error term.

Substituting Equation (52) into Equation (53) yields

$$s_{ijt} = (\beta_1 + \phi\beta_2)\Delta\theta_{gt} + X'_{it}(\pi + \beta_2\eta) + \varepsilon_{it}, \quad (54)$$

where

$$\varepsilon_{it} = \beta_2 R_{it} + u_{it}.$$

The household controls enter only through the combined coefficient vector $(\pi + \beta_2\eta)$. Consequently, the coefficient on $\Delta\theta_{gt-1}$ remains

$$\beta_{\text{tot}} = \beta_1 + \phi\beta_2$$

which captures the total effect of weather-induced agricultural productivity on household labor allocation.

By construction,

$$\mathbb{E}[R_{ht} \mid \Delta\theta_{gt}, X_{ht}] = 0,$$

and the exogeneity assumption implies

$$\mathbb{E}[u_{ht} \mid \Delta\theta_{gt}, X_{ht}] = 0.$$

Therefore,

$$\mathbb{E}[\varepsilon_{ht} \mid \Delta\theta_{gt}, X_{ht}] = 0,$$

which implies that the coefficient on $\Delta\theta_{gt}$ consistently identifies

$$\beta_{\text{tot}} = \beta_1 + \phi\beta_2.$$

Hence, the coefficient on weather shocks consistently identifies the total effect of weather-induced agricultural productivity on household labor allocation, incorporating both the direct effect of weather and the indirect effect through weather-induced adjustments in household productivity.

D.3 Other Endogeneity Concerns

The reduced-form analysis exploits variation in weather-induced agricultural productivity, $\Delta\theta_{gt}$, which is constructed from exogenous rainfall and temperature conditions at the community level. Because weather conditions are determined by nature rather than household decisions, they provide plausibly exogenous variation in agricultural productivity that helps mitigate several common endogeneity concerns.

First, the approach addresses simultaneity between agricultural productivity and labor allocation. Observed agricultural productivity and household labor decisions are jointly determined, since labor input affects agricultural output while expected returns influence labor allocation. By using weather-induced productivity rather than observed productivity, the estimated variation is predetermined with respect to contemporaneous household labor decisions.

Second, the specification reduces omitted-variable bias. Household characteristics such as farming ability, wealth, land quality, infrastructure, credit access, and other persistent local conditions may jointly affect agricultural productivity and labor allocation. Household fixed effects absorb all time-invariant household heterogeneity, while year fixed effects capture common macroeconomic shocks and other aggregate factors affecting all households. The remaining identifying variation therefore comes from within-household changes in weather-predicted agricultural productivity over time.

Third, the use of weather shocks alleviates concerns about reverse causality. Household production decisions, such as irrigation, input use, or labor management, may affect observed agricultural outcomes or vegetation indices, but they cannot influence rainfall or temperature. Moreover, weather-induced productivity is predicted out-of-sample using a Random Forest model trained on satellite-based weather and vegetation data, reducing mechanical feedback between household decisions and the constructed productivity measure.

Finally, the use of satellite-based weather and remote-sensing data minimizes classical measurement error arising from household surveys. Because the weather variables are measured independently of household responses, measurement error in reported agricultural outcomes or labor allocation does not contaminate the construction of the weather-induced productivity measure.

E Empirical Robustness Check

E.1 Alternative definition of “idle labor”

One concern is that the estimated increase in idle labor following weather productivity shocks may reflect changes in educational participation rather than labor market outcomes. Although our baseline definition of idle labor already excludes individuals engaged in employment and focuses on those who are available for work or exhibit weak labor-force attachment, some school-age individuals may still be classified as idle. To address this concern, we construct a more restrictive measure of idle labor that excludes individuals whose years of education increase between survey waves.

The first set of columns in Table 18 reports the results using this “no students” definition. The results are nearly identical to the baseline estimates. Weather-induced productivity shocks continue to increase the idle labor share significantly in Ethiopia, while the estimated effects remain small and statistically insignificant in Niger and Malawi. This suggests that the increase in idle labor observed in Ethiopia is not driven by school-

ing decisions.

As a further robustness check, we restrict the sample to prime-age individuals aged 25–60. This restriction removes nearly all potential schooling transitions while also excluding retirement-age individuals, thereby focusing on individuals with the strongest attachment to the labor market. The second set of columns in Table 18 again produces results that are broadly consistent with the baseline findings. Weather productivity shocks continue to generate a large and statistically significant increase in idle labor in Ethiopia. In Niger, the estimated effect remains small and statistically insignificant. In Malawi, the estimated coefficient becomes positive and marginally significant under the prime-age definition, although it’s magnitude remains substantially smaller than that observed in Ethiopia.

Overall, these robustness checks indicate that the baseline findings are not sensitive to alternative definitions of idle labor. Although the prime-age specification yields a small positive idle-labor response in Malawi, the effect is modest compared with the consistently large and statistically significant increase observed in Ethiopia across both alternative definitions. These findings continue to support the interpretation that labor absorption constraints are substantially more severe in Ethiopia, whereas workers released from agriculture in Niger and, to a much greater extent, Malawi are generally more successfully absorbed into productive non-agricultural employment.

Table 18: Alternative Idle Workers and Weather Productivity Growth

	(1)	(2)	(3)	(4)	(5)	(6)
	Niger		Malawi		Ethiopia	
Idle Workers	No Students	Age 25-60	No Students	Age 25-60	No Students	Age 25-60
$\Delta\theta_{gt-1}$	0.0591 (0.55)	0.0869 (0.75)	0.139 (1.59)	0.237* (1.87)	1.064** (2.20)	1.107** (2.08)
_cons	0.0280*** (6.34)	0.0266*** (6.04)	0.0114*** (4.81)	0.0156*** (4.68)	0.170*** (8.86)	0.164*** (8.47)
Community Fixed Effect	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effect	Yes	Yes	Yes	Yes	Yes	Yes
Mean dep. var	0.028	0.028	0.005	0.006	0.196	0.191
SD dep. var	0.134	0.134	0.053	0.062	0.380	0.376
Observations	3448	3254	1584	1450	1880	1604
R-squared	0.611	0.613	0.507	0.511	0.685	0.692

* t statistics in parentheses

** $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

E.2 Pre-trend test for weather shocks

A potential concern is that weather productivity shocks may proxy for underlying development trends rather than represent exogenous short-run shocks. For instance, areas with improving climate conditions may also experience long-run development changes—such as infrastructure expansion, policy reforms, or improved market integration—that independently facilitate structural transformation. In this case, the estimated rela-

tionship between weather shocks and labor reallocation could reflect these concurrent trends rather than the causal effects of agricultural productivity changes.

To address this concern, we conduct a placebo test using the weather productivity measure from the current agricultural year, denoted as $\Delta\theta_{gt}$. Although this measure is constructed from weather conditions occurring in the same calendar year as the household survey, it serves as a placebo because agricultural productivity is only fully realized at the end of the harvest season. The LSMS labor module records individuals' labor allocation over the preceding 12 months, implying that the reported labor decisions are largely determined before the current year's harvest outcomes are observed. Consequently, from the perspective of household labor allocation, $\Delta\theta_{gt}$ represents information that has not yet been realized and therefore should not systematically predict current labor allocation. A significant coefficient would instead suggest that the weather productivity measure is capturing underlying trends, such as gradual economic development or policy changes, rather than exogenous weather-induced productivity shocks.

Table 19 reports the results. Across all three countries and all sectoral outcomes, including agricultural, manufacturing, service, and idle labor shares, the coefficients on $\Delta\theta_{gt}$ are uniformly small and statistically insignificant. These findings provide reassurance that our baseline results are not driven by spurious trends or omitted factors correlated with weather productivity over time.

The absence of any systematic relationship between future weather shocks and current labor allocation suggests that differential development trends—such as policy changes or infrastructure improvements—are unlikely to drive the main results. These findings support the interpretation that lagged weather productivity shocks capture exogenous variation in agricultural productivity and provide causal evidence on labor reallocation responses.

Table 19: Household Sectoral Share and Future Weather Productivity Growth

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
	Niger				Malawi				Ethiopia			
	Agriculture	Manufacturing	Service	Idle	Agriculture	Manufacturing	Service	Idle	Agriculture	Manufacturing	Service	Idle
$\Delta\theta_{gt}$	0.416 (1.00)	0.134 (0.49)	-0.500 (-1.26)	-0.0661 (-0.40)	0.530 (1.57)	-0.282 (-1.22)	-0.0102 (-0.06)	-0.112 (-1.24)	-0.0290 (-0.06)	-0.0521 (-0.49)	-0.0658 (-0.28)	0.0354 (0.09)
_cons	0.603*** (78.04)	0.133*** (24.91)	0.228*** (33.24)	0.0299*** (11.08)	0.802*** (116.61)	0.0356*** (7.18)	0.145*** (29.57)	0.00805*** (4.60)	0.701*** (24.87)	0.0224*** (3.36)	0.0922*** (7.87)	0.148*** (6.62)
Household Fixed Effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean dep. var	0.634	0.118	0.209	0.028	0.776	0.052	0.154	0.005	0.660	0.023	0.094	0.182
SD dep. var	0.406	0.276	0.333	0.135	0.357	0.188	0.301	0.054	0.462	0.134	0.265	0.371
Observations	3780	3780	3780	3780	1910	1910	1910	1910	2240	2240	2240	2240
R-squared	0.681	0.626	0.637	0.610	0.721	0.591	0.698	0.506	0.721	0.632	0.624	0.674

* t statistics in parentheses

** * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

E.3 Alternative weather measure

To further assess the validity of the weather productivity shock, we examine whether sectoral labor allocation responds to alternative weather measures using growing-season rainfall and temperature. Unlike the

constructed weather productivity shock, raw weather variables do not necessarily map monotonically into agricultural productivity. In particular, a monotonic increase or decrease in rainfall or temperature does not always translate into higher productivity, as both insufficient and excessive weather realizations may harm crop outcomes. To account for this non-monotonic relationship, we allow for nonlinear effects by including quadratic terms for rainfall and temperature. Table 20 reports the regression results, and the corresponding predicted relationships are illustrated in Figures 9-11.

The results show consistent nonlinear patterns across countries. In Niger, rainfall exhibits a statistically significant inverted U-shaped relationship with agricultural labor share: the linear rainfall coefficient is positive while the squared term is negative and significant. This implies that increases in rainfall raise agricultural labor participation at low levels but reduce it beyond a threshold, consistent with diminishing returns or adverse effects from excessive rainfall. Temperature also exhibits a nonlinear relationship, with a negative linear term and a positive quadratic term, implying a U-shaped relationship between temperature and agricultural labor share. This pattern suggests that the marginal effect of temperature depends on the initial temperature level, although the estimated relationship should be interpreted cautiously within the range of temperatures observed in the data.

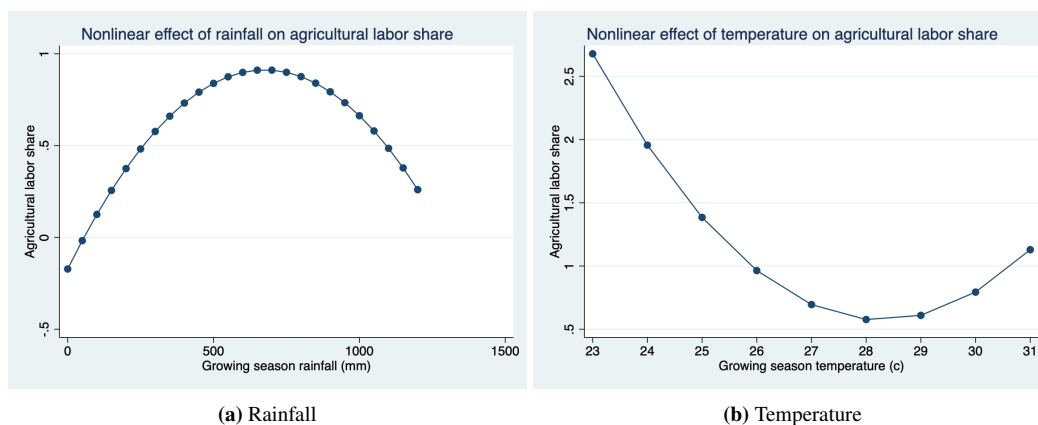
In Ethiopia, we observe similar nonlinear responses. Rainfall again follows an inverted U-shaped pattern, with both the linear and squared terms statistically significant. Temperature effects are also nonlinear, although the signs differ slightly, reflecting country-specific climatic conditions. Together, these results indicate that agricultural labor allocation responds systematically to weather conditions in Ethiopia.

In Malawi, the estimated coefficients for rainfall and temperature are statistically insignificant. Although the estimated relationships do not contradict the baseline specification, they provide weaker evidence of nonlinear weather effects than those observed in Niger and Ethiopia. The lack of statistical significance may reflect lower climatic variation or reduced statistical power in the Malawi sample, limiting the precision with which nonlinear responses can be estimated.

Overall, the results indicate that nonlinear responses to rainfall and temperature are most clearly evident in Niger and Ethiopia. For Malawi, the alternative specification provides weaker supporting evidence, but the estimates do not suggest a qualitatively different mechanism. Taken together, these findings are consistent with the interpretation that weather conditions affect sectoral labor allocation primarily through agricultural productivity, while highlighting that the strength of the evidence varies across countries.

Table 20: Agricultural Household Sectoral Share and Weather Variables

	(1)	(2)	(3)	(4)	(5)	(6)
	Niger		Malawi		Ethiopia	
	Rainfall	Temperature	Rainfall	Temperature	Rainfall	Temperature
rain_mm	0.00321*** (4.34)		0.000295 (0.42)		0.00267* (1.77)	
c.rain_mmc.rain_mm	-0.00000237*** (-3.11)		-0.000000105 (-0.34)		-0.00000123* (-1.73)	
temp_c		-4.273** (-2.34)		-0.0147 (-0.03)		0.526** (1.98)
c.temp_cc.temp_c		0.0755** (2.39)		-0.000763 (-0.06)		-0.00747 (-1.64)
_cons	-0.234 (-1.40)	60.96** (2.32)	0.644* (1.73)	1.496 (0.25)	-0.419 (-0.67)	-6.633* (-1.85)
Household Fixed Effect	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effect	Yes	Yes	Yes	Yes	Yes	Yes
Mean dep. var	0.634	0.634	0.793	0.793	0.660	0.660
SD dep. var	0.406	0.406	0.350	0.350	0.462	0.462
Observations	3780	3780	1174	1174	2240	2240
R-squared	0.689	0.682	0.730	0.730	0.726	0.729

* *t* statistics in parentheses** * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ **Figure 9:** Household Sectoral Share and Weather Variables in Niger

E.4 Alternative subsistence measure

To further assess the role of subsistence constraints, we construct an alternative proxy for subsistence based on household income. While the baseline analysis relies on a survey-based food subsistence indicator, income provides an additional measure of household resource availability. Specifically, we define a relative

Figure 10: Household Sectoral Share and Weather Variables in Malawi

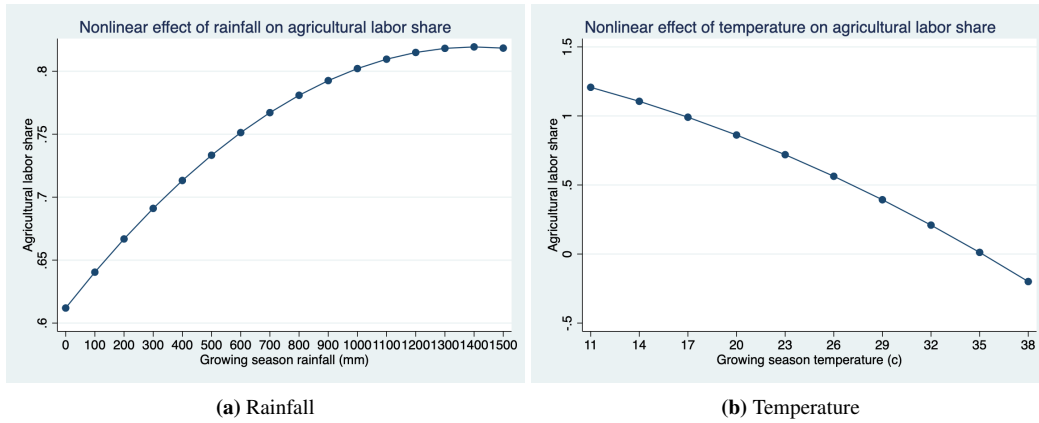
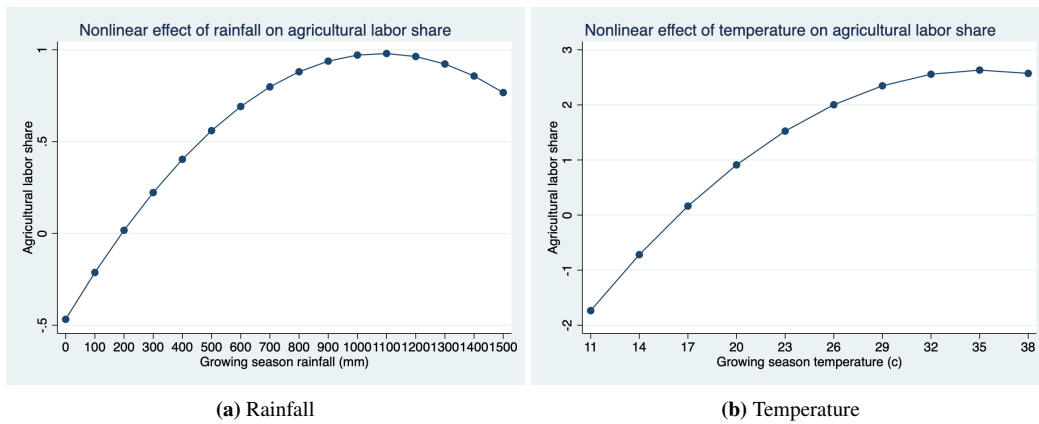


Figure 11: Household Sectoral Share and Weather Variables in Ethiopia



poverty threshold using the bottom 25 percent of the within-country distribution of per capita household income. Households below this threshold are classified as low-income and interpreted as being closer to subsistence, while those above the threshold are considered relatively unconstrained. We then estimate the effect of weather productivity shocks separately for these two groups.

Table 21 reports the results. In Niger and Malawi, productivity shocks lead to larger reductions in agricultural labor share among households above the relative poverty line, while the response among low-income households is weaker and statistically insignificant. These patterns are consistent with subsistence mechanism, whereby households below to subsistence are less able to release labor from agriculture following productivity improvements. In Ethiopia, however, the responses do not differ systematically by income group. Although the estimated effect for higher-income households is statistically significant, the point estimate for lower-income households is of similar magnitude but estimated less precisely. This pattern suggests that income proximity to subsistence is not the primary factor governing labor release in Ethiopia.

Overall, these findings reinforce the heterogeneous mechanisms identified in the main analysis. In Niger and Malawi, lower income, which implies stronger food subsistence constraints, limits households' ability to release labor from agriculture. By contrast, the appendix provides little evidence that income-based subsistence

constraints are the primary barrier in Ethiopia, consistent with the main results showing that Ethiopia's slower structural transformation is instead driven by downstream labor reallocation and labor absorption frictions.

The results are robust to alternative definitions of the relative poverty threshold. When we instead classify households in the bottom 10 percent of the income distribution as low-income, we obtain qualitatively similar patterns across countries. Productivity shocks continue to induce stronger labor reallocation among higher-income households, while the response among low-income households remains limited. Taken together, these results provide additional evidence supporting the subsistence mechanism and reinforce the interpretation that subsistence constraints play an important role in shaping labor reallocation, particularly in more constrained environments such as Ethiopia.

Table 21: Alternative Subsistence Criteria and Weather Productivity Growth

	(1)	(2)	(3)	(4)	(5)	(6)
	Niger		Malawi		Ethiopia	
	≤ 25% HH Income	> 25% HH Income	≤ 25% HH Income	> 25% HH Income	≤ 25% HH Income	> 25% HH Income
$\Delta\theta_{gt-1}$	0.404 (0.72)	-0.718* (-1.91)	-0.547 (-0.59)	-1.232** (-2.37)	-1.275 (-1.10)	-1.088* (-1.90)
_cons	0.805*** (35.40)	0.561*** (39.49)	0.781*** (29.26)	0.727*** (59.92)	1.001*** (25.56)	0.609*** (26.83)
Household Fixed Effect	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effect	Yes	Yes	Yes	Yes	Yes	Yes
Mean dep. var	0.817	0.567	0.855	0.723	0.954	0.594
SD dep. var	0.368	0.400	0.311	0.380	0.204	0.478
Observations	368	2218	108	1038	98	1622
R-squared	0.700	0.696	0.482	0.747	0.776	0.724

* *t* statistics in parentheses

** * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

E.5 Alternative Definition of Working-Age Population

In the baseline analysis, we define the working-age population as individuals aged 15-85, reflecting the fact that in developing countries many individuals remain economically active at older ages because of limited formal retirement systems. To ensure that our findings are not driven by this age definition, we conduct robustness checks using alternative age ranges.

Specifically, we re-estimate the main regressions by restricting the working-age population to ages 15–70 and 15–75. Tables 22 and 23 report the results.

Across both alternative age definitions, the estimated effects of weather productivity shocks remain highly consistent with the baseline results. In Niger, positive weather productivity shocks continue to reduce the agricultural labor share while increasing manufacturing employment, with no statistically significant effect on the service or idle labor shares. In Malawi, productivity improvements likewise reduce agricultural employment and increase manufacturing participation, while the service sector remains largely unaffected. Although the estimated increase in idle labor is positive, it is economically small and only marginally significant,

consistent with the baseline finding that most released workers are successfully absorbed into productive non-agricultural employment.

The Ethiopian results are also highly robust. Weather productivity shocks continue to reduce the agricultural labor share while increasing manufacturing employment and generating little systematic change in service employment. Most importantly, the increase in idle labor remains positive and statistically significant under both alternative age definitions. This confirms that workers released from agriculture are substantially less likely to be absorbed into productive non-agricultural employment in Ethiopia than in Niger or Malawi, consistent with the paper's interpretation that limited labor demand constrains structural transformation.

Overall, both the magnitude and statistical significance of the estimated coefficients remain remarkably stable across alternative definitions of the working-age population. These robustness checks demonstrate that the paper's central conclusions regarding labor release, labor reallocation, and labor absorption are not sensitive to the choice of working-age cutoff, reinforcing the validity of the baseline results.

Table 22: Household Sectoral Share and Weather Productivity Growth (ages 15–70)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
	Niger				Malawi				Ethiopia			
	Agriculture	Manufacturing	Service	Idle	Agriculture	Manufacturing	Service	Idle	Agriculture	Manufacturing	Service	Idle
$\Delta\theta_{gr-1}$	-0.692** (-2.20)	0.562*** (3.22)	0.0652 (0.25)	0.0983 (0.92)	-0.939* (-1.84)	0.712** (2.55)	-0.203 (-0.67)	0.197* (1.83)	-1.056** (-2.09)	0.200** (2.01)	-0.184 (-0.94)	0.899* (1.94)
_cons	0.622*** (49.72)	0.115*** (15.34)	0.220*** (19.76)	0.0264*** (6.47)	0.834*** (68.90)	0.0515*** (8.47)	0.155*** (20.46)	0.0124*** (4.60)	0.707*** (35.11)	0.0200*** (4.60)	0.0936*** (11.46)	0.139*** (8.05)
Household Fixed Effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean dep. var	0.624	0.118	0.206	0.028	0.815	0.055	0.168	0.005	0.668	0.021	0.101	0.170
SD dep. var	0.402	0.275	0.331	0.135	0.526	0.198	0.323	0.055	0.455	0.128	0.273	0.359
Observations	3722	3722	3722	3722	1844	1844	1844	1844	1860	1860	1860	1860
R-squared	0.675	0.632	0.634	0.611	0.846	0.597	0.703	0.509	0.734	0.615	0.625	0.690

Robust *t* statistics in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 23: Household Sectoral Share and Weather Productivity Growth (ages 15–75)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
	Niger				Malawi				Ethiopia			
	Agriculture	Manufacturing	Service	Idle	Agriculture	Manufacturing	Service	Idle	Agriculture	Manufacturing	Service	Idle
$\Delta\theta_{gr-1}$	-0.669** (-2.14)	0.578*** (3.32)	0.0489 (0.19)	0.0980 (0.92)	-0.917* (-1.82)	0.707** (2.54)	-0.191 (-0.63)	0.197* (1.84)	-0.996*** (-2.76)	0.168** (2.45)	-0.253* (-1.83)	0.955*** (2.90)
_cons	0.624*** (51.21)	0.114*** (15.46)	0.223*** (20.44)	0.0261*** (6.48)	0.838*** (69.08)	0.0507*** (8.28)	0.155*** (19.68)	0.0122*** (4.58)	0.689*** (48.77)	0.0219*** (7.28)	0.0880*** (16.40)	0.156*** (12.97)
Household Fixed Effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean dep. var	0.630	0.118	0.208	0.028	0.822	0.055	0.167	0.005	0.659	0.023	0.094	0.181
SD dep. var	0.405	0.276	0.332	0.134	0.522	0.197	0.322	0.054	0.461	0.135	0.265	0.369
Observations	3758	3758	3758	3758	1880	1880	1880	1880	2214	2214	2214	2214
R-squared	0.680	0.630	0.634	0.612	0.846	0.596	0.703	0.509	0.729	0.634	0.626	0.685

Robust *t* statistics in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

E.6 Multiple Distance Measures

To examine whether our results depend on how spatial frictions are measured, we conduct robustness checks using alternative distance indicators. In the baseline specification, mobility barriers are captured using dis-

tance to a specific economic center. However, different measures of remoteness may capture distinct aspects of market access and employment opportunities. We therefore re-estimate the model using three alternative distance measures: distance to the nearest road, distance to the nearest market, and distance to the nearest population center. Each measure is interacted with weather productivity shocks to capture how spatial frictions influence labor reallocation.

Tables 24–26 report the results for Niger, Malawi, and Ethiopia, respectively. Across countries, the qualitative patterns are broadly consistent with the baseline findings, although the precision of the estimates varies across alternative distance measures

In Niger, the interaction terms generally suggest that greater distance attenuates the response of non-agricultural employment to productivity shocks, although the estimates are imprecisely estimated across the alternative distance measures.

In Malawi, productivity shocks continue to increase non-agricultural participation, and the interaction terms remain broadly consistent in sign across the alternative measures of remoteness. However, statistical significance varies across specifications, suggesting that the estimated effects are sensitive to the particular distance measure and the relatively limited spatial variation in the Malawi sample.

In Ethiopia, the interaction terms generally indicate that remoteness weakens households' responses to productivity shocks, consistent with the presence of spatial mobility barriers. Although the estimated effects differ in precision across alternative measures of distance to roads, markets, and population centers, the overall pattern remains qualitatively consistent with the baseline specification.

Overall, these robustness checks suggest that the main conclusions are not driven by the particular measure of remoteness adopted in the baseline analysis. While the statistical significance of individual interaction terms varies across alternative distance measures, the estimated coefficients generally point in the same direction and continue to support the interpretation that spatial frictions can impede labor reallocation following productivity shocks.

Table 24: Multiple Distance Measures and Weather Productivity Growth in Niger

	(1)	(2)	(3)	(4)	(5)	(6)
	Road		Market		Population Center	
	Non-agriculture	Idle	Non-agriculture	Idle	Non-agriculture	Idle
$\Delta\theta_{gt-1}$	0.337	-0.0172	0.746	-0.0368	1.193**	-0.165
	(0.91)	(-0.17)	(1.41)	(-0.25)	(2.41)	(-1.28)
$\Delta\theta_{gt-1} \times \text{distance to road}$	0.0121	0.00434				
	(1.27)	(0.99)				
$\Delta\theta_{gt-1} \times \text{distance to market}$			-0.00165	0.00149		
			(-0.30)	(0.70)		
$\Delta\theta_{gt-1} \times \text{distance to population center}$					-0.00626*	0.00268
					(-1.77)	(1.58)
Constant	0.341***	0.0272***	0.340***	0.0270***	0.340***	0.0271***
	(28.23)	(6.34)	(27.82)	(6.38)	(27.69)	(6.31)
Household fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Mean dep. var.	0.327	0.028	0.327	0.028	0.327	0.028
SD dep. var.	0.392	0.135	0.392	0.135	0.392	0.135
Observations	3,780	3,780	3,780	3,780	3,780	3,780
R^2	0.679	0.610	0.678	0.610	0.679	0.611

Notes: The dependent variables are the household shares of labor allocated to non-agricultural employment and idle labor. Each pair of columns uses a different measure of spatial accessibility: distance to the nearest road, market, or population center. All regressions include household and year fixed effects. Standard errors are clustered at the community level. *t*-statistics are reported in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 25: Multiple Distance Measures and Weather Productivity Growth in Malawi

	(1)	(2)	(3)	(4)	(5)	(6)
	Road		Market		Population Center	
	Non-agriculture	Idle	Non-agriculture	Idle	Non-agriculture	Idle
$\Delta\theta_{gt-1}$	0.720	0.270*	0.686	0.0710	0.858	0.247
	(1.47)	(1.86)	(1.06)	(0.55)	(1.12)	(1.19)
$\Delta\theta_{gt-1} \times \text{distance to road}$	-0.0569*	-0.0124				
	(-1.77)	(-1.47)				
$\Delta\theta_{gt-1} \times \text{distance to market}$			-0.0317	0.0157		
			(-0.62)	(1.27)		
$\Delta\theta_{gt-1} \times \text{distance to population center}$					-0.0173	-0.00167
					(-0.88)	(-0.37)
Constant	0.189***	0.0123***	0.189***	0.0125***	0.186***	0.0121***
	(21.66)	(4.82)	(20.33)	(4.62)	(20.16)	(4.55)
Household fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Mean dep. var.	0.207	0.005	0.207	0.005	0.207	0.005
SD dep. var.	0.340	0.054	0.340	0.054	0.340	0.054
Observations	1,878	1,878	1,878	1,878	1,878	1,878
R^2	0.697	0.510	0.697	0.510	0.697	0.509

Notes: The dependent variables are the household shares of labor allocated to non-agricultural employment and underutilized labor. Each pair of columns uses a different measure of spatial accessibility: distance to the nearest road, market, or population center. All regressions include household and year fixed effects. Standard errors are clustered at the community level. *t*-statistics are reported in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 26: Multiple Distance Measures and Weather Productivity Growth in Ethiopia

	(1)	(2)	(3)	(4)	(5)	(6)
	Road		Market		Population Center	
	Non-agriculture	Idle	Non-agriculture	Idle	Non-agriculture	Idle
$\Delta\theta_{gt-1}$	0.142	0.116	0.176	1.244**	0.0129	0.331
	(0.47)	(0.21)	(0.54)	(2.12)	(0.04)	(0.49)
$\Delta\theta_{gt-1} \times \text{distance to road}$	-0.0119	0.0432*				
	(-1.34)	(1.84)				
$\Delta\theta_{gt-1} \times \text{distance to market}$			-0.00309	-0.00368		
			(-1.17)	(-0.65)		
$\Delta\theta_{gt-1} \times \text{distance to population center}$					-0.00318	0.0201
					(-0.66)	(0.85)
Constant	0.113***	0.144***	0.112***	0.159***	0.110***	0.152***
	(11.99)	(8.07)	(12.19)	(9.45)	(12.44)	(8.97)
Household fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Mean dep. var.	0.116	0.182	0.116	0.182	0.116	0.182
SD dep. var.	0.291	0.371	0.291	0.371	0.291	0.371
Observations	2,240	2,240	2,240	2,240	2,240	2,240
R^2	0.662	0.689	0.662	0.683	0.661	0.686

Notes: The dependent variables are the household shares of labor allocated to non-agricultural employment and underutilized labor. Each pair of columns uses a different measure of spatial accessibility: distance to the nearest road, market, or population center. All regressions include household and year fixed effects. Standard errors are clustered at the community level. *t*-statistics are reported in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

F Structural Estimation: Model Fit

This section evaluates the fit of the structural model by comparing empirical moments from the data with their simulated counterparts. The model is estimated using a Simulated Method of Moments (SMM) approach, targeting a set of moments that capture both aggregate labor allocation and heterogeneity across households.

The estimation targets three groups of moments. First, we match the average sectoral labor shares, including agricultural, manufacturing, and idle labor shares. These moments capture the overall distribution of labor across sectors. Second, we match moments by distance bins, which proxy for spatial frictions and access to non-agricultural opportunities. These moments capture how labor allocation varies with geographic access and help identify mobility barriers in the model. Finally, we match moments by climate bins, which capture heterogeneity in weather-driven productivity shocks.

Figures 12– 14 compare the empirical and simulated moments for Niger, Malawi, and Ethiopia. Overall, the model reproduces the main empirical patterns across the three countries. The simulated moments closely match the empirical moments used in estimation, including agricultural labor shares, idle labor shares, and

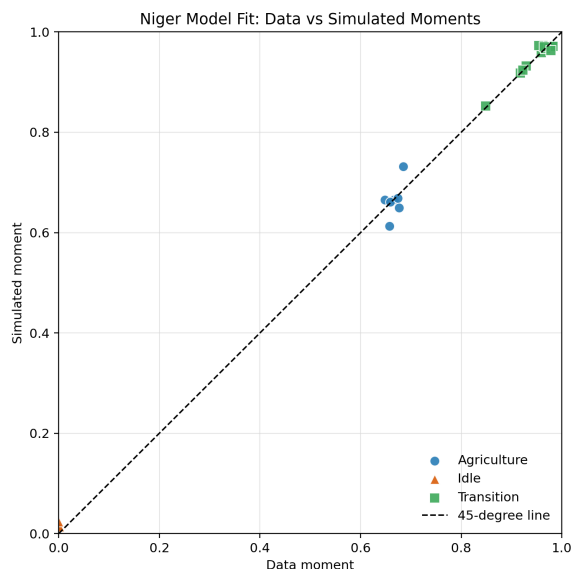


Figure 12: Model Fit in Niger

transition probabilities into productive non-agricultural employment.

In Niger, the model closely matches the aggregate labor-allocation moments and reproduces the observed variation across both distance and climate bins. In particular, it captures the relatively low idle labor share and the strong responsiveness of transitions into productive non-agricultural employment following favorable productivity shocks.

In Malawi, the model also fits the data well, particularly in terms of aggregate labor shares and their variation across climate conditions. The simulated moments closely match the empirical distribution across both distance and climate bins. While there are small deviations in some bins, especially for manufacturing shares, the model successfully captures the overall responsiveness of labor allocation to productivity and spatial conditions.

For Ethiopia, the model reproduces the main empirical patterns of labor allocation, including relatively high idle labor and weaker transitions into productive non-agricultural employment. Although the simulated moments deviate from the empirical moments for some agriculture and idle-labor bins, the model captures the qualitative relationships targeted by the estimation and provides a reasonable approximation for the counterfactual analysis.

Taken together, the model fit indicates that the structural framework is able to jointly match aggregate labor allocation and key dimensions of heterogeneity across households. In particular, the model successfully captures cross-country differences in labor reallocation patterns, including the stronger absorption of labor in Niger and Malawi and the accumulation of idle labor in Ethiopia. These results provide support for the model's ability to capture the underlying mechanisms driving structural transformation and justify its use for counterfactual policy analysis.

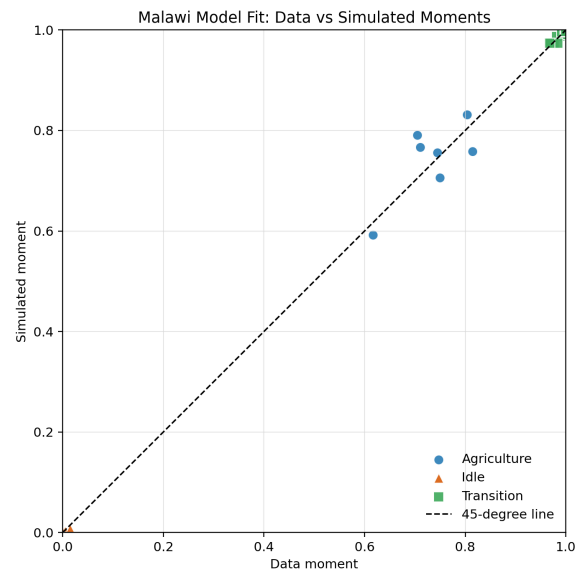


Figure 13: Model Fit in Malawi

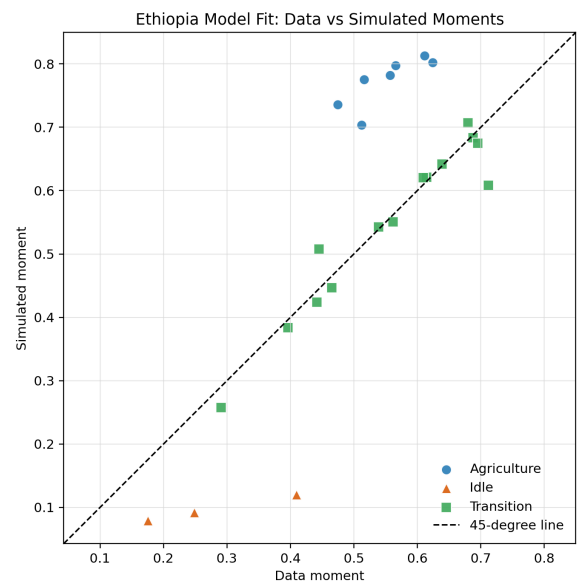


Figure 14: Model Fit in Ethiopia

G Policy Calibration

This appendix documents the calibration of the five counterfactual policy interventions. Each policy is assigned the same benchmark budget of USD 500 million to facilitate comparisons across interventions. The calibration follows a common two-step procedure.

First, the policy budget is converted into a country-specific implementation intensity using external information on policy costs together with country-specific demographic statistics. Depending on the intervention, the implementation intensity measures the share of households covered, kilometers of roads constructed, or jobs created.

Second, the implementation intensity is mapped into the corresponding structural parameter of the model. Food subsidies reduce household subsistence requirements, agricultural input programs increase agricultural productivity, rural road investments reduce migration costs, land reforms convert households from weaker to stronger land rights, and industrial investment increases the labor absorption probability. All demographic statistics are obtained from the World Bank World Development Indicators and the LSMS household surveys, while policy costs are collected from published policy evaluations and World Bank project documents.

G.1 Policy 1: Targeted Food Subsidy

The first policy represents targeted food transfers to working-age individuals in rural households whose food consumption falls below the estimated subsistence threshold. The objective is to relax food subsistence constraints and thereby reduce the amount of household labor that must remain in agriculture.

Let B denote the common policy budget. Let s_c^T denote the share of the baseline survey population that is simultaneously working age, rural, and living in a household below the food-subsistence threshold. This is a population-weighted target share and therefore differs from the household-level food-subsistence share reported in Table 1. The number of targeted individuals is calculated as

$$N_c^T = POP_c \times s_c^T, \quad (55)$$

where POP_c denotes the national population in the baseline year.

The transfer per targeted individual is

$$T_c^F = \frac{B}{N_c^T}. \quad (56)$$

Let F_c denote the annual food-subsistence expenditure benchmark per person. The policy intensity is defined as the transfer relative to this benchmark:

$$\lambda_c^F = \min \left\{ 1, \frac{T_c^F}{F_c} \right\}. \quad (57)$$

The policy is implemented for households classified as food-subsistence constrained by proportionally reducing the additional subsistence wedge:

$$\bar{c}'_i = c_0 [1 + (1 - \lambda_c^F) \delta Z_i] h_i, \quad (58)$$

where $Z_i = 1$ denotes a food-subsistence-constrained household and $Z_i = 0$ otherwise.

Table 27 reports the calibrated values of λ_c^F . The calibrated policy intensity differs across countries because both the annual food subsistence expenditure and the number of targeted households vary substantially. Countries with higher subsistence costs or a larger subsistence-constrained population receive a smaller transfer per household under the common policy budget, resulting in a smaller proportional reduction in the subsistence requirement.

Table 27: Calibration of Policy 1: Targeted Food Subsidy

Country	Target Share	Target Population N_c^T (million)	F_c (USD)	T_c^F (USD)	λ_c^F
Niger	0.120	2.058	273.75	242.90	0.887
Malawi	0.273	4.789	262.80	104.40	0.397
Ethiopia	0.022	2.527	427.05	197.88	0.463

Target share is the share of the baseline survey population that is simultaneously working age, rural, and living in a household below the food-subsistence threshold. It is therefore a person-level joint share rather than the household-level food-subsistence share reported in Table 1. The targeted population is calculated as $N_c^T = POP_c s_c^T$, where POP_c is national population in the baseline year. F_c denotes the annual food-subsistence benchmark per person. The transfer per targeted individual is $T_c^F = B/N_c^T$, where $B = \$500$ million, and the calibrated policy intensity is $\lambda_c^F = \min\{1, T_c^F / F_c\}$.

G.2 Policy 2: Agricultural Productivity Improvement

The second policy represents an agricultural input program that increases farm productivity through fertilizer, improved seeds, or related input packages. The policy is calibrated as a budget-constrained productivity intervention targeted to rural households.

Let B denote the common policy budget, C_A denote the cost of providing an input package to one rural household, and N_c^R denote the number of rural households in country c . The share of rural households reached by the policy is

$$s_c^A = \min \left\{ 1, \frac{B}{C_A N_c^R} \right\}. \quad (59)$$

If each treated household experiences a proportional productivity gain g_A , the average productivity increase is

$$\lambda_c^A = s_c^A g_A. \quad (60)$$

In the benchmark calibration, I set the cost per household to $C_A = 80$ USD and the productivity gain among treated households to $g_A = 0.15$. These values are consistent with evidence from agricultural input programs in Sub-Saharan Africa, including fertilizer and improved seed subsidy programs.

The number of rural households is constructed from national population statistics and LSMS survey data. Let POP_c denote the national population, $\bar{h}_c$ denote average household size in the baseline survey, and s_c^R denote the share of households classified as rural in the baseline LSMS survey. Total households are

$$HH_c = \frac{POP_c}{\bar{h}_c}, \quad (61)$$

and rural households are

$$N_c^R = s_c^R HH_c. \quad (62)$$

The policy is implemented as a proportional increase in agricultural productivity,

$$A_i' = (1 + \lambda_c^A) A_i. \quad (63)$$

Table 28 reports the country-specific calibration. Because the common budget covers all rural households in Niger and Malawi under the benchmark cost, the calibrated productivity increase equals the treated-household gain of 15 percent in both countries. In Ethiopia, the rural target population is much larger, so the same budget covers only part of the rural household population and implies a smaller average productivity increase.

Table 28: Calibration of Policy 2: Agricultural Productivity Improvement

Country	Total Households HH_c	Rural Share s_c^R	Rural Households N_c^R	Coverage	λ_c^A
Niger	3.953	0.637	2.517	1.000	0.150
Malawi	4.302	0.708	3.046	1.000	0.150
Ethiopia	35.420	0.419	14.844	0.421	0.063

Notes: Household counts are reported in millions. Total households are computed as national population divided by average household size in the baseline survey. Rural share is calculated from the LSMS baseline sample. Rural households are $N_c^R = s_c^R HH_c$. Coverage is $B/(C_A N_c^R)$, capped at one. The calibrated productivity increase is $\lambda_c^A = \min\{1, B/(C_A N_c^R)\} g_A$, with $B = \$500$ million, $C_A = 80$ USD per household, and $g_A = 0.15$.

G.3 Policy 3: Rural Road Infrastructure

The third policy represents investments in rural roads and transport infrastructure. In the structural model, these investments reduce spatial mobility barriers by lowering the effective cost of reaching non-agricultural opportunities.

Let B denote the common policy budget and let C_c^R denote the country-specific cost per kilometer of rural road construction. The number of kilometers financed by the policy is

$$K_c^R = \frac{B}{C_c^R}. \quad (64)$$

This calibration is intended to reduce the magnitude of the estimated distance friction toward zero. For countries where the estimated distance coefficient is positive but statistically insignificant, such as Malawi, the proportional adjustment has a negligible mechanical effect, reflecting that spatial mobility barriers are not

empirically important in the estimated model.

Let $\bar{K}_c^R$ denote the country-specific rural road gap measured in kilometers. For Niger and Malawi, the World Bank transport assessments report rural accessibility gaps as percentages rather than missing road length. To obtain a road-gap measure in kilometers, I first approximate each country's rural road stock and then multiply it by the reported accessibility gap. Specifically, for Niger, the rural road stock is proxied by the sum of regional roads (2,075 km) and unclassified roads (6,952 km), giving 9,027 km. Since approximately 40 percent of agricultural and agro-pastoral zones are reported to be isolated, the rural road gap is calculated as

$$\bar{K}_{NGR}^R = 0.40 \times 9,027 = 3,611 \text{ km.}$$

For Malawi, the classified road network is 15,551 km, of which secondary and tertiary roads account for approximately 75 percent, yielding an estimated rural road stock of 11,663 km. Because only 23.1 percent of the rural population has access to an all-season road, the rural road gap is calculated as

$$\bar{K}_{MAL}^R = (1 - 0.231) \times 11,663 = 8,965 \text{ km.}$$

For Ethiopia, the road gap is measured directly from the Roads Development Program as the difference between the planned and existing rural road network:

$$\bar{K}_{ETH}^R = (63,542 + 108,797) - (33,618 + 79,806.9) = 58,914.1 \text{ km.}$$

The share of the rural road gap closed by the policy is

$$s_c^R = \min \left\{ 1, \frac{K_c^R}{\bar{K}_c^R} \right\}. \quad (65)$$

Because road investments reduce only the transportation component of mobility barriers, I assume that complete closure of the rural road gap removes only a fraction $\bar{\rho}^R$ of the estimated distance friction. In the benchmark calibration, I set $\bar{\rho}^R = 0.5$. The resulting policy intensity is therefore

$$\lambda_c^R = s_c^R \bar{\rho}^R. \quad (66)$$

The policy is implemented by proportionally shrinking the estimated distance-friction parameter toward zero:

$$\kappa'_{1c} = (1 - \lambda_c^R) \kappa_{1c}. \quad (67)$$

Road construction costs are obtained from the World Bank Road Costs Knowledge System (ROCKS), using country-specific average costs for rural gravel reconstruction and closely related road works.

Table 29: Calibration of Policy 3: Rural Road Infrastructure

Country	Road Cost C_c^R (USD/km)	Road Gap $\bar{K}_c^R$ (km)	Financed Roads K_c^R (km)	Gap Closed s_c^R	$\bar{\rho}^R$	λ_c^R
Niger	15,960	3,611	31,335	1.000	0.500	0.500
Malawi	37,617	8,965	13,291	1.000	0.500	0.500
Ethiopia	81,500	58,914	6,135	0.104	0.500	0.052

Notes: $B = \$500$ million. Financed roads are calculated as $K_c^R = B/C_c^R$. The share of the road gap closed is $s_c^R = \min\{1, K_c^R/\bar{K}_c^R\}$. The benchmark calibration $\bar{\rho}^R = 0.5$, so that full closure of the rural road gap removes one-half of the estimated distance friction. The final policy intensity is $\lambda_c^R = s_c^R \bar{\rho}^R$, and the counterfactual migration-cost parameter is $\kappa'_{1c} = (1 - \lambda_c^R) \kappa_{1c}$.

G.4 Policy 4: Strengthening household land-rights

The fourth policy represents land certification and tenure reforms that strengthen the land rights. These reforms reduce institutional barriers to structural transformation by allowing households to retain or transfer the value of agricultural land while participating in non-agricultural activities. Unlike the previous policies, land reform changes households' institutional environment rather than directly modifying an estimated structural parameter.

Let B denote the common policy budget and let C_L denote the cost of providing land certification or tenure reform to one household. Let N_c^{NT} denote the number of households with weak land rights in country c . The policy coverage is

$$\lambda_c^L = \min \left\{ 1, \frac{B}{C_L N_c^{NT}} \right\}, \quad (68)$$

where λ_c^L is interpreted as the share of weak land-right households that receive the reform.

The number of households with weak land is calculated as

$$N_c^{NT} = HH_c \times s_c^{NT}, \quad (69)$$

where HH_c denotes the total number of rural households and s_c^{NT} is the share of households classified as having weak land rights in the baseline LSMS survey.

The benchmark calibration follows the World Bank's estimates of land certification costs. The average implementation cost is approximately USD 50 per household, including cadastral mapping, registration, certificate issuance, and administrative costs. Under the common policy budget of USD 500 million, the available resources are sufficient to provide land certification to all households with weak land rights in the three study countries. Consequently,

$$\lambda_c^L = 1$$

for Niger, Malawi, and Ethiopia.

The policy is implemented by converting households with weak land rights into stronger land rights:

$$T'_i = \begin{cases} 1, & \text{if } T_i = 0, \\ T_i, & \text{otherwise.} \end{cases} \quad (70)$$

For each household, the probability of accessing non-agricultural opportunities is then recalculated as

$$q'_i = \Lambda(\mu_0^{NT}(1 - T'_i) + \mu_0^T T'_i + \kappa_1 dist_i). \quad (71)$$

The policy therefore changes household outcomes by expanding the share of households with stronger land rights while leaving the estimated structural parameters unchanged.

Table 30: Calibration of Policy 4: household land-rights

Country	Weak land-right Share (s_c^{NT})	Weak land-right Households (N_c^{NT} , million)	Coverage (λ_c^L)	Converted Share
Niger	0.941	2.37	1.000	100%
Malawi	0.126	0.38	1.000	100%
Ethiopia	0.281	4.17	1.000	100%

Notes: Weak land-right households are identified as households with land type 0 in the baseline LSMS survey. Land types 1 and 2 are classified as stronger rights. The number of weak land-rights households is calculated as

$$N_c^{NT} = s_c^{NT} N_c^R,$$

where N_c^R is the total number of rural households reported in Table 28. The benchmark implementation cost is USD 50 per household. Under the common budget of USD 500 million, the available resources are sufficient to provide land certification to all households with weak land rights in each country. Consequently, the policy coverage is $\lambda_c^L = 1$, and all Weak land-right households are converted to stronger land rights before the household-specific probability of accessing agricultural opportunities is recalculated.

G.5 Policy 5: Increasing Labor Absorption Capacity

Policy 5 represents industrial investment that expands the ability of the non-agricultural sector to absorb workers. Unlike the previous policies, which relax supply-side constraints faced by households, this intervention targets the demand side of structural transformation by increasing the probability that workers who successfully reallocate into the non-agricultural labor market obtain productive employment.

The calibration proceeds in two steps. First, the common policy budget is translated into the number of jobs created using evidence from recent industrial investment projects. Second, the implied job creation is converted into an increase in the labor absorption probability in the structural model. The structural model does not explicitly model vacancy creation or job matching. Instead, the effect of industrial investment is summarized by an additive increase in the probability that a worker seeking employment is successfully absorbed into productive non-agricultural employment.

Following the World Bank Competitiveness and Job Creation Project, I use the construction of the Bole Lemi II and Kilinto Industrial Parks in Ethiopia as the benchmark investment. The project provided approximately USD 425 million in financing and created approximately 19,000 direct jobs. Scaling this benchmark to the

common policy budget of USD 500 million implies

$$J_c = \frac{500}{425} \times 19,000 = 22,353 \quad (72)$$

new jobs.

To account for differences in labor market size across countries, the same investment is expressed relative to the unemployed population in each country. Let U_c denote the number of unemployed workers in the benchmark labor market. The calibrated increase in labor absorption capacity is

$$\lambda_c^D = \frac{J_c}{U_c}. \quad (73)$$

Because the model-implied pool of reallocated workers seeking employment is not directly observed, I use the observed unemployed population in the benchmark labor market as its empirical proxy. Unemployed workers constitute the closest observable group of individuals who have entered the labor market but have not yet been absorbed into productive employment. Under the stylized assumption that newly created jobs are allocated proportionally among workers seeking employment, the ratio J_c/U_c approximates the increase in the probability that a reallocated worker successfully obtains a productive non-agricultural job.

$$p'_c = \min \{1, p_c + \lambda_c^D\}, \quad (74)$$

where p_c denotes the estimated labor absorption probability from the structural model.

The objective of the counterfactual analysis is to isolate the contribution of each structural friction to the process of structural transformation. Accordingly, each policy perturbs a single model parameter while holding all other mechanisms fixed. The labor-absorption policy should be interpreted as a stylized, non-spatial simulation that abstracts from the geographic placement of industrial parks and their interaction with household-specific migration costs. Instead, the policy is modeled as a uniform increase in the labor-absorption probability faced by workers who have successfully entered the non-agricultural labor market.

For Ethiopia, the benchmark labor market is Addis Ababa because both industrial parks are located in the city. The unemployed population is obtained from the 2020 Urban Employment and Unemployment Survey (UEUS). For Niger and Malawi, the unemployed population is obtained from the most recent ILOSTAT labor force statistics. Using country-specific labor market sizes allows the same investment budget to generate different increases in labor absorption capacity across countries.

Table 31: Calibration of Policy 5: Labor Absorption Capacity

Country	Investment	Jobs Created	Reference Labor Market	λ_c^D
Niger	\$500 million	22,353	...	...
Malawi	\$500 million	22,353	...	...
Ethiopia	\$500 million	22,353	Addis Ababa UEUS	0.056

G.6 Interpretation of Policy Calibration

The purpose of the policy calibration is to compare the effectiveness of alternative interventions under a common fiscal budget rather than to evaluate the exact impacts of specific government programs. To ensure comparability across policies and countries, each intervention is assigned the same benchmark budget of USD 500 million. Country-specific policy intensities therefore arise from differences in demographic structure, infrastructure needs, land institutions, and labor market conditions rather than differences in public expenditure.

The calibration follows a common two-step procedure. First, the benchmark budget is translated into a policy implementation intensity using external information on program costs together with country-specific demographic or geographic statistics. Second, the implementation intensity is mapped into the corresponding structural parameter of the model. Food subsidies reduce the household subsistence requirement, agricultural input programs increase agricultural productivity, road investments reduce migration costs, land reforms convert households from weak to stronger land rights, and industrial investment increases the probability that workers are absorbed into productive non-agricultural employment.

The resulting policy parameters are summarized in Table 8 and are held fixed throughout the counterfactual analysis. The common policy budget is translated into country-specific changes in the relevant structural parameters using the calibration procedures described above. Consequently, differences in simulated policy outcomes reflect both the estimated structural parameters and the calibrated policy intensities implied by country-specific baseline conditions. Because the same calibration framework is applied consistently across countries under a common fiscal budget, the counterfactual experiments provide a comparable basis for assessing how country-specific structural constraints influence the effectiveness of alternative development policies.